

RokDoc Seismic Data Conditioning & Inversion Training

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Renaissance Africa Energy Company Limited





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Overview

This workbook provides an introduction to simultaneous inversion and seismic data conditioning (SDC) in RokDoc. It begins with an overview of inversion and data conditioning in general, followed by a step-by-step progression through the procedures involved in preparing for and carrying out an inversion. Each element of the workflow is presented with guidelines on best practice and recommended procedure. At the end there is a brief explanation of an alternative workflow using elastic impedance and coloured inversion.



WHY INVERT DATA?

Simultaneous inversion

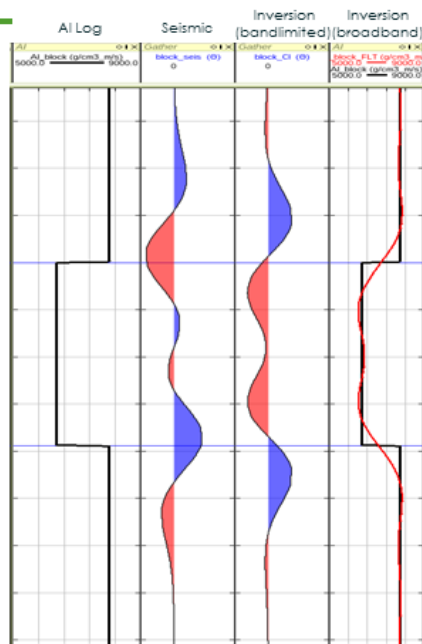


What does it do?

- Input: pre-stack seismic trace data
 - Angle gathers or angle stacks
 - Output: impedance volumes
 - Acoustic impedance
 - Shear impedance
 - (Density)
- Note that this course is about a deterministic inversion that gives a single result. Other types of inversion can give probability distributions for subsurface quantities, or multiple equally probable realisations.

Why would this be a useful thing to do?

Benefits of Inversion



Consider an acoustic contrast as shown in track 1;

Track 2 shows the **seismic** response; amplitudes occur at the interfaces.

Track 3 shows a response from a **bandlimited inversion**; amplitudes occur within the intervals. However the response is still relative (i.e. zero mean).

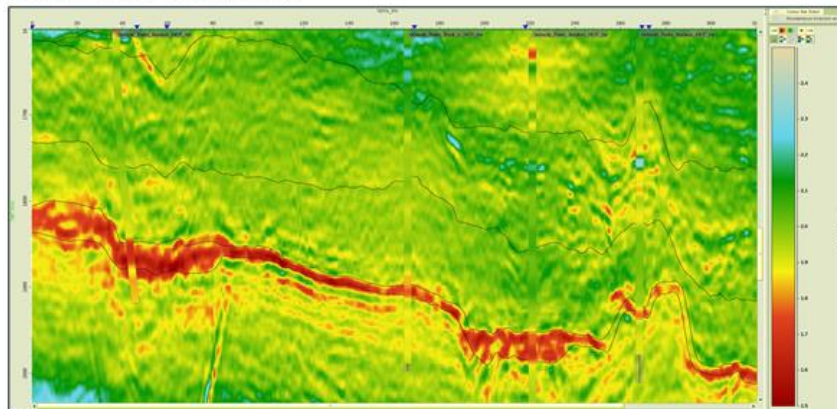
Track 4 show the response from a **broadband inversion** against the original log; the seismic amplitudes have been transformed into impedance values and are no longer relative.

Benefits of inversion



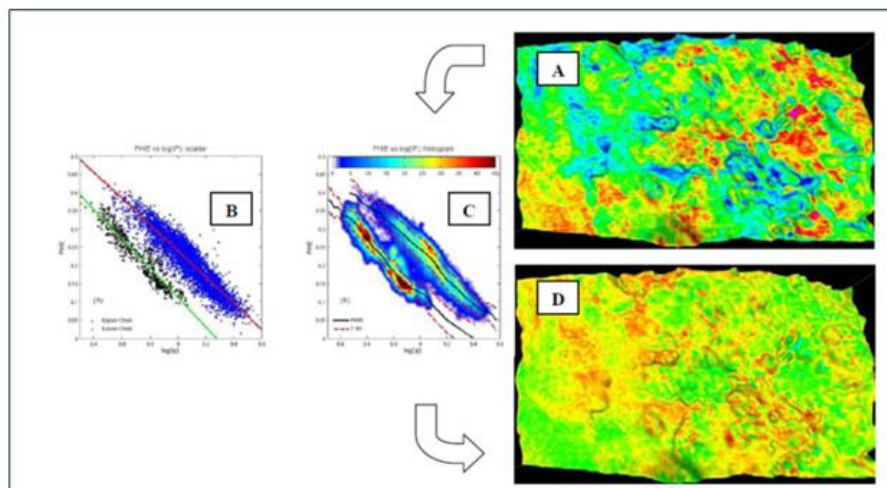
Layer-based properties

- Much easier to understand than seismic wiggles
- More direct input to reservoir models



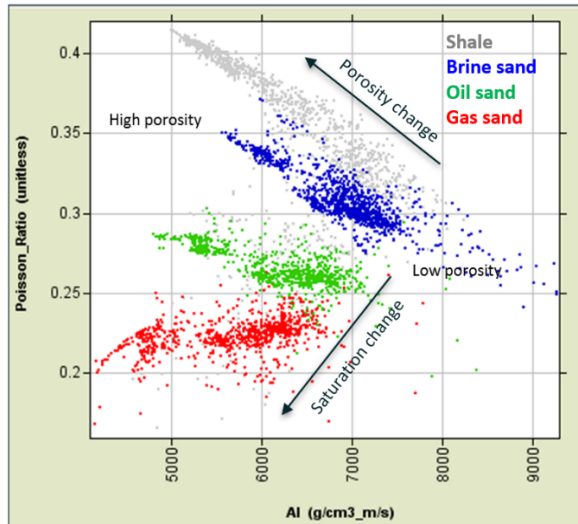
Sand distribution = red layer (low V_p/V_s)

Benefits of inversion

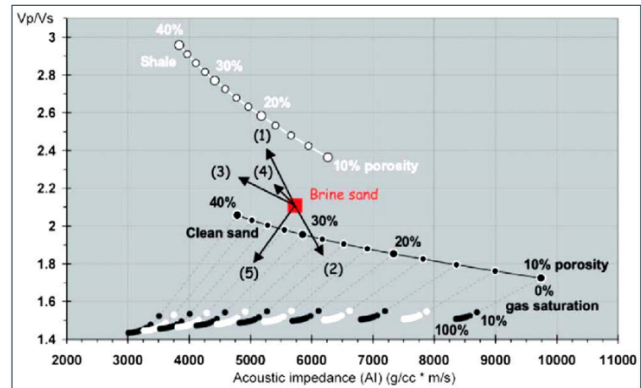


Map of acoustic impedance (A) is transformed to reservoir porosity (D) using well-based transforms (B,C). Hansen et al (2011).

Acoustic impedance vs Poisson's ratio and Velocity Ratio



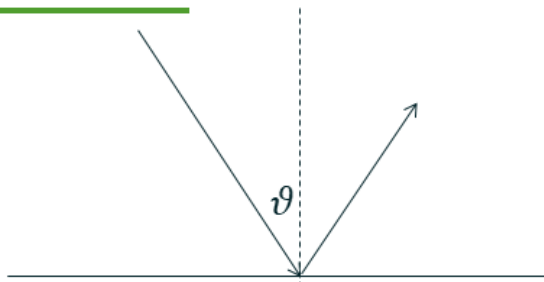
If we can work out AI and PR from seismic data, we could use them to predict sand presence and fluid fill.



Typical example of rock physics template (RPT) superimposed onto VP/Vs versus Acoustic impedance cross plot (Avseth et al. 2005).

HOW DO INVERSION ALGORITHMS WORK?

How simultaneous inversion works



The reflection coefficient can be calculated from the contrast in properties across the interface.

$$R = (1 + \tan^2 \theta) r_{pp} - 8 \left(\frac{V_s}{V_p} \right)^2 \sin^2 \theta r_{ss} - \left(\frac{1}{2} \tan^2 \theta - 2 \left(\frac{V_s}{V_p} \right)^2 \sin^2 \theta \right) \frac{\Delta \rho}{\rho}$$

(Fatti et al 1994)

where
 $r_{pp} = \frac{\Delta Z_p}{2Z_p}$ and $r_{ss} = \frac{\Delta Z_s}{2Z_s}$

The first term depends on the contrast in acoustic (Pwave) impedance $Z_p (= \rho V_p)$ and the second term depends on the contrast in the Swave impedance $Z_s (= \rho V_s)$. The third term depends on the density contrast, but is small if (V_s/V_p) is close to 0.5 as it usually is. Therefore, amplitude variation with angle can tell us about Z_p , Z_s and to a lesser extent density.

The Shuey equation

A further simplification, which forms the basis of much practical AVO analysis, is to write

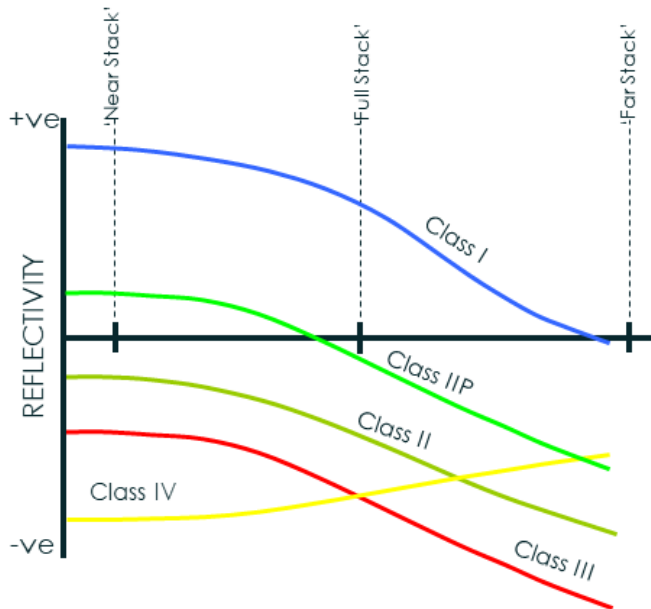
$$R = A + B \sin^2 \theta.$$

This is typically valid out to incidence angles of around 35° or so. A is called the AVO intercept and B the AVO gradient.

For a reflection from a particular interface, we could measure the amplitude as a function of incidence angle and thus calculate intercept and gradient.

If we have data for reflections from a number of interfaces, one way to analyse them is to plot them on an intercept-gradient cross-plot, an idea we'll come back to later.

AVO response



The way amplitude changes as a function of angle depends on the contrast in V_p , V_s and density across the interface.

The **AVO intercept A** depends on the contrast in P impedance (acoustic impedance) across the interface

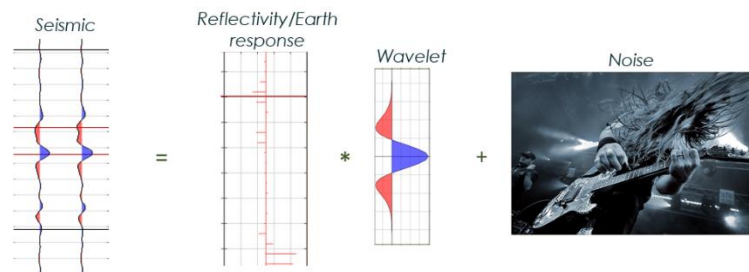
$$AI = \rho V_p$$

The **AVO gradient B** depends on the contrast in Poisson's ratio across the interface

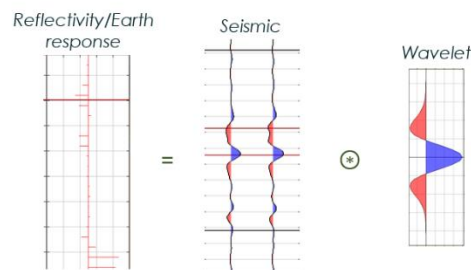
$$\sigma = \frac{0.5 - \left(\frac{V_s}{V_p}\right)^2}{1 - \left(\frac{V_s}{V_p}\right)^2}$$

Among minerals, quartz is anomalous in having a very low Poisson's ratio. Therefore sandstones have a low Poisson's ratio compared to shales, so a shale/sand interface may have a large PR contrast and a high AVO gradient. Hydrocarbon porefill in a sandstone lowers the PR still further so gives a bigger AVO gradient.

The convolution model of seismic

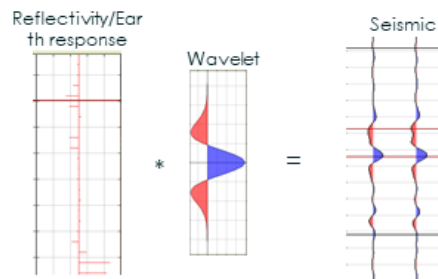


In practice we ignore the noise (to some extent), so in theory we could try to rearrange or "invert" this equation to derive the earth response from seismic like so:

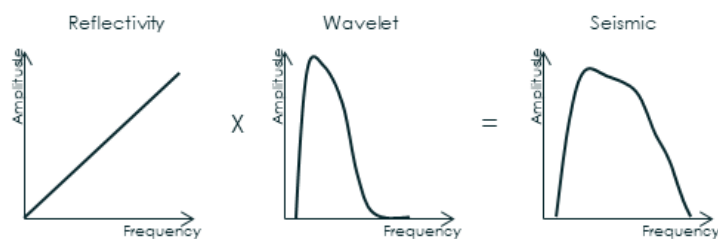


In reality this doesn't work out...

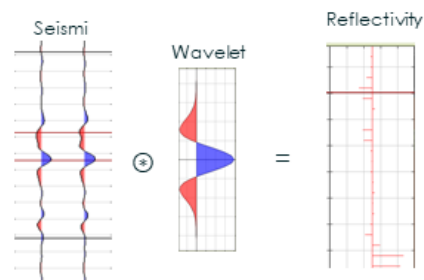
Convolution in Fourier Domain



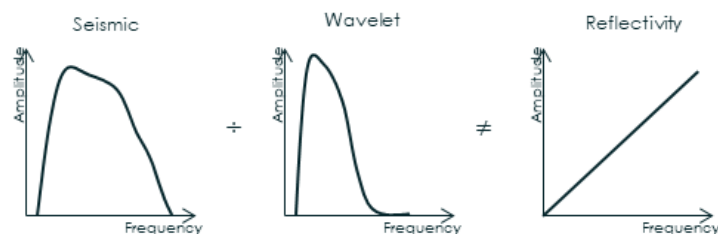
Convolution in the time domain is equal to multiplication in the Fourier domain



Deconvolution



Deconvolution in the time domain is equal to division in the Fourier domain



Attempting this would result in dividing by zero. The bandlimited nature of seismic means that we cannot reconstruct reflectivity using deconvolution.

Impedance from Reflectivity

There are some mathematical 'tricks' that could be used in order to remove zeros and perform deconvolution. However recovering impedances from reflectivity is still problematic as it is non-unique. Consider the relationship between reflectivity and impedance at a given boundary k :

$$R_k = \frac{Z_{k+1} - Z_k}{Z_{k+1} + Z_k}$$

We would get the same answer for R_k if Z_k and Z_{k+1} were equal to 1 and 4, or 10 and 40 or 200 and 800, and so on... The answer is non-unique.

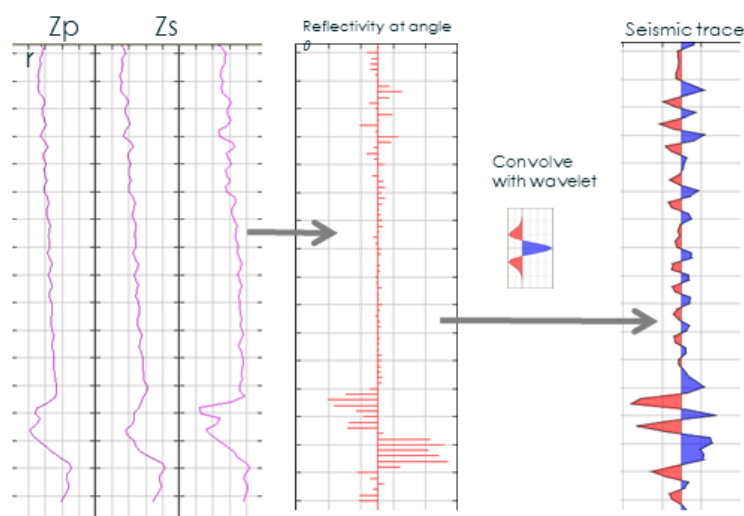
If we could recover reflectivity and in put an *a priori* top layer (Z_k in this case) then we could work out Z_{k+1} and then recursively work out Z_{k+2} and Z_{k+3} ...

However for this to return the correct result, the following conditions must be met:
The source wavelet must be completely removed, noise must be absent, all multiple reflections must be removed and all spherical and transmission losses – including out of plane reflections – must be completely removed. Without these conditions met the calculated impedance would quickly "drift" from the correct result.

Given that the above conditions are never likely to be met when dealing with seismic data; model based approaches to inversion are preferred.

Calculating synthetic response

If we have a stack of layers and know Z_p , Z_s and r in each one we can calculate for a given angle the reflection coefficients. If we convolve the reflection coefficients with a wavelet then we model the seismic trace at that particular incidence angle.



The idea of inversion is to reverse this process. Given seismic traces at several angles, and knowing the wavelet, we get back to the layer properties Z_p , Z_s and r .

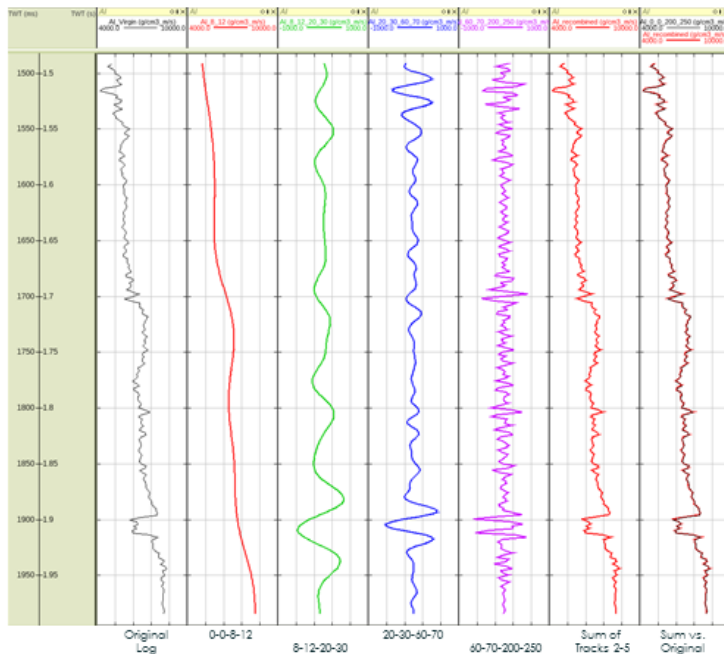


This has several implications:

- We need to know the wavelet.
- AI determines the AVO intercept, the reflection coefficient at zero angle of incidence. This is easily measured so we can hope for accurate Z_p values.
- SI determines the AVO gradient, the slope of the line if we plot amplitude against $\sin^2\theta$. This is harder to measure because of the effect of noise. We expect Z_s to be less well determined than Z_p therefore.
- Density enters the equation in the third term, multiplied by a small coefficient. The seismic trace is therefore relatively insensitive to density, and conversely it will be hard to estimate density from seismic.

THE BANDWIDTH LIMITATION

Deconstructing logs into frequency components



The original AI log is shown in track 1.

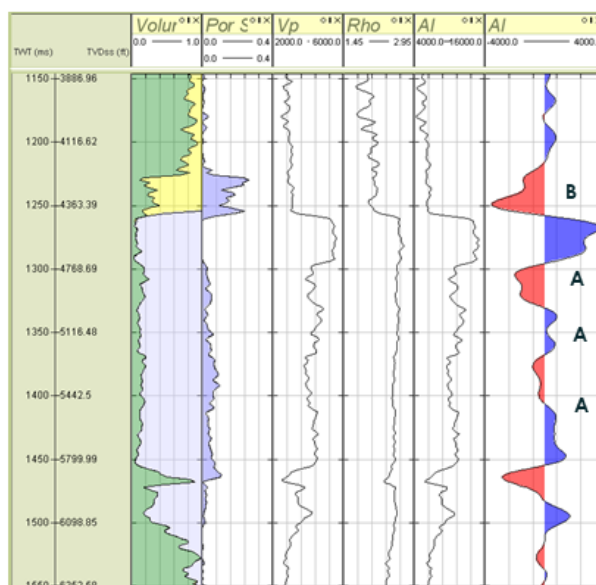
In tracks 2-5 different filters have been applied to the AI log. Each log displays a different band of the original AI log.

In track 6 the sum of tracks 2-5 is displayed. With all frequency components added together the original log has been reconstructed.

In track 7 the original log overlays the reconstructed log. Note that the two are identical (apart from the log thickness which was altered for display).

The bandwidth limitation

A fundamental issue is the limited bandwidth of the seismic traces.



Left AI track is rV_p from logs

Right AI track is filtered 5-10-40-60 Hz
Note it is zero mean.

At points A the filtered AI has a strong resemblance to actual AI

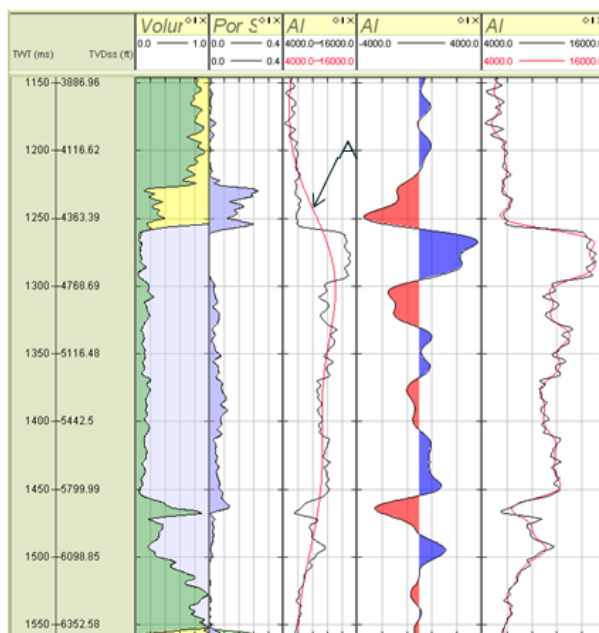
This is because the background is more or less constant

At point B the relative AI is a doublet, not the real step change.



In some cases we may be quite happy to work with this band-limited impedance, if we are interested in beds whose thickness is similar to that of a seismic loop, set in a fairly constant background. In that case we might use simple and robust approaches such as coloured inversion (Lancaster and Whitcombe 2000). However, if we are working with thick layers, or gradational changes, or needing to compare the impedance against a threshold value (for example to distinguish pay sand from non-pay), then we need to invert to absolute impedance values.

Adding back the low frequencies



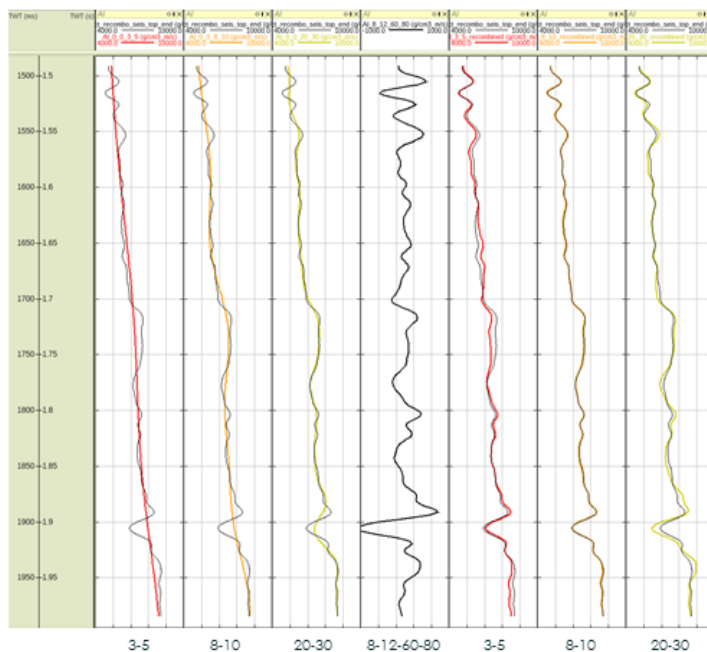
If we can somehow create the low-frequency trend (curve A) and add it to our band-limited impedance, we get the red curve on the right. This is in good agreement with the original data (black curve), though we have lost resolution because the seismic is band-limited at the high-frequency end too.

This means that we have to find the low-frequency trend from somewhere other than the seismic traces. Possible inputs are well data, geological models and seismic velocities.



The starting model for the inversion has to supply the low-frequency background that is missing from the seismic data. Techniques for expansion of the seismic bandwidth at the low-frequency end are under active development by seismic acquisition contractors and will reduce the uncertainty in inversion output, but the problem cannot be eliminated.

Recombining Z from LFBM and Seismic



In all tracks black log is high cut filtered (60-80 corner points) to simulate typical high end of seismic bandwidth.

Track 4: AI log has been filtered 8-12-60-80 to simulate seismic bandwidth.

In tracks 1-3 the coloured logs are filtered versions of the black log using high cut filter (corner points shown below log).

In tracks 5-7 the low frequency component of tracks 1-3 have been added to the bandlimited log in track 4.

For perfect recombination a low frequency of 0-0-8-12 would need to be added to track 4. If the frequencies added are two low the AI will be underestimated and visa versa.



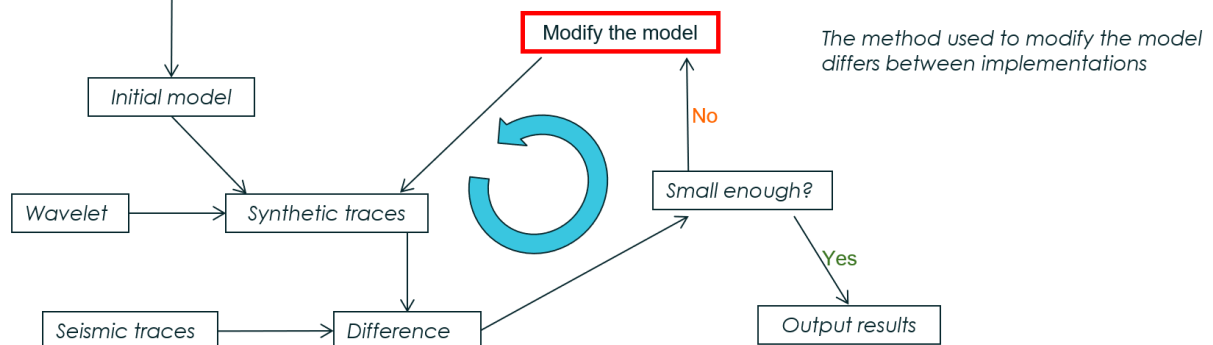
Different inversion packages have different ways of calculating the required impedance model. The method used in RokDoc is to start with an initial model (based for example on extrapolating impedance data away from well control) and modifying it so as to give a seismic response that agrees with the real trace data. The real seismic data contain noise, of course, so trying to fit the traces exactly may lead to unrealistic impedance results. This is kept under control by specifying the extent to which the impedance can deviate from the initial model. This is not set as a hard constraint, but rather as a penalty for deviation from the model which is balanced against a penalty for not matching the trace data exactly.

Inversion algorithms



Even if we know the wavelet exactly, there is no simple solution to get from the observed seismic traces back to the layer properties. Some form of iterative process is needed, where we start with a model and modify it until the response matches the observed seismic traces.

Horizons, Well log data, Seismic velocities, variograms





THE TRUE-AMPLITUDE ASSUMPTION

The true amplitude assumption



Inversion assumes that the seismic trace represents the convolution of a wavelet with the reflectivity series at the appropriate incidence angle.

There are lots of complications in practice:

- Acquisition artefacts: variation in source strength/receiver sensitivity from one shot to another; array directivity
 - Can be removed in processing
- Multiples
 - Can be removed in processing but perhaps with damage to primary amplitudes
- Absorption: wavelet changes with travel time, not only in amplitude but also in frequency content and phase
 - Can be removed in processing but needs to be checked against well ties
- Spherical divergence
 - Can be removed in processing but needs to be checked against well ties

The true amplitude assumption II



- Thin-bed tuning
 - Causes change in amplitude with offset that may be very hard to allow for
- Anisotropy
 - VTI anisotropy is common in shale; the effect on synthetic response is easily calculated but usually the parameters to describe the anisotropy are not known
 - HTI anisotropy is potentially signal: it can be used to infer fracture density and orientation, though not usually through inversion
- Near-surface effects
 - Shallow gas will cause amplitude shadowing; the effect can be partly removed by AGC but will cause problems if the effect is severe or varies rapidly within a spread length (full stack only)



ELEMENTS OF AN INVERSION WORKFLOW

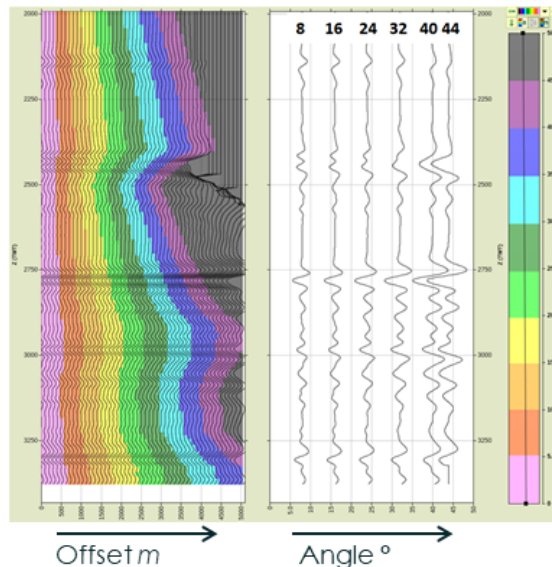
Elements of an inversion workflow



- Input data: angle gathers
- Radon de-multiple, trim statics, angle limited stacking
- Well ties and wavelets
- SDC:
 - zero phasing
 - spectral balance
 - aligning events
 - correcting amplitude scaling
- Well ties and wavelets (post SDC)
- Background model from wells, seismic
- Inversion parameter testing
- Inversion QC
- Derived outputs e.g. Poisson's ratio

INPUT SEISMIC DATA

Input data: angle gathers



Some PSDM algorithms are able to output angle gathers or angle stacks directly.

How do we calculate the offset to angle relationship?

Several approximate formulae are available, but we need to know the velocity structure in the subsurface. This can come either from well data or from seismic stacking velocities. Inaccuracy in offset to angle conversion is a common source of uncertainty in AVO studies.

Knowing the offset – angle relation, we can output angle gathers (eg at 1° angle increment) or angle stacks.

Quite often, the only input available to a prestack inversion will be angle stacks. These are usually produced with the needs of the interpreter in mind, to obtain an overview of AVO effects in relation to structure or mapped prospects. For this purpose only a few angle stacks are needed, because it is not possible to view more than two or three simultaneously on the average interpretation workstation. It is also usual to include data out to quite far angles, up to the point where critical angle effects start to become obvious. Thus a typical dataset might be four substacks: near $0 - 12^{\circ}$, mid $12 - 24^{\circ}$, far $24 - 36^{\circ}$, ultrafar $36 - 48^{\circ}$. In some cases the very near traces might be excluded if they show obvious evidence of multiple contamination (or damage to primaries due to aggressive multiple removal). For quantitative AVO studies, such a dataset is not ideal. An approximate equation such as Fatti's is likely to be adequate only out to 35 or 40° , and the rather wide angle range in each substack introduces some smearing of the AVO effect. Given the choice, one might therefore prefer to have angle stacks with ranges such as $0 - 7^{\circ}$, $7 - 14^{\circ}$, $14 - 21^{\circ}$, $21 - 28^{\circ}$, $28 - 35^{\circ}$ and $35 - 42^{\circ}$.



WELL TIES AND WAVELETS

Well ties and wavelets



Inversion needs good well ties

We are going to create a model of the subsurface elastic properties that honours the seismic trace data. If the well logs lead to a synthetic trace that doesn't match the real data, we can guarantee that the inversion result won't match either the well data or the seismic traces (or both).

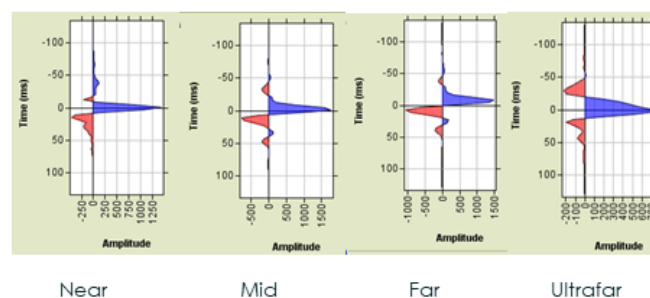
If the well ties are poor at the objective level, consider postponing inversion: revisit log and seismic quality issues instead.

The well tie is also key to establishing the time-depth relation. This is important if we are going to build a background model (in TWT) from well log data (in depth).

The wavelet



Accurate knowledge of the wavelet is critical for inversion. It will generally vary with offset:



Note change in frequency content, phase rotation, amplitude change with offset.

In addition there will be change with increasing travel time, and maybe lateral changes in amplitude (which will be apparent if we have several wells to tie).

Seismic Data Conditioning

ZERO PHASING

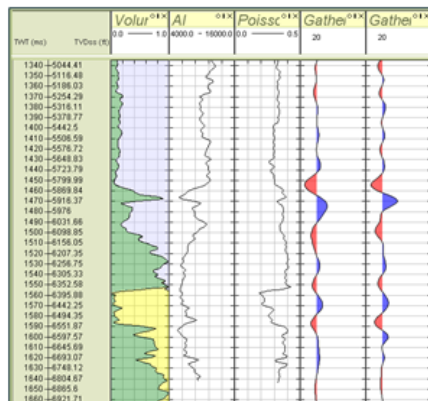
Seismic Data Conditioning



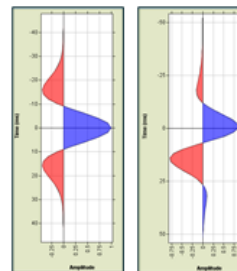
A typical sequence will be as follows:

- (1) Zero phase the input seismic

This is not necessary for the inversion itself, as the wavelets are explicit inputs, but it is needed prior to time alignment.

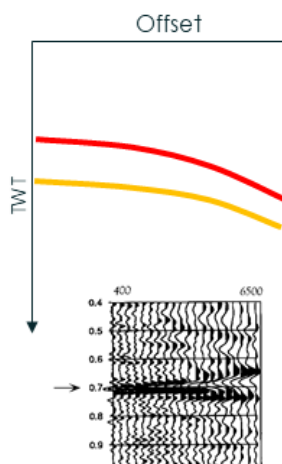


The two synthetic traces look as though they need a time shift for alignment, but in fact the difference is just in the phase of the wavelet:



SPECTRAL BALANCING

Seismic Data Conditioning

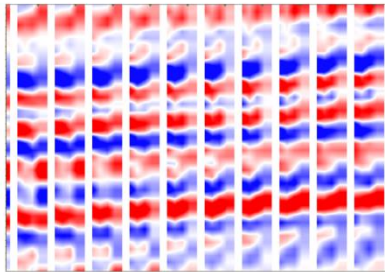


- (2) Spectral balancing

The reflections in a gather show approximately hyperbolic normal moveout (NMO). These curves converge towards the far traces. When NMO correction is applied to line up reflections horizontally across the gather, the traces are stretched at the far offsets. A further contribution to stretch of the far offsets comes from the effect of migration on dipping reflectors.

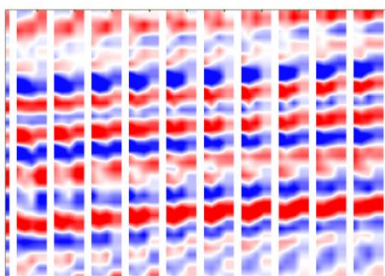
The frequency change from near to far can be corrected by equalisation of the spectrum across the gather. This is not needed for inversion where the wavelet is explicitly taken into account, but it matters for other types of AVO analysis (eg AVO gradient calculation) because interference between reflections from adjacent beds will change depending on the frequency, potentially causing false AVO anomalies.

Spectral balancing

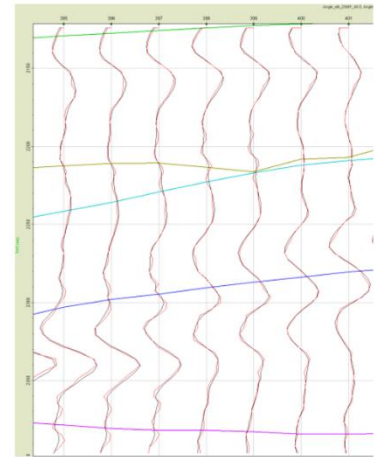


Before

The differences are subtle.



After



Ultrafar stack **before** and
after spectral balancing

ALIGNING REFLECTIONS ACROSS THE GATHER

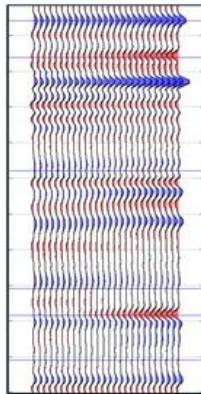
Seismic data conditioning



(3) Aligning reflections across the gather

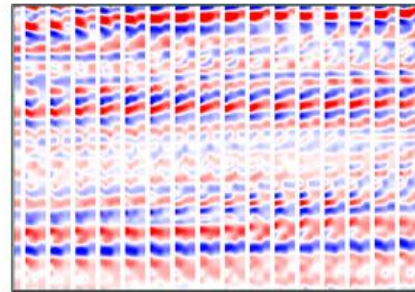
The seismic processor should have removed NMO so that most reflections are flat across the gather. This ideal is often not achieved: the velocity field used to calculate NMO correction may be incorrect, the algorithm may be inadequate (fourth order in offset is needed at large offset), or anisotropy may be an issue.

Because of interference and AVO effects, some reflections are not expected to be flat even after perfect NMO removal; the first step is to calculate an offset synthetic.



In this case, the synthetic shows that the reflections should be very nearly flat, but.....

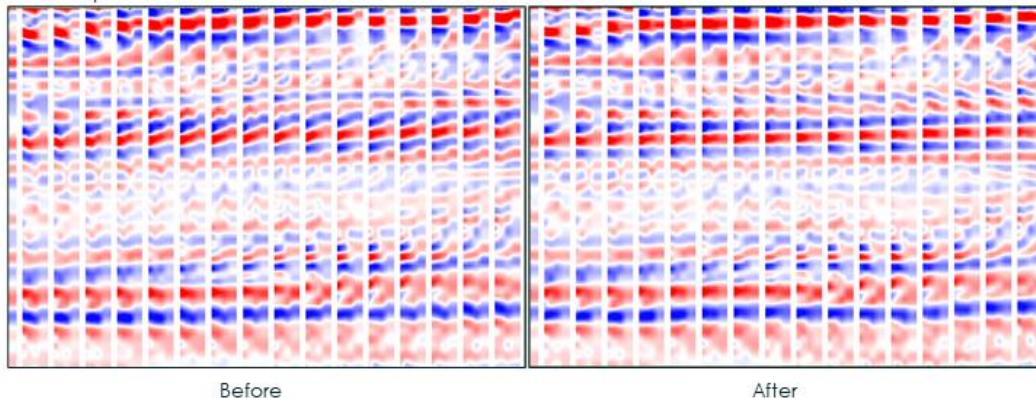
...in fact they aren't.



Aligning reflections



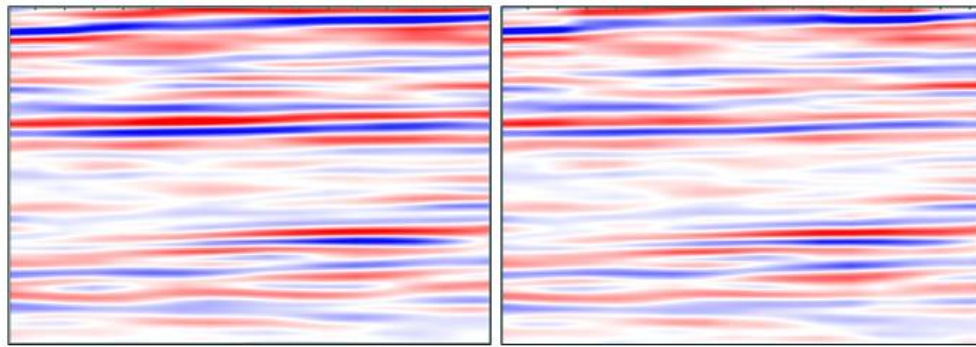
Trim static solution: in a window sliding down the trace, cross-correlation is calculated between (say) far trace and near trace, and hence the shift required to line up far with near. Interpolating between shifts at different TWTs gives a continuous shift that lines the traces up.



There's a balance to be struck: aligning the important events makes others worse.

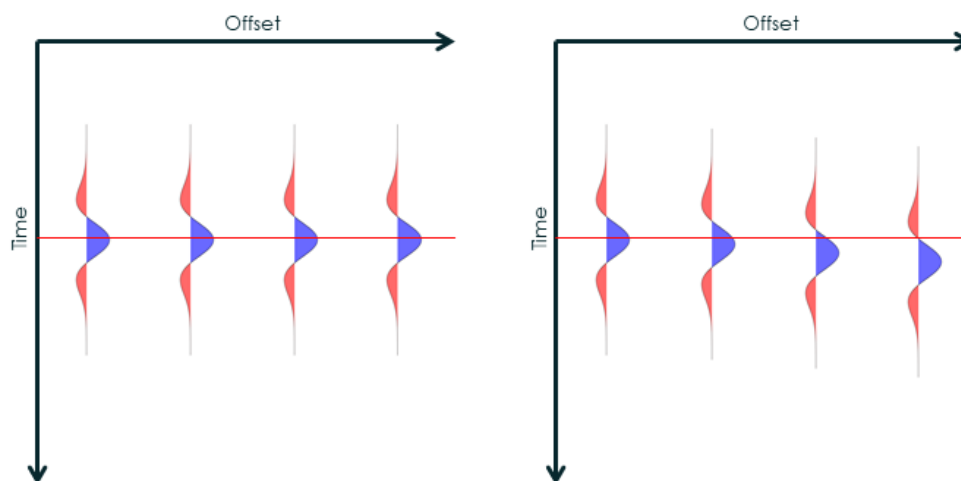
Aligning reflections

The AVO gradient is reduced in amplitude by trace alignment:



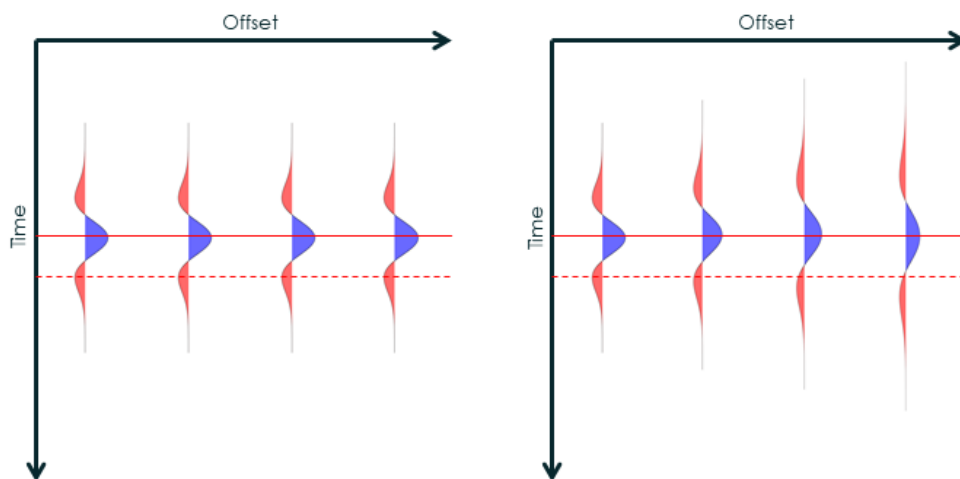
The effect of misalignment on AVO gradient is to increase its amplitude, perhaps quite markedly, so this amplitude reduction is an indicator of data improvement. The downside of the method is that (though not visible here) steps are likely to be generated in the shifted reflections. This effect can be removed by lateral smoothing of the time-shifts.

Time Alignment



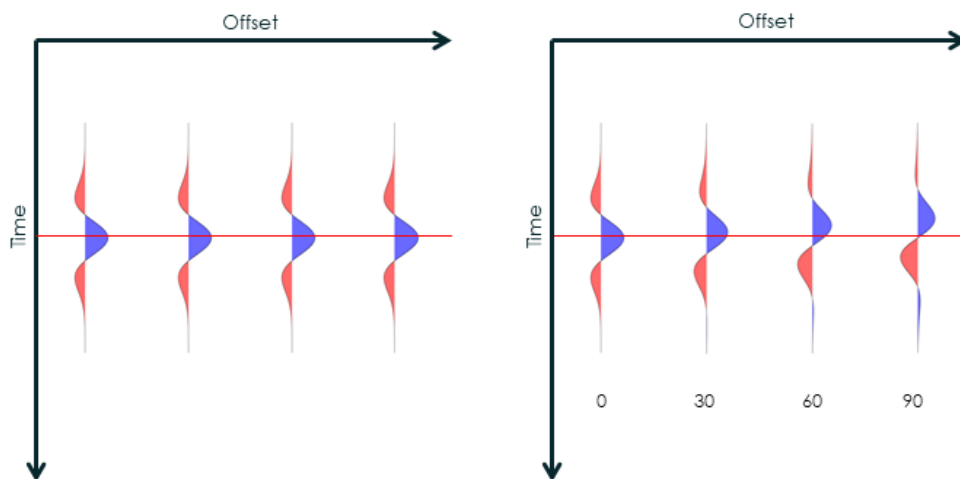
This slide illustrates the influences of poor time alignment.

Frequency



This slide illustrates the influence of spectral differences.

Phase



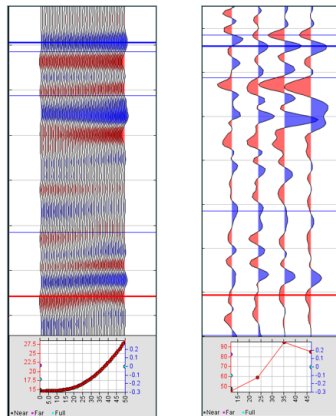
This slide illustrates the influence of phase differences.

CORRECTING AMPLITUDE SCALING

Seismic data conditioning: Amplitude Scaling

(4) Correcting amplitude scaling

The seismic processor will have tried to correct for spherical divergence (energy spreading over a larger wavefront area as distance from the source increases) and for absorption (Q compensation). However, these corrections are only approximate. For example, overburden VTI anisotropy can have a big effect on spherical divergence, and Q is rarely known accurately. Therefore, we need to compare amplitude variation across the offset range with a well synthetic.



One method is to compare well synthetic to seismic by measuring RMS amplitude in a large window containing some strong events (or perhaps along an event of well-known AVO characteristics) and comparing ratios of mid:near, far:near and Ultrafar : Near for synthetic and seismic.

In this case the ratios were:

Mid/near	Seismic	1.18	Synthetic	1.10
Far/near		1.86		1.33
Ultrafar/near		1.70		1.87

so the seismic scalars would be mid $1.10/1.18 = 0.93$, far 0.72 , Ultrafar 1.10

Correcting amplitude scaling

- Should calculate scaling for all the wells on the survey:
 - better average scalars per stack
 - well-to-well variation is useful to know
 - if it is large, try to understand why (near-surface effects?)
 - consider trace-by-trace inversion scaling or long gate AGC
- Much the same information is contained in the extracted wavelet amplitudes: which is the better way to calculate scalars? We shall return to this in the exercises.
- Scaling is critical for AVO gradient calculation. It is not strictly needed for simultaneous inversion as the software will calculate scalars for you, but they still need to be checked and variations understood.

Other pre-processing issues (outside SDC)

Other pre-processing issues (outside SDC)

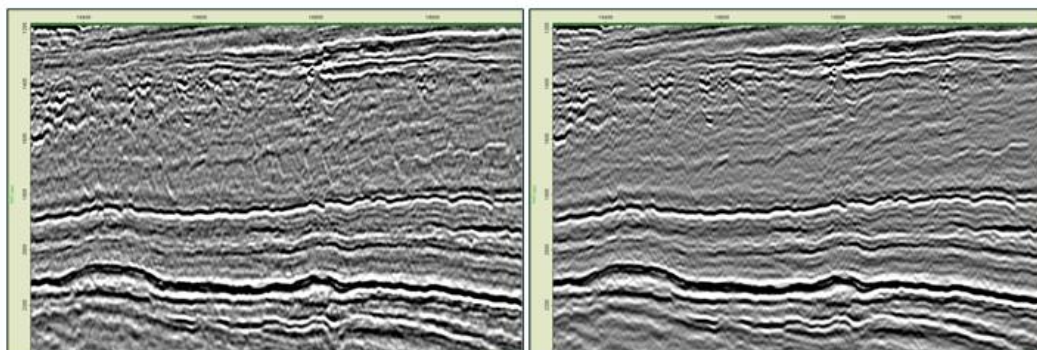
Multiples

- Need to be removed; inversion will happily treat them as data.
- Radon demultiple is often used. It recognises multiples by their curvature in NMO-corrected gathers, then isolates and subtracts them.
 - This does not work well on the near traces where the multiple curvature is very low
 - Therefore residual multiple energy may be left on the near traces. Near traces should not be scaled down because of this.
 - Alternatively, aggressive multiple removal can damage primary amplitudes on the near traces.
- SRME is increasingly used. It predicts the multiple train from the seismic trace itself and then subtracts it. It works least well on the far traces, where some of the required data are missing because of the finite maximum offset.

Other preprocessing

Noise reduction: important for improving AVO gradient estimate

- Frequency slice filtering: idea is that low frequencies are over-sampled and therefore can be smoothed laterally



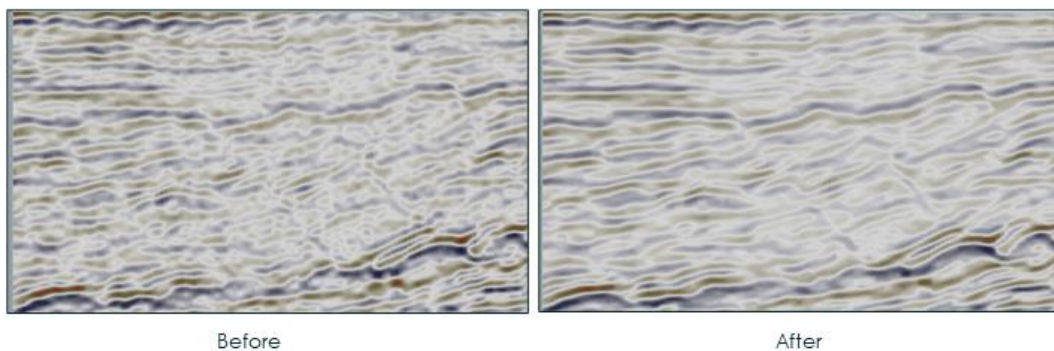
Before

After

Other pre-processing

Noise reduction: dip-steered median filter

- Automated calculation of local dip from correlation of neighbouring traces
- Replacement of amplitude value at each time sample of a trace with the median amplitude of traces within a footprint (eg 3x3 or 5x5 traces) around it, extracted along the local dip direction

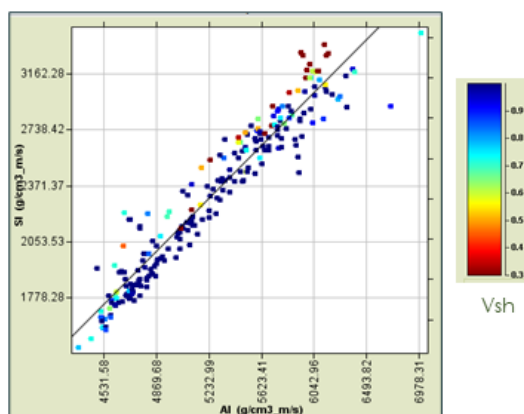


BACKGROUND TRENDS

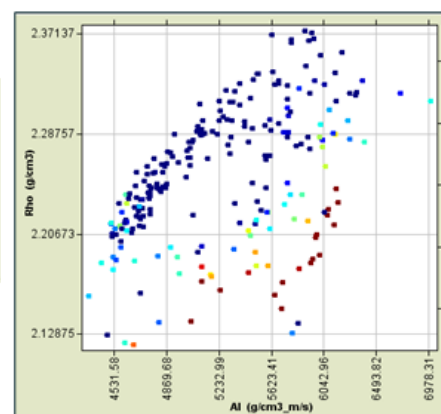
Background trends

In order to stabilise the process, simultaneous inversion does not invert for AI, SI and r directly but instead inverts for deviations from two background trends, applicable to the brine-fill case:

(a) Between $\ln SI$ and $\ln AI$



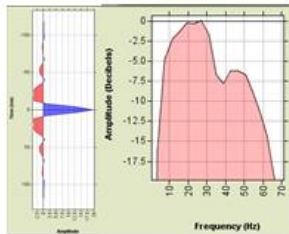
(b) Between $\ln r$ and $\ln AI$



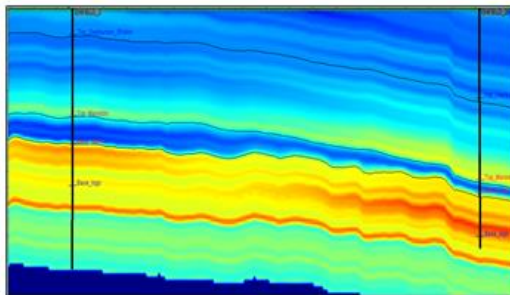
For this example, it's easy to fit an SI – AI trend, but the r – AI trend is clear only for the shale, with the sands being quite scattered.

The low-frequency background model

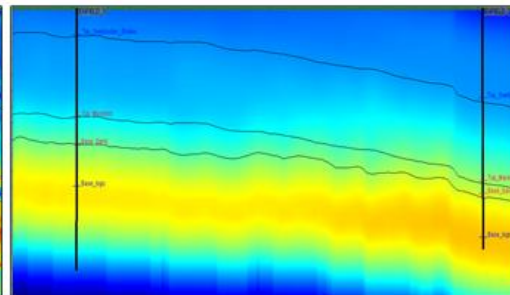
The low frequency background model



The spectrum of this typical seismic wavelet is 6dB down at about 8 Hz. It is this gap in the spectrum that we are trying to fill, so in this case from 0 - 8 Hz.



Model created by extrapolating impedance values away from a well, stretching and squeezing conformably between horizons



Model after high-cut filter, retaining only the frequencies needed to fill in the range not present in the seismic trace data.

The well logs need to be edited to remove any spurious values at top and base of data. If there are multiple wells, the impedances can be interpolated between them. This could be a simple inverse-distance weighting but could also take account of what is known of the geology, for example by using horizontal and vertical variograms in a kriging approach. Seismic processing velocities are another source of information about lateral variation away from the wells, perhaps best used by converting them to smoothed interval velocities and using them in a collocated co-kriging process. When using the results from an inversion, it's necessary to keep in mind what features come from the seismic data and what features come from the background model. Therefore, good practice is to output the low frequency model, and where derivative volumes are created from the inversion (e.g. Poisson's ratio from AI and SI) to create the same derivative volumes from the low-frequency model. Then inverted result and model can be compared side-by-side. Sometimes it is useful to subtract the volume calculated from the model from the volume calculated from the inversion result, to give for example a "relative Poisson's ratio".



SETTING INVERSION PARAMETERS



Setting inversion parameters



The parameters that usually need to be tested prior to running the production inversion are:

- How many substacks to use. The default would be to use all the available data, but sometimes it may give better results to exclude ultrafars (perhaps very low frequency or noisy) or nears (multiple contamination).
 - Well ties should already have given an indication of which substacks have doubtful usefulness.
- Scaling. The default is the scaling from comparison with well data explained previously. If it is found or suspected that scaling is laterally variant, eg because of near-surface effects, then laterally variable scalars may be needed.
- Choice of high-cut frequency applied to the background model
 - The initial estimate from the wavelet may need to be refined.



Setting inversion parameters

- Model weight. The inversion process tries to minimise both the variation of impedance away from the starting model and the misfit between synthetic traces and the real seismic data. If the starting model has been filtered so that only the very low frequencies remain, it is often not a close fit to the real well impedances over the interval of interest. In that case, it will give better results to use a low model weight
 - Because sensitivity to seismic noise may be different for AI, SI and density, it may be useful to set the model weight differently for each of these three outputs.
- To decide on parameter settings, the key step is to compare results at the wells for a range of parameter choices. When parameters are close to optimal, testing on selected seismic lines is needed; away from the wells it may be hard to say which result is better, though sometimes obviously incorrect behaviour can be seen.



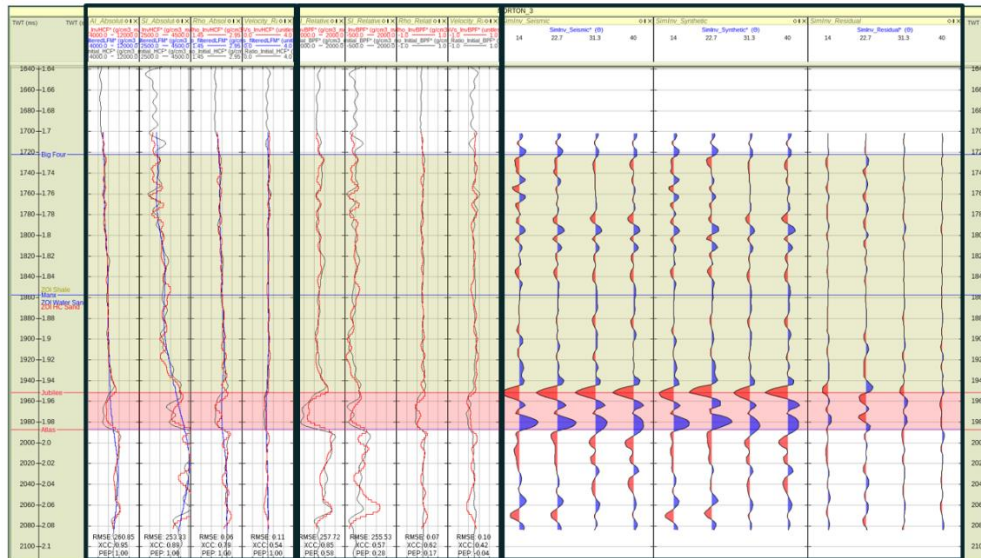
Testing at wells



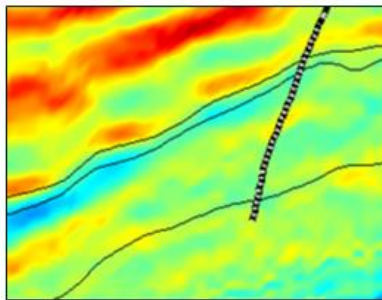
Left tracks are AI, SI, Rho, Vp/Vs: red = inversion result, black = well log blue = LFM

Middle tracks are AI, SI, Rho, Vp/Vs: red = relative inversion result, black = relative well log data.

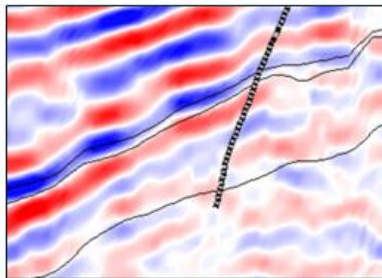
Right tracks are seismic, synthetic and their difference (residual), Near, Mid, far Ultrafar, traces in each one.



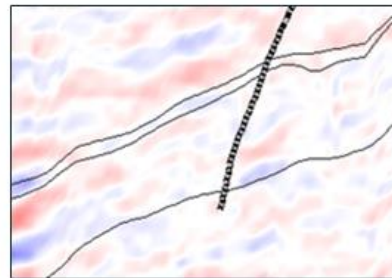
Testing along a line



AI inversion result



Near trace: Synthetic



Residual



INVERSION QC



Inversion QC

The main points that need to be checked and documented are:

- If available, inspection of gathers for recognition of multiples and other forms of coherent noise
- Well-tie quality, especially at the objective level
- Wavelets: well determined, similar between wells, logical change across the offset range
- Reflection alignment across the offset range
- Amplitude scaling versus offset checked against well synthetics
- Low frequency model: no artefacts due to spurious log values; variation between wells plausible geologically
- Good match to AI, SI, Vp/Vs at wells
- Low synthetic residual (synthetic – actual seismic) everywhere in the inversion window
- For features of particular interest (eg low Poisson's ratio bodies that might be indicative of hydrocarbon sand), comparison of inverted result with the equivalent quantity calculated from the background model

DERIVATIVE PRODUCTS

Derivative products



Simple calculations combine AI and SI volumes to give products that are potentially useful for fluid and lithology discrimination:

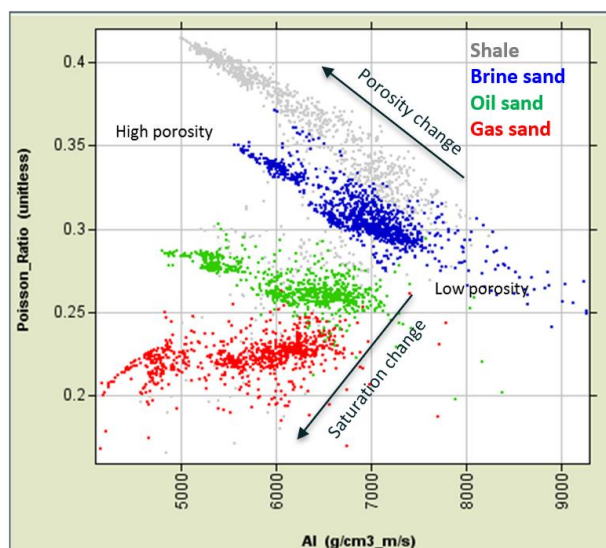
$$\frac{V_p}{V_s} = \frac{AI}{SI}$$

$$\sigma = \frac{0.5 - \left(\frac{SI}{AI}\right)^2}{1 - \left(\frac{SI}{AI}\right)^2}$$

$$\lambda\rho = AI^2 - 2SI^2$$

$$\mu\rho = SI^2$$

Acoustic impedance and Poisson's ratio

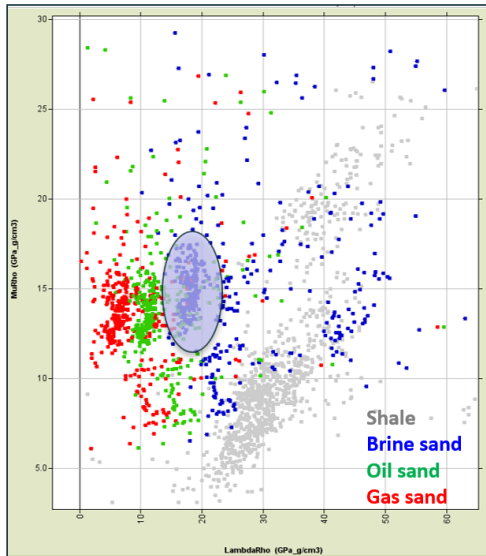


A typical case:
grey = shale,
blue = brine sand
green = oil sand
red = gas sand.

If we can work out AI and PR from seismic data, we could use them to predict sand presence and fluid fill.



Using derivative products



$\lambda \rho - \mu \rho$ plot, grey = shale, blue = brine sand, green = oil sand, red = gas sand.

Polygons can be used to separate the individual fluid/lithology classes, e.g. rectangle as shown to identify (most of) the brine sands.

2D Probability Density Functions can be used for Bayesian classification.

These could then be applied to the inversion output volumes.

Well-ties and wavelets

Well ties and wavelets

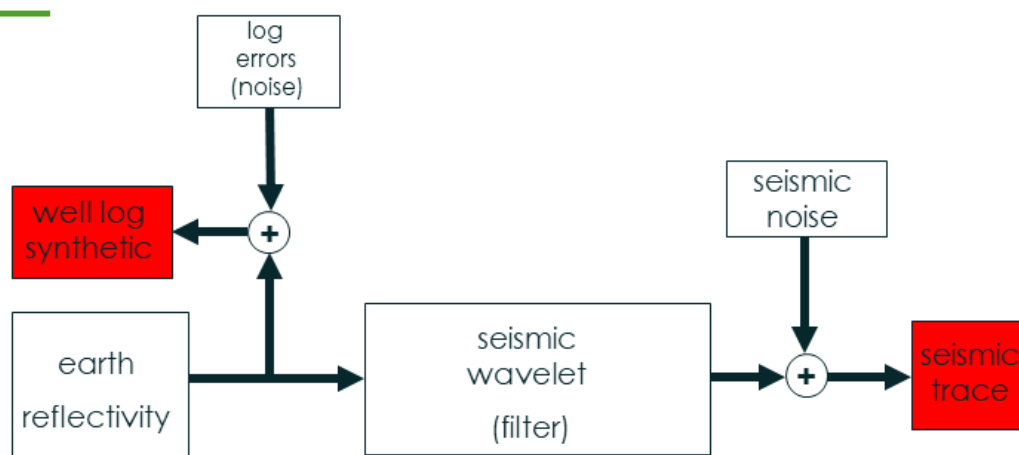


We want to achieve several things with the well-ties:

- Satisfy ourselves that there is a reasonable match between synthetic and seismic data at the objective level, without which inversion will not be useful
- Establish the time-depth relation that will be needed for building the starting model
- Determine the offset-dependent wavelets that are needed as input to the inversion.

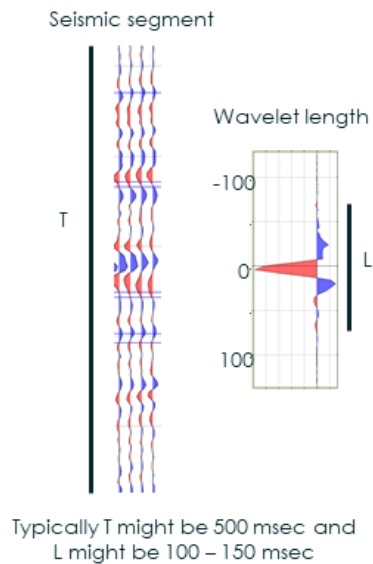
In general we should establish the well ties at the start of the inversion preparations, but seismic data conditioning will change the time-depth relations and the wavelets, so the well ties should be repeated on the final conditioned seismic.

Quantitative well ties: the model



The Roy White methodology (White 1980) allows us to quantify how good a well tie is (and hence, demonstrate improvement after log editing or seismic conditioning) and to put error bars on the seismic wavelet spectrum (so we can see whether the wavelet is well determined).

Accuracy and goodness-of-fit



The goodness of fit is measured by PEP, the proportion of trace energy predicted by the synthetic seismogram:

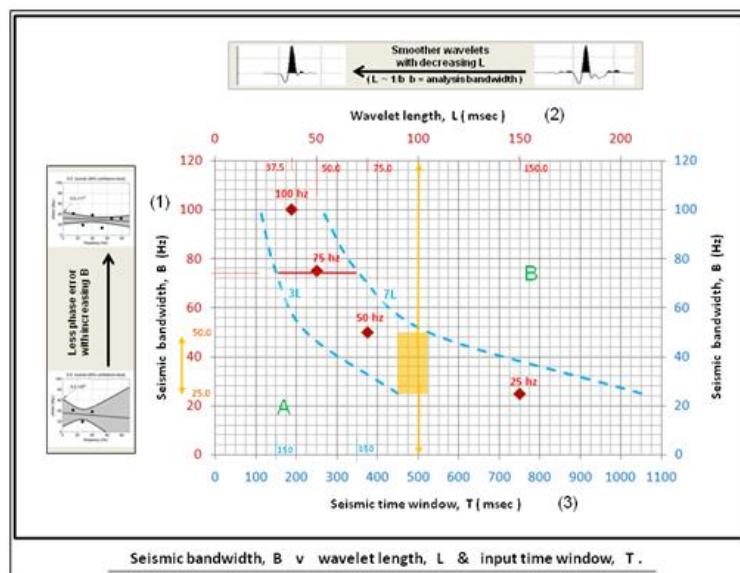
$PEP = 1 - (\text{energy in residuals} / \text{trace energy})$.
It is approximately the same as the R², where R is the cross-correlation coefficient. A good well tie has a PEP greater than about 0.7.

Goodness of fit is not a measure of the accuracy of the wavelet. The cross-correlation increases with increase in wavelet length, but the longer the wavelet is the more chance that noise is being matched. A measure of accuracy is NMSE, the normalized mean squared error, calculated from:

$$NMSE = \frac{0.59L}{T} \times \frac{1 - PEP}{PEP}$$

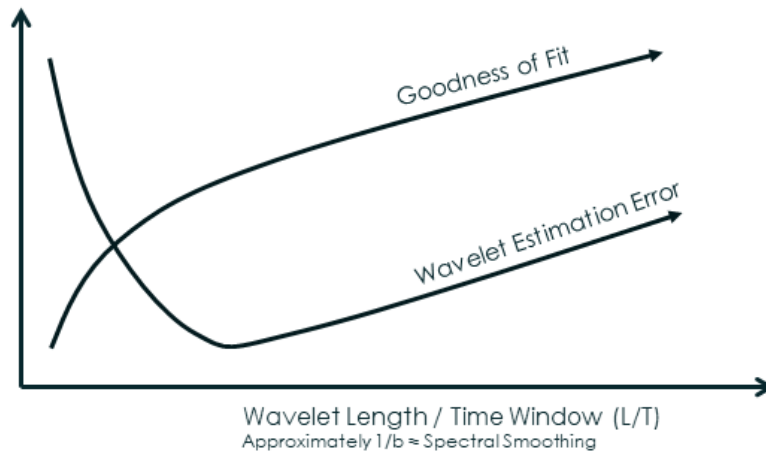
Where L is the wavelet length and T is the length of the time window from which the wavelet is extracted. A good well tie has NMSE less than about 0.2.

Well Ties & Wavelets



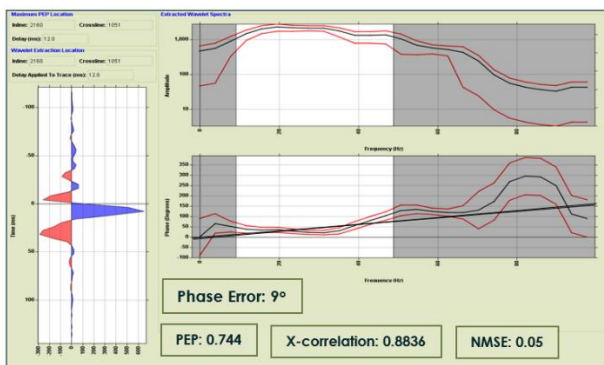
Wavelet length estimation guidance.

Wavelet Length Considerations



This is a wavelet length vs quality diagram.

Wavelet spectrum with error bars



B = bandwidth (Hz) of seismic data

b = analysis bandwidth

T = duration of data segment, bT = smoothing

PEP = "predictability"
or "proportion of energy predicted"
 $PEP = 1.0 - (\text{relative energy of misfit})$
 $= 1.0 - (\text{relative energy of residuals})$

NMSE = normalised mean square error
in the filtered synthetic
 $\approx (\text{error energy in the wavelet}) / (\text{wavelet energy})$

Time zero of the wavelet

controls temporal (vertical) resolution

wavelet length $\propto (1/b)$

bT is an averaging factor
estimation errors are $oc (1/\sqrt{bT})$

a measure of goodness-of-fit,
not by itself a measure of accuracy

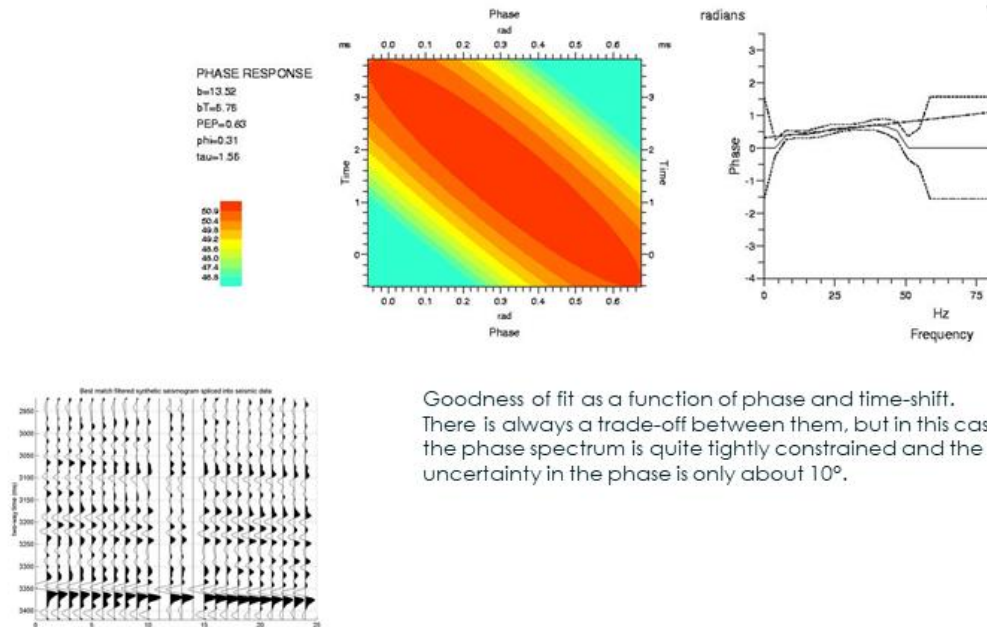
a measure of accuracy of the fit
(and, approximately, of the wavelet)

does it agree with acquisition
and processing time shifts?

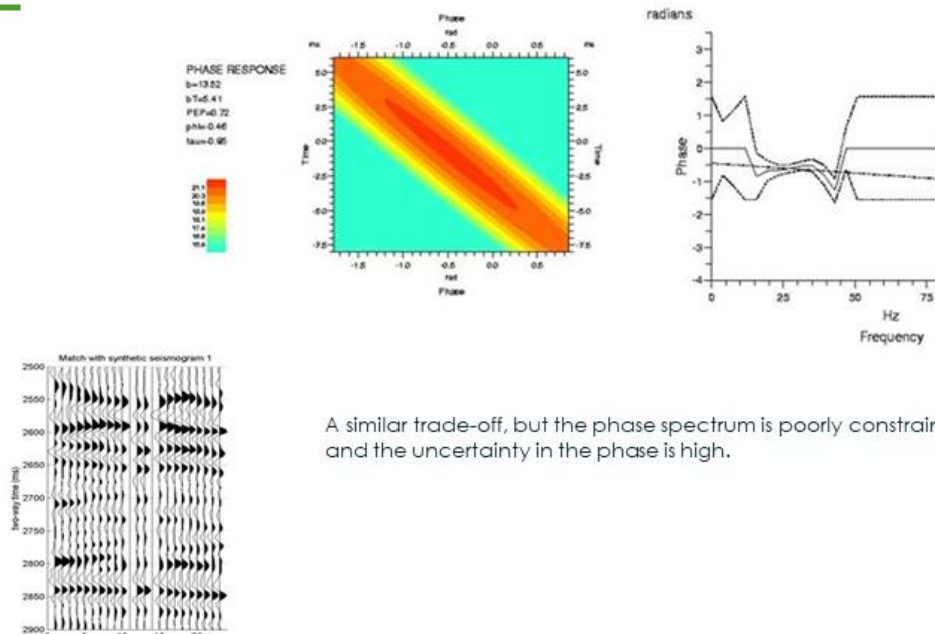
The straight-line fit to the well-determined part of the phase spectrum gives the wavelet phase (from the intercept) and timing (from the slope). The uncertainty in the wavelet phase is approximately given by

$$40\sqrt{NMSE} \text{ degrees.}$$

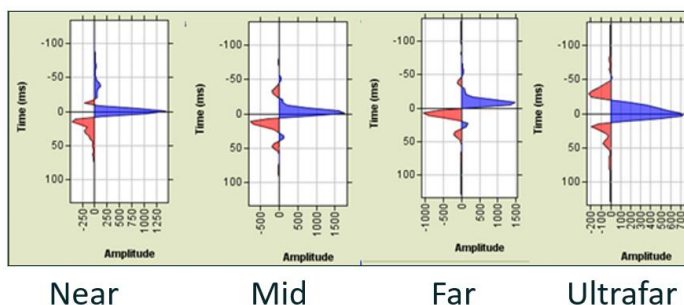
Phase determination: a good tie



Phase determination: a poor tie



Wavelet consistency

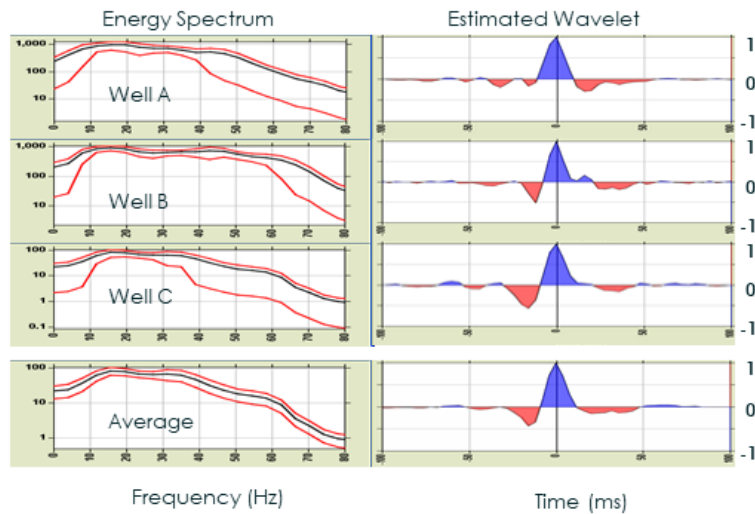


Wavelets should show consistency across the offset range. We expect a gradual phase rotation and loss of high frequencies from near to far.

Are we happy with these wavelets? The near to far progression looks reasonable but the jump in phase and frequency content to the ultrafars is suspicious. A problem for the Roy White method is that it is very sensitive to timing issues, i.e. any stretch/squeeze of the trace. This might be caused by events not being flat across the gather, with a shift up or down of the ultrafar trace that varies with TWT. The ultrafar wavelet might be more consistent after trace alignment with SDC.

Wavelets should also be consistent with the interpreter's common-sense views. For example, the seabed reflection gives a good idea of the approximate phase and polarity of the wavelet at that level. In many geological sections there are some isolated hard and soft reflections (i.e. from boundaries between thick homogeneous sequences) that can be used in a similar way.

Wavelet consistency

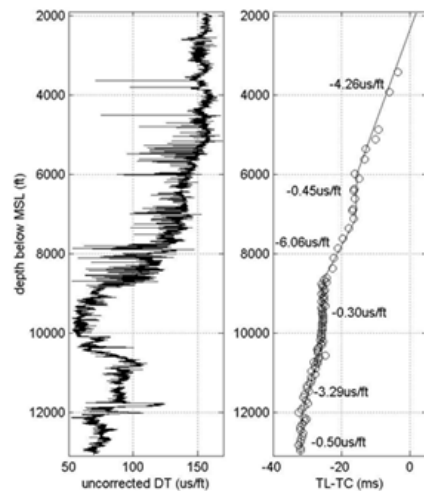


Wavelets should also be consistent between wells. Where they are, they can be averaged in the frequency domain, with weights according to their error bars.

Another possible complication is that VSP or checkshot times may be read at the first break or at the first peak/trough of the signal. Because seismic velocity in the earth is frequency-dependent (an inevitable consequence of absorption), there will be a systematic difference between the results, as the first method measures the group velocity and the second measures the phase velocity. Because of this, a small stretch may be needed to match the synthetic to the seismic even where the check-shot data are otherwise of good quality.

Time-depth relations

Since the Roy White method is quite sensitive to stretch or squeeze of the synthetic, it's important to have the best possible time-depth curve as input.



On the left is the sonic curve, on the right the integrated log time minus the check shot time, here fitted using straight lines with knee points. Where there is dense checkshot coverage (eg from a VSP) this is much better than using all the checkshot times individually, because of possible timing errors at each individual level.

A practical problem sometimes encountered is that in some companies the curation of checkshot/VSP data is poor, and time/depth data of unknown provenance is provided. This may have been created by an interpreter to match well data to interpreted horizons on a particular dataset, or it may have been imported from a neighbouring well. Data that can't be traced back to a VSP or checkshot report need to be treated warily; it may be better to ignore them.

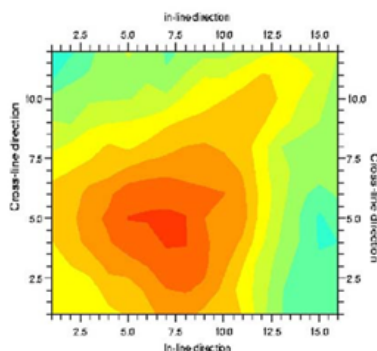
When the time-depth relation is incorrect, the software may point to a location where the principal reflections match the times predicted by the synthetic. If a large lateral shift (say more than 150 m) is found in an area of flat-lying geology, it is a good idea to re-examine the $t - z$ relation.

Well location

It is common for the best tie to be found some distance from the as-surveyed well location. There are two possible reasons for this:

- Well surface location or (more often) deviation survey is inaccurate; this should only be an issue for legacy wells
- The seismic reflections may not have been correctly positioned by the migration process, because of an inaccurate velocity field or migration algorithm (eg using time migration where depth migration is needed).

Therefore it is useful to scan locations around the well for the best-matching trace.



Contours of PEP (goodness of fit) around well (located at centre of square) show the best-fit location is shifted to the SW.

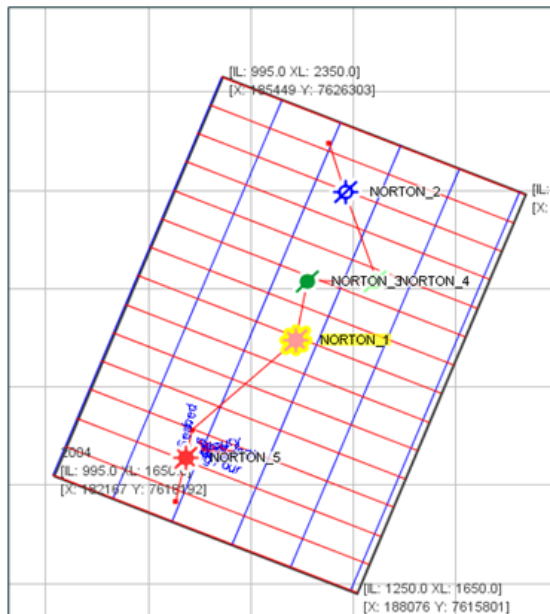
Note:

- If the cause of the shift is error in the well location, the shift should be much the same for all sub-stacks.
- If the shift is caused by inadequate seismic processing, it may be different from one sub-stack to another.

Exercise 1: well ties on the Norton dataset

INTRODUCTION TO NORTON

Introduction to Norton



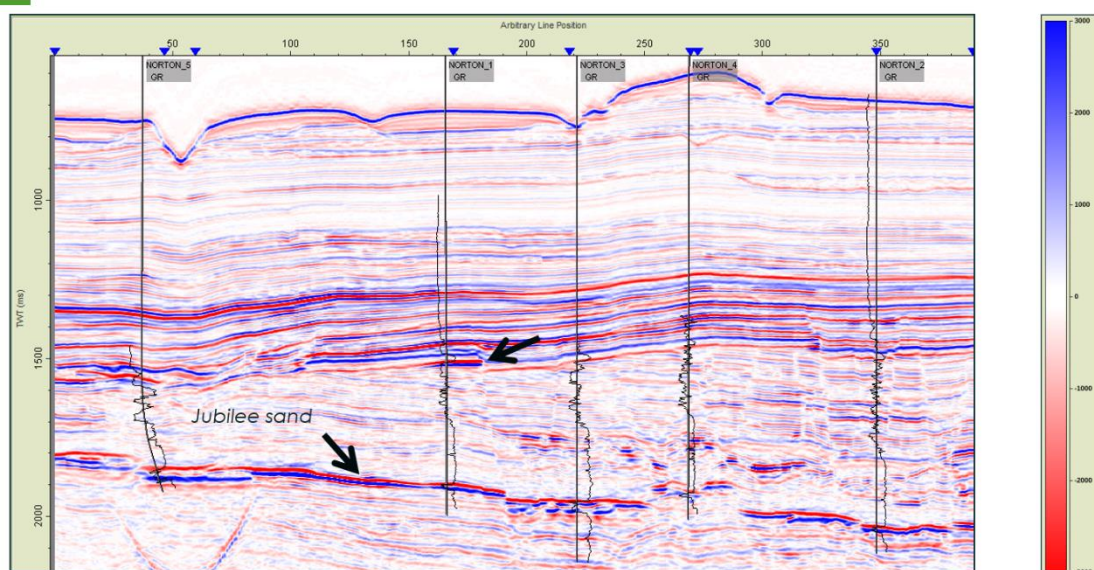
Seismic data:

Nears 8 - 19°
Mids 19 - 30°
Fars 30 - 41°
Ultrafars 34 - 45°

Polarity: SEG normal

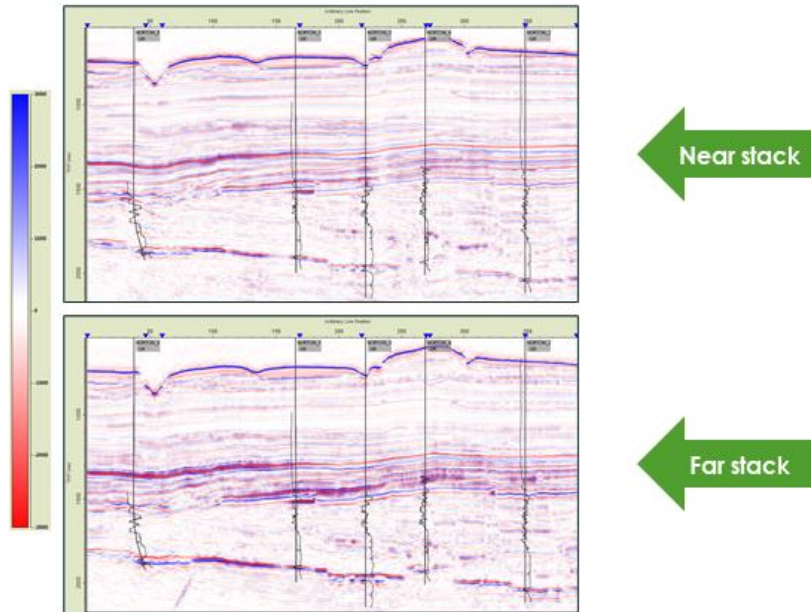
Gas and oil accumulation.

Arbitrary line, full stack



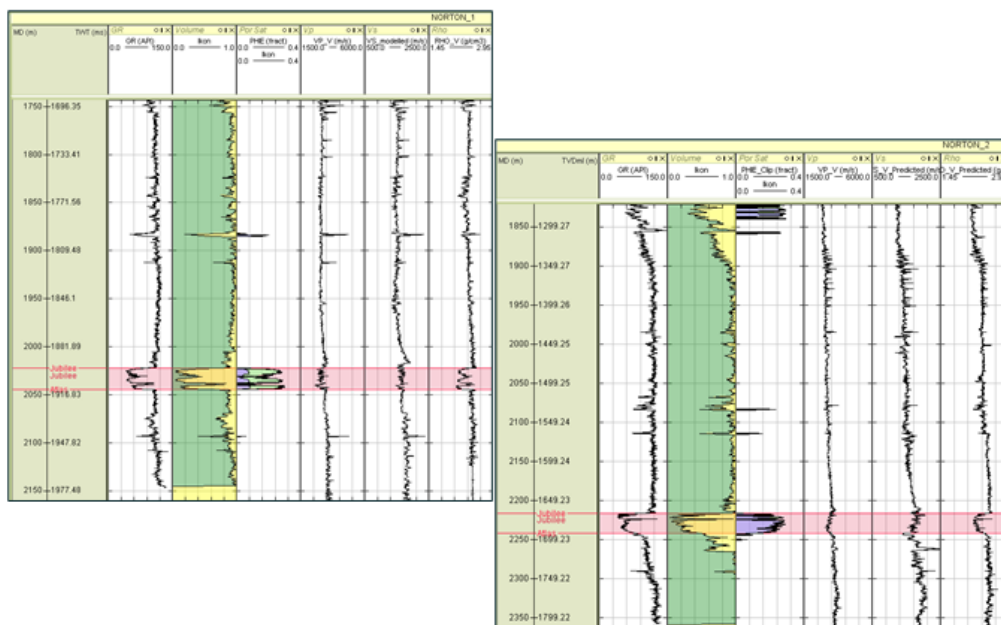
For location see previous slide. Target is gas and oil in the Jubilee sand. There appears to be gas at a shallower level in Norton-1. Well curve is GR.

Arbitrary line, near and far stack

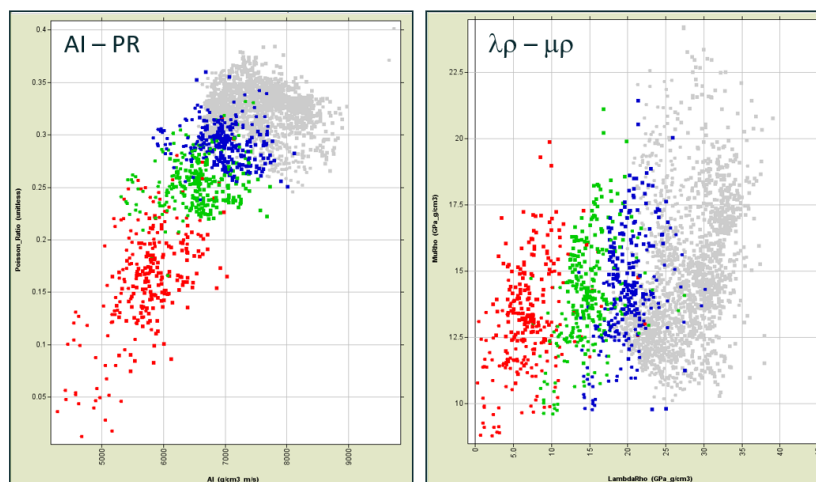


Just from these sections, what can you say qualitatively about likely wavelet phase and frequency behaviour?

Wells Norton-1 and 2



Why might we want to invert the dataset?



Grey = shale, blue = brine sand, green = oil sand, red = gas sand (Gassmann substitution from in situ sand saturations)

Excellent sand/shale discrimination and brine/oil/gas discrimination. Therefore a simultaneous inversion to AI and SI should give us derived volumes that will be good lithology and fluid discriminators, which can be used as input to a reservoir model.



Well-tie workflow

In this workflow you will carry out well-ties on Norton-1 for all four substacks, then compare and average the wavelets with those for Norton – 2 and - 3.

BACKUS AVERAGING

The first step is Backus averaging. We shall be creating well synthetics using the Zoeppritz equations. These are exact solutions, but because they have no concept of the seismic disturbance as a wave, they will predict large reflection amplitude from very thin layers that would in reality not be seen by the seismic. This is particularly a problem at large offset. The solution is to smooth the input data, in such a way that the smoothed logs give the correct seismic response as calculated from wave theory. Backus averaging does this by taking harmonic averages of the elastic moduli and an arithmetic average of the density in a sliding window (Liner and Fei 2007). Liner & Fei recommend that the window size should be set to less than one-third of the minimum shear velocity divided by the dominant frequency (*ibid*). Folstad & Schoenberg (1992) state that fine layering below '1/10 of the smallest wavelength ... does not affect seismic wave propagation'. They recommend a minimum Backus window of one tenth of the minimum compressional velocity divided by the maximum effective frequency. For Norton – 1, the minimum shear velocity is about 1100 m/s and the minimum compressional velocity is about 2100 m/s, the maximum effective frequency for the near stack is about 69 Hz and the dominant frequency is about 35 Hz, so a suitable window size lies in the range 3-8 m.

Open RokDoc by double-clicking on the RokDoc icon on the desktop.

Click 'File' > 'Open' and navigate to the 'Norton_SDC_course.rok' file. Click 'Open'

In the map, double-click on the symbol for Norton – 1. This brings up the well viewer. The in-situ Vp, Vs and ρ logs are displayed. Make sure that the z-stick is reading 'MD'

In order to calculate Backus averaging filter length an amplitude spectrum of the seismic data need to be estimated over the target zone from which we can find the highest frequency of the data (value @ -6 dB down) and use it to calculate 1/10th of the smallest wave length. **Click on 'Seismic Management' > 'Spectral plot'** In spectral plot window **Click on '+' button > 'Volumes' and select the Near stack, set the time range, by clicking on 'Horizon', from Jubilee -400 to Atlas 200 ms and click 'OK'** This will bring up amplitude spectrum of the near angle stack around the target zone.

Click on 'Well Ops' > 'Synthetics & Filters' > 'Backus Averaging'(Alt+B). In the Backus Averaging dialogue that appears, check that the selected Vp-Vs-Rho set is 'Insitu_conditioned' which consists of the in situ logs (if it is not, click on the Vp-Vs-Rho button, then in the pop-up list click on the line with name



'Insitu_conditioned', which turns yellow, then click OK). **Set the Backus window size to 4 m and then click 'Calculate'**. The Backus average logs are displayed in red over the insitu logs.

Enter the suffix 'Backus_V_4m' in the box next to the 'Save' button, then click 'Save'. To close the dialog, click on the 'X' in the top margin. For further information or help with this module, click on the '?' icon.

The temporary logs will be saved and displayed in black over input logs.

CREATE A STARTING GATHER

The well-tie module requires that you create an initial gather as a starting point in comparing the synthetic and seismic. In this case we shall use a zero-phase 35 Hz ricker wavelet to do this. This has been created for you under the name 'R35'.

Click on 'Well Ops' > 'Synthetics & Filters' > 'Synthetic Generation (AVO)' (Alt+A). In the dialogue that appears, click on 'Vp-Vs-Rho', in the pop-up list click on the line with name 'Backus' which turns yellow, then click 'OK'. In a similar way click on the 'Wavelet' button and choose the wavelet with the name 'R35'.

The default parameters 'PP', 'Zoeppritz' and '60' degrees will be fine.

Click on 'Calculate'. Enter the name of the Gather as 'V Near' and click 'Save Gather'. To close the dialog, click on the 'X' in the top margin.

To display the gather, right-click in the last track of the display, and in the pop-up click on 'New Track' > 'Add Gather Track'. A Gather Track appears. Right click in it, and in the pop-up click on 'Select Gather 1'. In the pop-up list click on the line with the name 'V Near' which turns yellow, then click 'OK'.

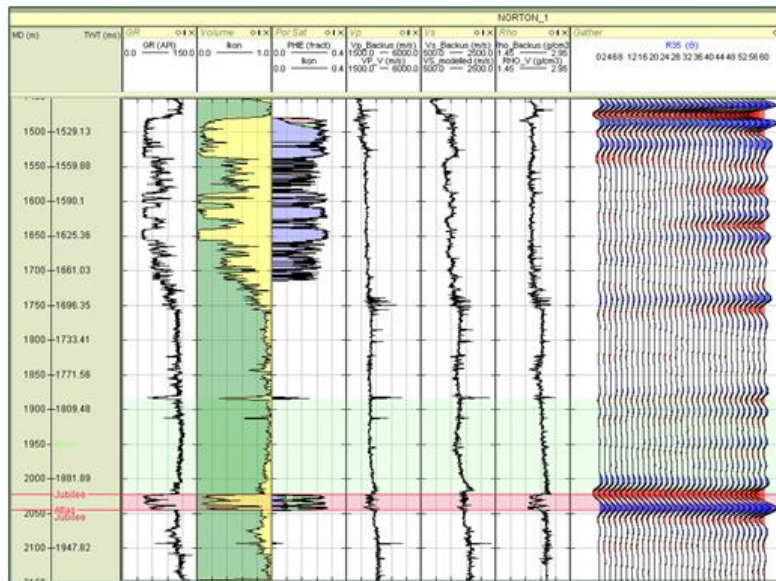
The gather will display.

To set the display parameters, right-click in the gather track, in the pop-up click on 'Display Options'. In the dialogue that appears uncheck 'Show Stack Traces', change 'Increment' to 2 and click 'OK'. Right-click in the gather track again and uncheck 'Show Attribute Plot'.

The display will look as follows. Note the AVO effects at angles beyond about 35°.



Well-tie workflow



Norton-1: Logs and initial gather

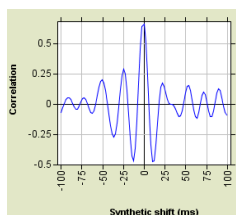
WELL TIE DIALOGUE

Change the z axis to be TWT rather than depth by clicking on 'ZStick' then on the button next to 'TWT'. To set the visible range, right-click on the TWT scale, in the pop-up click on 'Set z-scale', enter 1400 and 2000 for 'Visible top' and 'Visible bottom' and click OK.

Clean up the display by closing (click on X in top margin) all tracks except GR and Gather. Now we can start on the well ties, beginning with the near trace volume.

Click on 'Well Ops' > 'Wavelet Estimation and Stretch and Squeeze' (Alt+W). In the dialogue click on 'Select Synthetic Gather', select 'V Near' from the list. Click on 'Select Volume', select 'Near_Stack_8_19' from the list. Click on 'Select Angles/Offsets', check 'Stack log', enter 1 for 'Increment' and 8 for 'Near', 19 for 'Far', and click 'OK'. The range of angles in the near stack is 8 - 19°.

In the bottom part of the dialogue tab, enter 1490 for 'Start (ms)' and 1960 for 'Stop (ms)' and 100 for 'Correlation Window (ms)'. We are using the entire logged interval for the tie. You may need to press the return key after entering the last number, and then a display as shown appears for the correlation between the synthetic created with the Near stack and the seismic trace at the well. The correlation is quite high for a starting tie with no adjustment made to the wavelet or lateral position.





In the next tab, 'Wavelet estimation', toggle on 'Use cross-correlation range' under 'Well Synthetic Time Window', then click 'Estimate Wavelet'. On the next tab, 'Results', under 'Wavelet extraction statistics' observe that ' b/B ' is displayed in red and is unacceptably low. This is telling us that the wavelet is too long and is in part fitting noise as well as signal. The solution is to reduce the wavelet length; trial and error has shown that a wavelet length of 72 msec gives acceptable results.

The accuracy of well ties

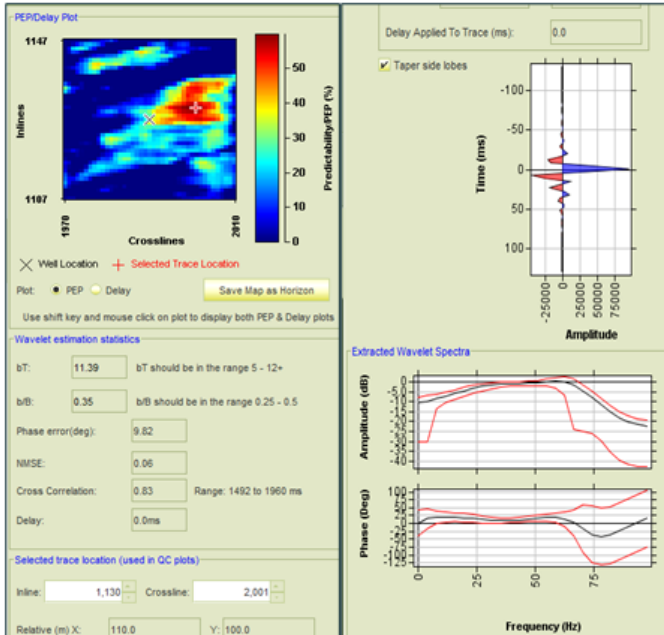
TABLE 1. Parameters and Quality Control in Quantitative Matching

B = bandwidth (Hz) of seismic data	<i>controls temporal (vertical) resolution</i>
b = analysis bandwidth	<i>wavelet length $\propto (1/b)$</i>
T = duration of data segment, bT = smoothing	<i>bT is an averaging factor estimation errors are $oc (1/\sqrt{bT})$</i>
PEP = "predictability" or "proportion of energy predicted"	<i>a measure of goodness-of-fit, <u>not by itself</u> a measure of accuracy</i>
PEP = 1.0 - (relative energy of misfit) = 1.0 - (relative energy of residuals)	
NMSE = normalised mean square error in the filtered synthetic	<i>a measure of accuracy of the fit (and, approximately, of the wavelet)</i>
$\simeq$ (error energy in the wavelet)/(wavelet energy)	
Time zero of the wavelet	<i>does it agree with acquisition and processing time shifts?</i>

Return to the previous tab, 'Wavelet estimation' and under 'Input Parameters' set 'Effective wavelet length (ms)(L)' to 72 then click 'Estimate Wavelet'. View the results in the 'Results' and 'Save wavelet' tabs. In the 'Save wavelet' tab, under 'Save Wavelet', change the name to 'NORTON 1 near' and click 'Save Wavelet'.

Try estimating a wavelet but this time in 'Wavelet Estimation', under 'Input Parameters', toggle on 'Advanced' and toggle on 'Noise Free' and click on 'Calculate'. Is this wavelet better than wavelet calculated in previous steps?

Norton-1 : nears wavelet

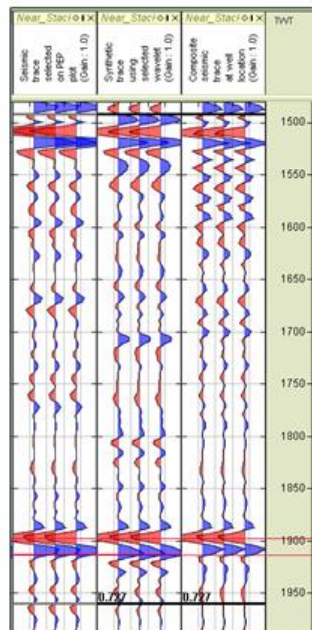


The best fit location is 110m away from the well position. PEP is fair at 0.60, NMSE is good at 0.06. The wavelet is therefore well determined even though the synthetic fit is only fair.

The wavelet is close to zero phase but with a slightly larger postcursor than precursor.

1. To view the synthetic/seismic match, expand the well track space if necessary: hover the mouse on the vertical bar at its right hand edge to get a double-headed arrow, click and drag, and do the same for the horizontal bar at the top to make the labels fully visible.

Norton-1 : nears wavelet

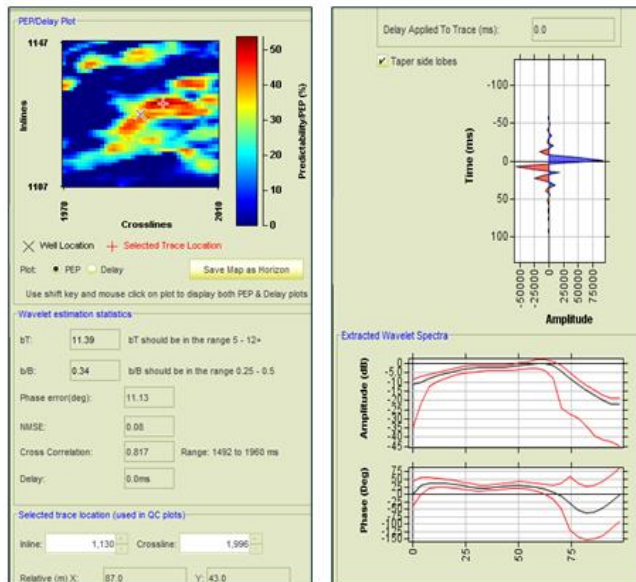


Tracks are seismic at 'best fit' location, synthetic with extracted wavelet, seismic at well location.

Why do you think the tie is poor for the weaker loops?

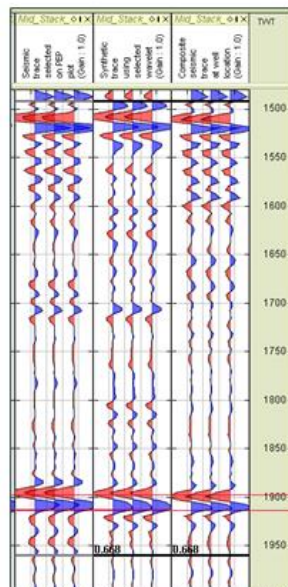
2. Expand back the 'Wavelet extraction and S & S' dialogue, return to the 'Input and S & S' tab, and select data for the mids (Mid_Stack_19_30) whose angle range is 19 - 30°. Change the angle/offset range to match.
3. Using the same 72 ms wavelet length as before, estimate the wavelet and view the results. Save the wavelet as 'NORTON 1 mid'.

Norton-1 : mids wavelet



The result is similar to that for the nears but PEP is worse at 0.54, NMSE is higher at 0.08. The location is 100 m away from the well.

Norton-1 : mids wavelet

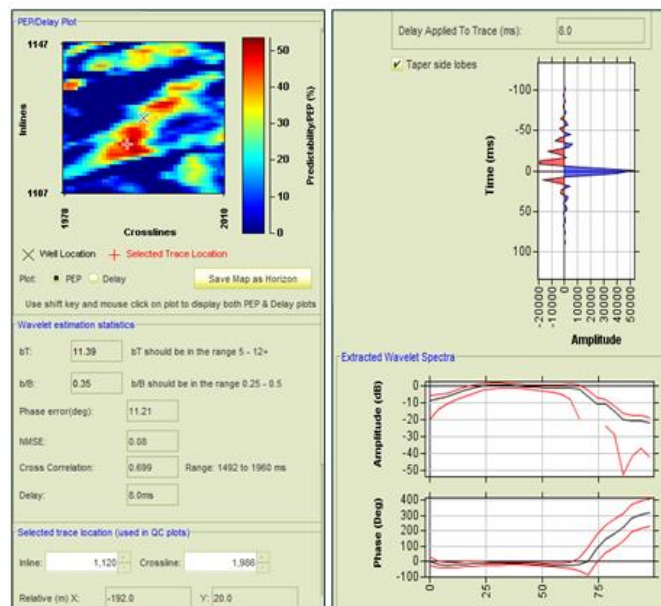


Tracks as before. Again the weaker loops are a poor match.



4. Repeat the process for the fars: 'Far_Stack_30_41'. Save the wavelet as NORTON 1 far.

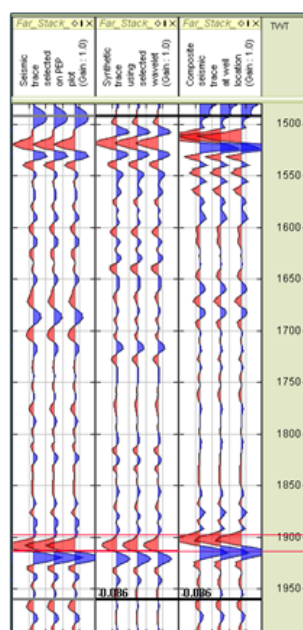
Norton-1 : fars wavelet



PEP and NMSE are reasonable but the location is 190 m away from the well, in the opposite direction to the location for the near and mid wavelets.

Are you happy with this? If not, what could you do about it?

Norton-1 : fars wavelet



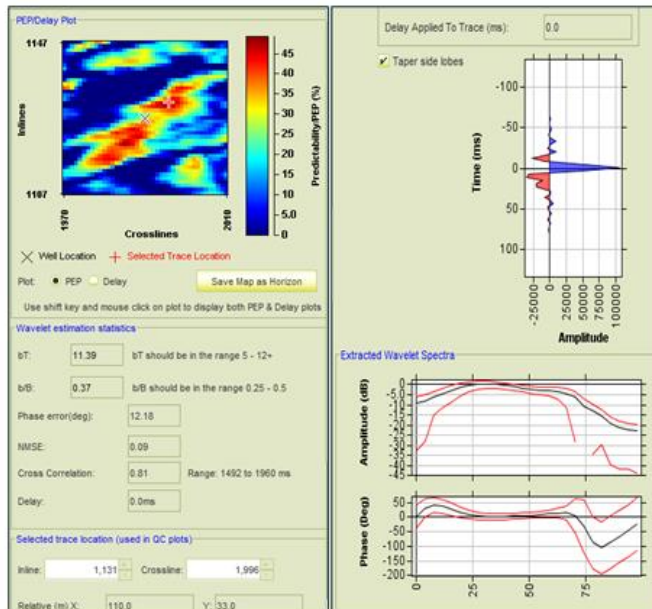
Tracks as before.

Note that there is an 8ms lag applied to the wavelet (toggle it on in the 'Save Wavelet' window to get the alignment between synthetic and seismic as shown).

Are you happy with this? What might be the cause?

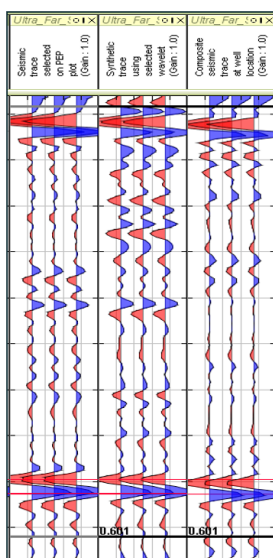
5. Repeat the process for the ultrafars: 'Ultra_Far_Stack_34_45'. Save the wavelet as NORTON 1 ultrafar.

Norton-1 : ultrafars wavelet



The lateral shift from the well is 110m and is now back to being towards the north-east, as it was for near and mid wavelets. Does this affect your thoughts about the far trace tie?

Norton-1: ultrafars wavelet

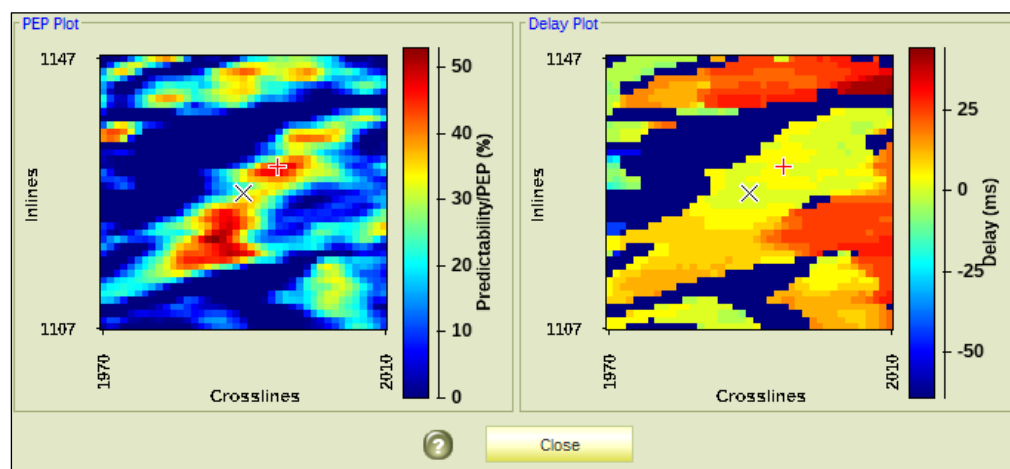


Tracks as before. Note that no lag has been applied, unlike for the fars.



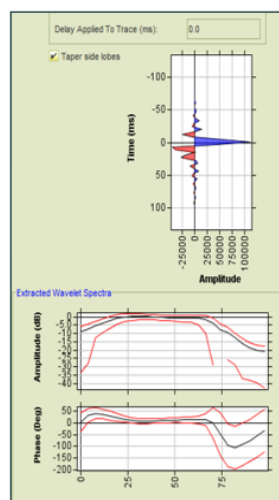
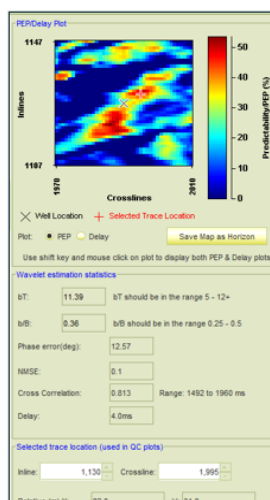
The anomalous far trace wavelet can be rectified by forcing an extraction location to a trace to the north-east of the well, close to where the other wavelets are extracted. This gives a slightly less good tie but a wavelet similar to that for the ultrafars.

To pick the new extraction location set the input parameters for the far stack and click 'Estimate wavelet'. In the results tab hold shift and click on the map. Expand the map and click to select a new wavelet extraction location in line with the locations of the other wavelets, use the PEP and delay maps to guide your selection. Click the 'Extract from selected trace' button to extract the new wavelet.



Since the time window and some other parameters differ in the e-course, the map in the e-course is different to the one shown here.

Norton-1: fars revisited



Forcing the tie to a location just north-east of the well gives a lower PEP but a wavelet similar to the ultrafars, with some post-cursor energy but no lag.

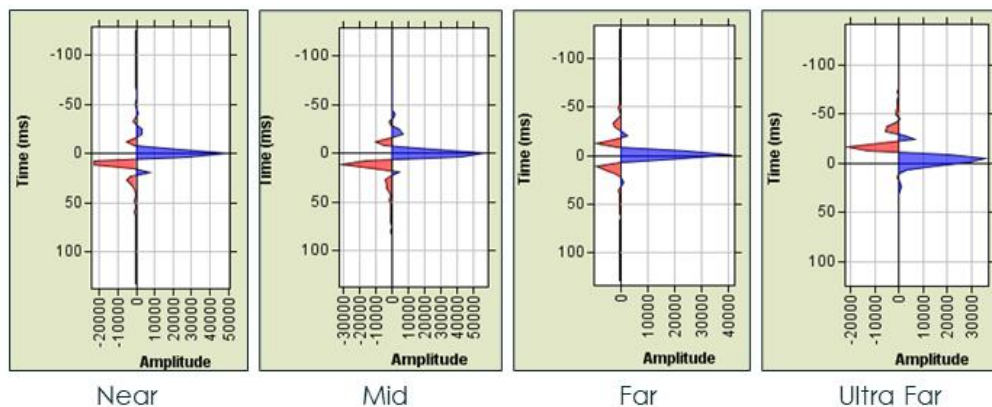
This wavelet is preferable to the one extracted previously; we expect to get consistency between wavelets across the angle range, with only gradual variation in the form of a decrease in bandwidth and modest phase rotation from near - far.



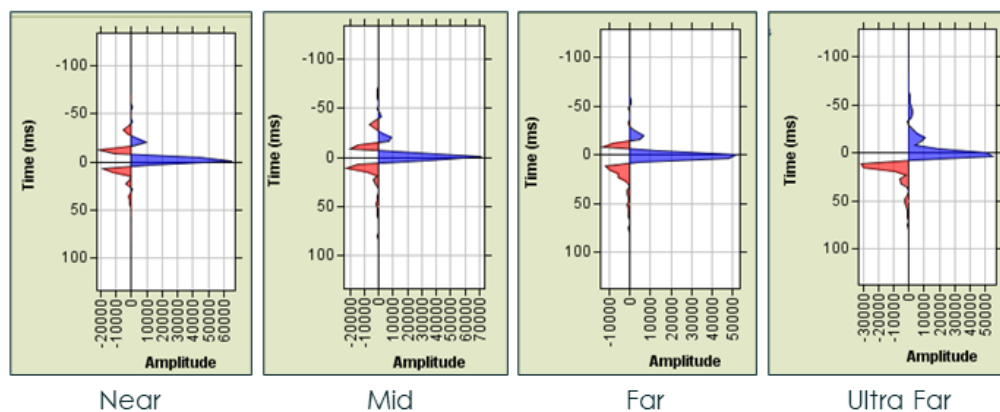
WAVELET AVERAGING

6. In the tree on the left-hand side of the viewer window, right-click on 'Wavelets', click on the 'Edit' popup. The wavelets for Norton-2 and Norton-3 have already been created by the same process as for Norton-1.

Norton-2



Norton-3

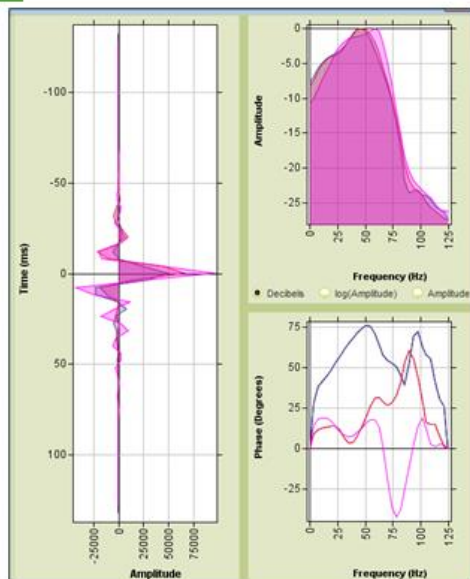


Ultrafar has a suspiciously large phase rotation (intercept phase is about 60°), but tie is quite good.

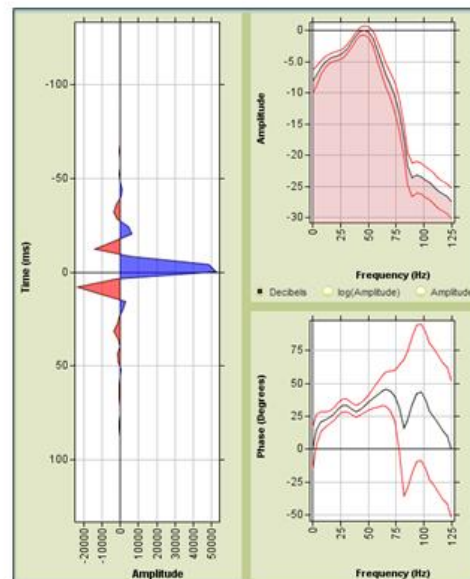


7. In the wavelet dialogue, click on 'Norton_1_mid' then 'Norton_2_mid' then 'Norton_3_mid', while observing the wavelet display. Finally hold down the Ctrl key while clicking the names so as to display all three simultaneously.
8. Click on wavelet 'Norton_1_mid' then click on 'Average' and in the pop-up list click on 'Norton_2_mid' and 'Norton_3_mid' while holding down the Ctrl key. Click 'Calculate', enter the name as 'Mid_average' and click 'Save' then 'Close'. View the average wavelet.

Norton – near wavelet



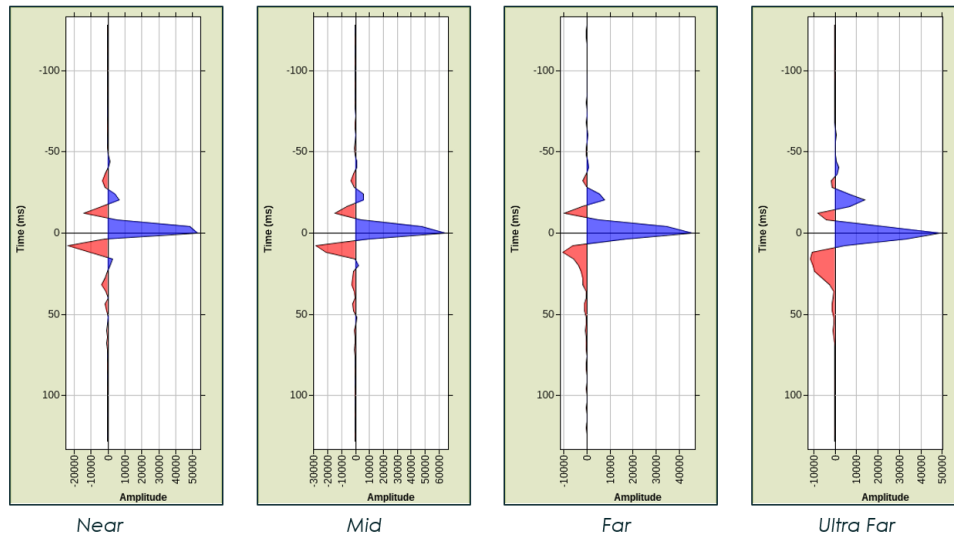
Wavelets from wells 1,2,3 are fairly similar but have quite different amplitudes – what does this tell us?



Average wavelet

9. Repeat the average procedure for the mid, far and ultrafar wavelets.

Norton – average wavelets



Fairly consistent wavelets. The amplitude varies a bit from one stack to another – what does this tell us?

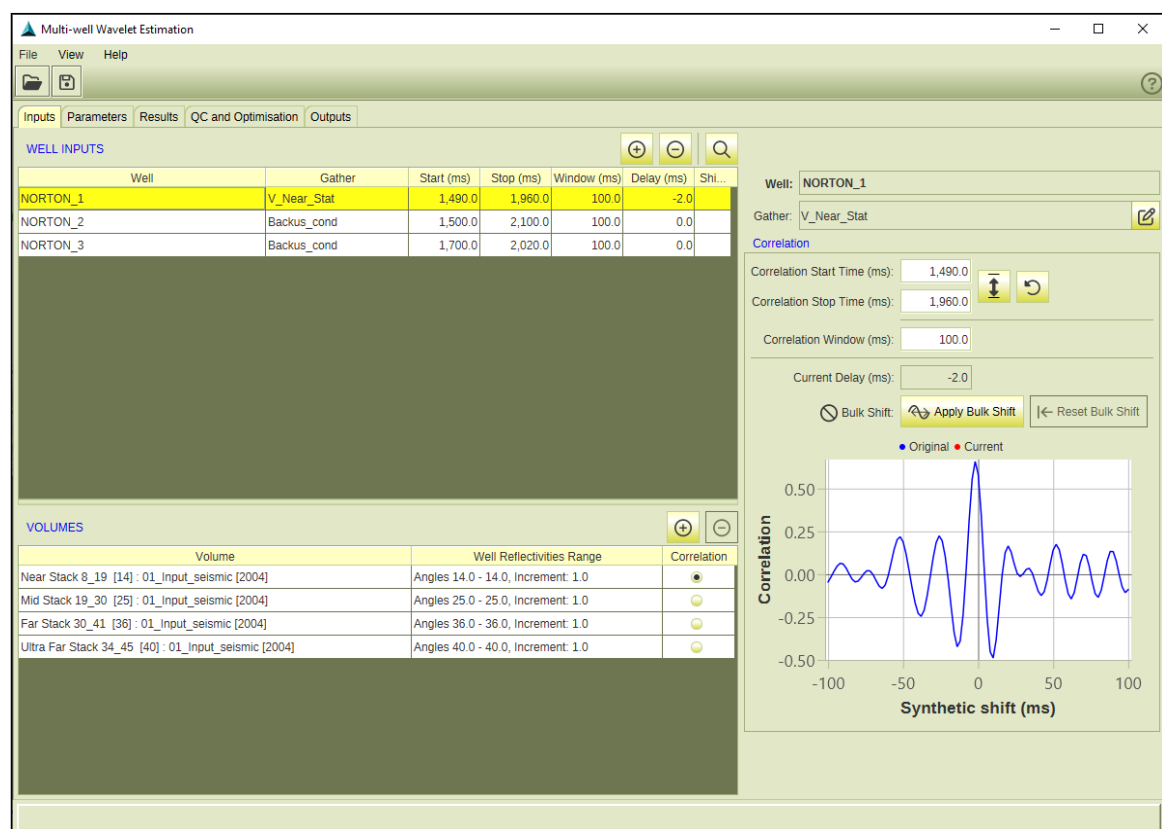
10. Click on 'Well' > 'Close' in the menu bar to close the well viewer window.



MULTI-WELL WAVELET ESTIMATION

The multi-well wavelet estimation module requires that there is a gather, correlation window or angle stacks set up for each well.

18. In the main project window top menu click on 'Multi-well Ops' > 'Multi-well Wavelet Estimation'. In the dialogue that appears, click on '+' adding a well data. The right hand-side window click Gather selection with following correlation window 'Correlation Start Time (ms)' and 'Correlation Stop Time (ms)' and 'Correlation Window (ms)' In the section 'Volumes', click on '+' and add angle stacks. You can add them one by one or select a prestack volume.



11. In the same dialog window, in the tab 'Parameters' you can chose 'Trace Selection' > 'Vertical' or 'Composite', selected trace extracted from seismic data with selected 'Inline Step Out' and 'Crossline Step Out'. Below the 'Wavelet Estimation Location' can be selected as 'Best Fit' or 'Well'. The 'Well Synthetic Time Window' inherited the cross-correlation from the previous tab or can be changed here. 'Wavelet Estimation Parameters' located at the bottom left where Sample interval of the wells or Seismic (ms) should be selected. 'Location' of the seismic extraction should be selected as a 'Top Hole' or 'Bottom Hole' with following 'Effective Wavelet Length (L) (ms)' and a type of tapering 'Spectral Analysis Method' > 'Papoulis' or 'Multitaper'. 'Multitaper' can be used to improve low frequency part of the wavelet especially in case if well logs cover only short interval. On the right 'Advanced Parameters' where 'Maximum Delay in Scan (ms)' can be selected with 'Relative Noise' > 'Noise Free' or



'Variance Ratio' and 'White Noise Factor (fraction)'. Click 'Estimate Wavelets' button before you proceed to the next tab.

Multi-well Wavelet Estimation

File View Help

Inputs Parameters Results QC and Optimisation Outputs

WELL PARAMETERS

Well	Location	Start (ms)	Stop (ms)	Link	T (ms)	T/L	Length (ms)	Trace Selection
NORTON_1	Best Fit	1,490.0	1,960.0	☒	470.0	6.9118	256.0	Trace Location: Vertical, Step outs: IL:10, XL:10
NORTON_2	Best Fit	1,500.0	2,100.0	☒	600.0	8.8235	256.0	Trace Location: Vertical, Step outs: IL:10, XL:10
NORTON_3	Best Fit	1,700.0	2,020.0	☒	320.0	4.7059	256.0	Trace Location: Vertical, Step outs: IL:10, XL:10

Well: **NORTON_1**

Trace Selection

Vertical **Composite**

Inline Step Out:

Crossline Step Out:

Wavelet Estimation Location

Best Fit **Well** User-defined

Well Synthetic Time Window

Start Time (ms):

Stop Time (ms):

Time Window [T] (ms):

Effective Wavelet Length [L] (ms):

Time Window / Wavelet Length [T/L]:

Total Wavelet Length (ms):

WAVELET ESTIMATION PARAMETERS

Sample Interval: ☐ Wells (ms) ☒ Seismic (ms)

Location: ☒ Top Hole ☐ Bottom Hole

Effective Wavelet Length [L] (ms):

Spectral Analysis Method: ☒ Papoulis ☐ Multitaper

ADVANCED PARAMETERS

Maximum Delay in Scan (ms):

Relative Noise: ☐ Noise Free ☒ Variance Ratio

White Noise Factor (fraction):

Estimate Wavelets



12. In the same dialog window, in the tab 'Results' and 'QC and Optimisation' statistics well-tie QC is displayed including additional QC windows at the well location with multiple QC tracks per angle stack/seismic volume – 'Elastic QC', 'Seismic Section QC', 'Standard QC', 'Petrophysics', 'AVO' and 'Gather'



Multi-well Wavelet Estimation						
File View Help						
Inputs Parameters Results QC and Optimisation Outputs						
WELL OUTPUTS						
Input	Wavelet Length	Sample Interval	Average Wav...	Save Wavelet	Save Bulk Shift	
▼ 01_Input_seismic_Near Stack 8_19 [14]						
NORTON_1	65	4.0	✓	✓	□	
NORTON_2	65	4.0	✓	✓	□	
NORTON_3	65	4.0	✓	✓	□	
▼ 01_Input_seismic_Mid Stack 19_30 [25]						
NORTON_1	65	4.0	✓	✓	□	
NORTON_2	65	4.0	✓	✓	□	
NORTON_3	65	4.0	✓	✓	□	
▼ 01_Input_seismic_Far Stack 30_41 [36]						
NORTON_1	65	4.0	✓	✓	□	
NORTON_2	65	4.0	✓	✓	□	
NORTON_3	65	4.0	✓	✓	□	
▼ 01_Input_seismic_Ultra Far Stack 34_45 [40]						
NORTON_1	65	4.0	✓	✓	□	
NORTON_2	65	4.0	✓	✓	□	
NORTON_3	65	4.0	✓	✓	□	
ADDITIONAL OUTPUTS OUTPUT SUMMARY OUTPUTS						
☒ Save Wavelet Set	☐ Save Gathers	☒ Average Wavelet	☒ Well Wavelets	☐ Well Bulk Shifts	Name: Multi-well Wavelet	Save

13. In the 'Outputs' tab generated wavelets and updated gathers are saved.



Seismic Data Conditioning

INTRODUCTION TO ROKDOC-3D

We are going to set up a RokDoc-3D session, which contains the functionality to work with 3D seismic including SDC and simultaneous inversion.

In the Project Viewer, select 'RokDoc 3D/4D' > 'New 3D Session'. In the popup, under 'Geometry' toggle on 'Arbitrary', click on 'Select Line' and click on 'Well section' in the new popup, then on 'OK'. Set the number of traces to 390 to force 25m trace spacing. The initial view is the arbitrary line through the wells that was used above to give an overview of Norton.

Change the session name, at the bottom left corner from default one to 'SDC & Inversion'. At the top right, add the horizons to the session by clicking in turn on each horizon except 'GOC' and 'OWC' horizons and the arrow.

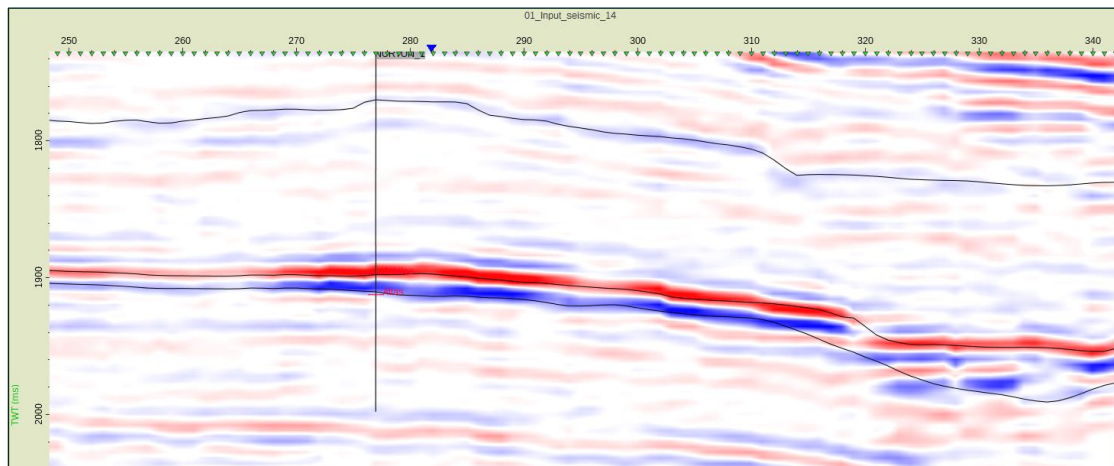
Under 'Volumes' click on 'Add Volume' to add Full stack and then click on 'Add Pre/Partial Stack'. The four angle stacks have been loaded as components of a pseudo-gather. In the pop-up, toggle on 'Add Pre-stack Volume'. Select '01_Input_seismic', then click 'OK'. You can add data to the session at any point using 'Data' menu.

In the 'Wells' panel click on Norton – 1, - 2, - 3, - 4 and – 5 while holding down the Ctrl key, then click the arrow. Finally, click 'OK' at the bottom of the dialogue. The RokDoc-3D session is created and displays a line of section without any seismic trace data.

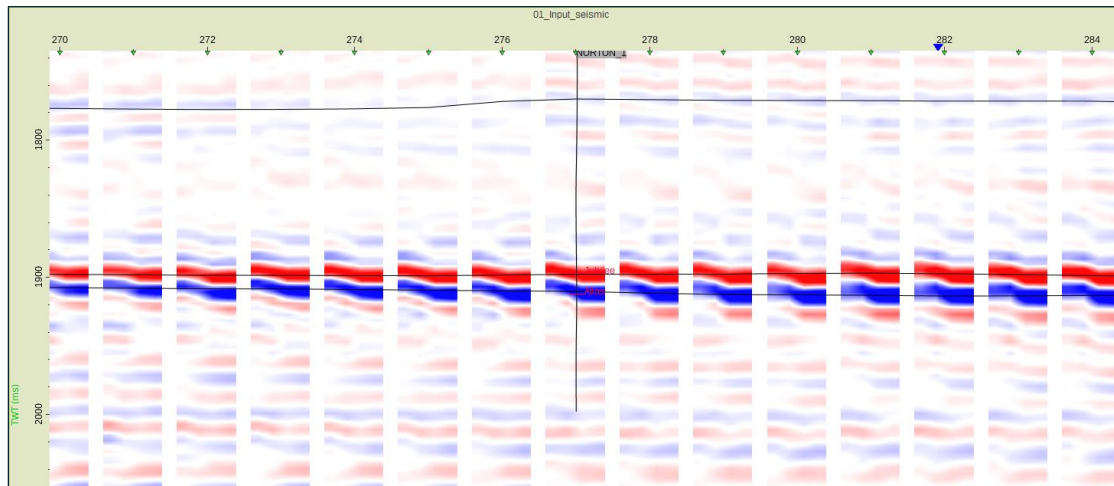
In the data tree on the left hand side, click the folder to the left of 'Seismic and Other Quantities', then to the left of 'P stack', then to the left of '01_Input_seismic'. We can now display individual angle stacks or the set of angle stacks as pseudo-gathers.

Toggle on 'Full_Stack_8_41'. The default gain may need adjustment. Right click on 'Full_Stack_8_41', then click on 'Set Scale'. Select the 'Session trace range' option, set the min and max to -10000 and 10000, then click 'OK'.

Zoom in on the area around Norton – 1 at the Jubilee sand level: hold down the Ctrl key, click in the top-left hand corner of the desired display area and hold down the mouse button to drag a rectangle to the bottom-right of the desired display window, release the mouse button and then release the Ctrl key. The result should look something like this:

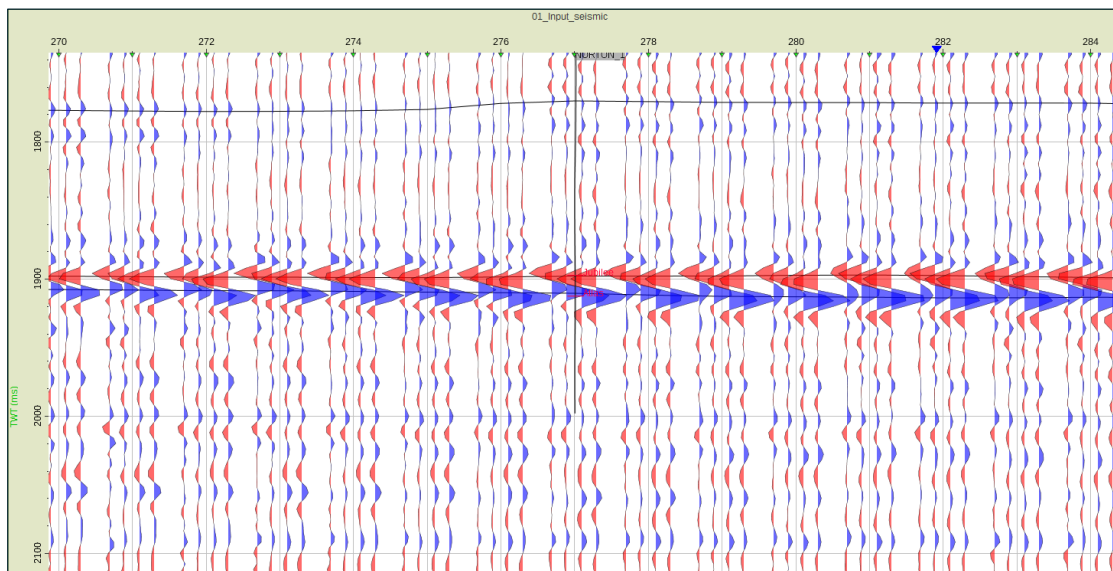


Toggle off 'Full_Stack_8_41' and toggle on '01_Input_seismic' in 'P Stack' folder in the tree. The data are now displayed as pseudo-gathers. Zoom in still further so there are only a dozen pseudo-gathers in the view.



In the e-course, the Petrel colour bar is used.

Right-click on '01_Input_seismic' in the tree and toggle on 'Plot as Wiggle'. The result should look like this:



ZERO PHASING

Zero Phasing

Zero phasing is needed as a preprocessing step for relative impedance inversion (eg coloured inversion) because such methods assume the input data are zero phase. For simultaneous inversion and other absolute inversion methods where the wavelets are an explicit input, there is no need to zero-phase the input data. The only reason to do so is that time alignment of events across the gather will be affected by rotation of wavelet phase with offset, so zero-phasing is a preprocessor to time alignment. In the Norton example, the average extracted wavelets are close enough to zero phase already.

If we do want to zero phase the data, it can be done in SDC.



SPECTRAL BALANCING

Spectral Balancing

Again, this is not necessary for simultaneous inversion, though it is needed before AVO gradient calculation and relative inversion of EEI projections.

The aim is to compensate for the progressive loss of high frequencies from near to far offsets, due to NMO stretch and other effects. The approach taken in RokDoc SDC is to calculate an operator to match the spectra of the various angle stacks to the spectrum of a reference trace.

It is not quite obvious which angle stack should be the reference trace. The safest thing is to use the far trace, which will have the lowest frequency content, but this means throwing away higher-frequency information on the near and mid traces. Using the near or mid trace as reference runs the risk of amplifying noise on the far traces.



Exercise 2: spectral balancing

For the purpose of this exercise, we shall experiment with using the mid trace as reference, although we have seen that this may not be a sensible thing to do as it may amplify noise on the far traces.

Click on 'Utilities > Spectra plot' to start the spectra plot tool

Click on '+' button > 'Quantities' and select the four partial stacks, set the time range, by clicking on 'Horizon', from Jubilee -400 to Atlas 200 ms and click 'OK'

We can apply a spectral balancing correction to match the spectra of all 4 partial stacks to a chosen reference stack. Here we choose the mid stack as our reference to avoid unduly amplifying noise while matching frequency content

Click on 'Seismic Data Conditioning > 'SDC Workflow' in the main menu bar. Give the workflow the name 'SDC'. Click the down-arrow at 'Add new step', click on 'Intercept&Gradient' from the dropdown list, and then click on the green '+' sign to the right of the list. Click the down-arrow at 'Add new step', click on 'Spectral Balancing' from the dropdown list, and then click on the green '+' sign to the right of the list. Spectral Balancing is added to the workflow list.

In the 'Controls' panel set the 'Reference Stack' to 2. This means that the traces will be matched to the mid trace. **Select 'Global' as the method.** This means that average spectra per stack are calculated for the dataset as a whole, operators are calculated for each stack, and then the same operator is applied to all traces with a given angle stack.

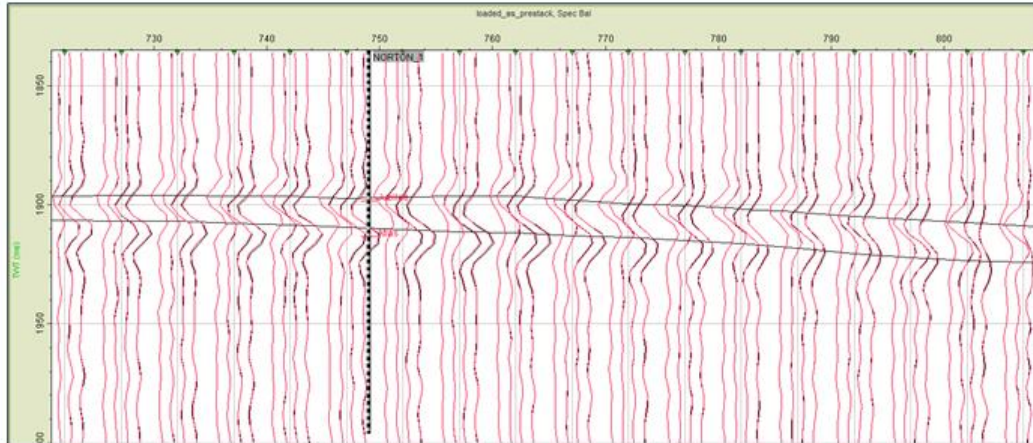
The default seismic range and smoothing parameters are acceptable, so proceed to specifying the TWT range to be used. Enter '-400' next to 'Jubilee' for the Upper Horizon, and '200' next to 'Atlas' for Lower Horizon. The operation is going to be applied over a TWT range similar to that used for the well ties.

Click 'Calculate'. After a short wait the 'Spec Bal' dataset is added to the left-hand data tree in the 'SDC > Pre-inversion' folder. **To view the result, click on the 'Spec Bal' checkbox. Then right-click on 'Spec Bal' and check on 'Plot as Wiggle'. Right-click again and click on 'Set Scale'. Check that the amplitude range is set to 'session trace range', between -10000 to 10000 and click 'OK'.**

To make the before–after trace comparison easier, right-click on '01_Input_seismic' under 'P Stack' and check on 'No Wiggle Fill'. Do the same for 'Spec Bal'. Right-click on 'Spec Bal' again and click on 'Change Line Colour'. On the pop-up colour chooser, click on a red square and then click 'OK'.

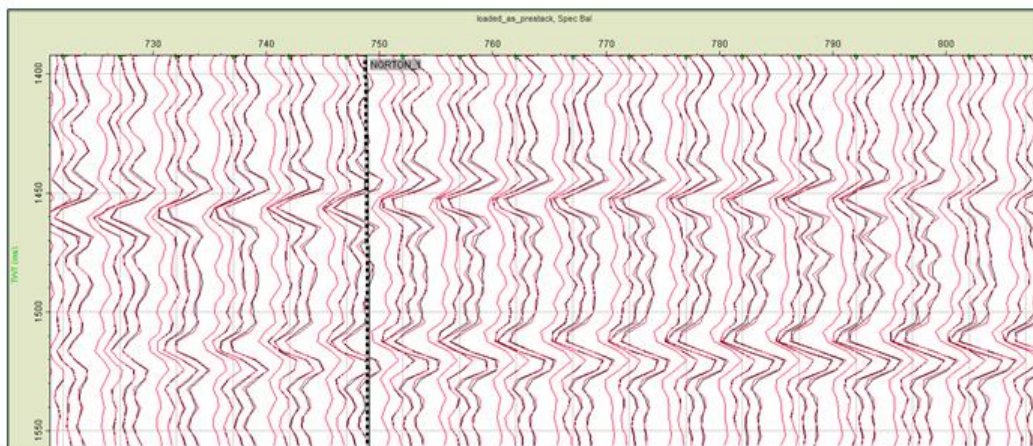
Add the spectrally balanced quantities to the spectra plot and check that the spectra have been balanced.

Spectral balance at the objective level



Pseudo-gathers made from near, mid, far and ultrafar traces. Black = original data, red = spectrally balanced. Differences are slight at this level.

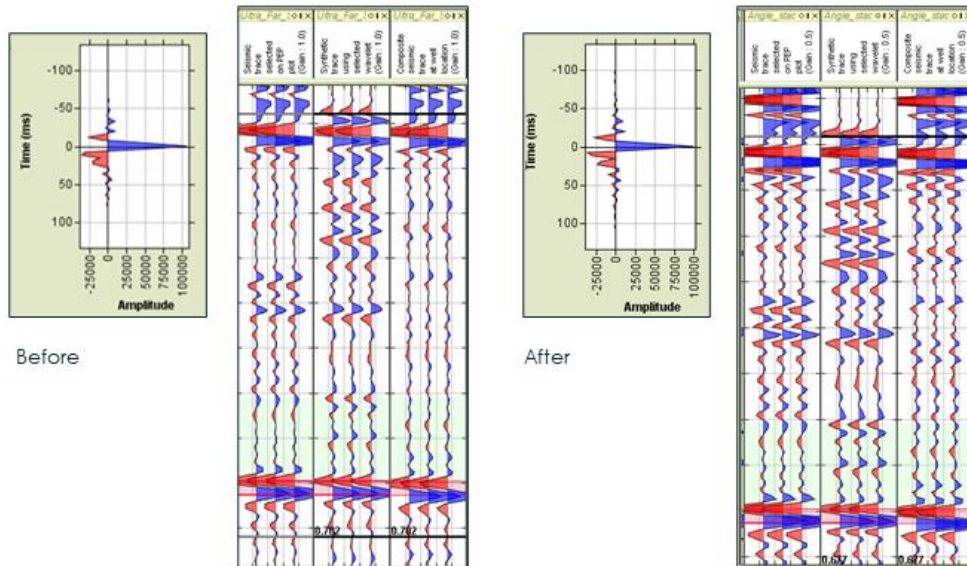
Spectral balance at a shallower level



Black = original data, red = spectrally balanced. Small but visible changes, especially on ultrafars.

1. Explore the effect of spectral balancing at different locations along the section. Are you happy with what it has done?

Ultra Far well tie at Norton-1, before and after spectral balance



Spectral balance: do we want to use it in this particular example?

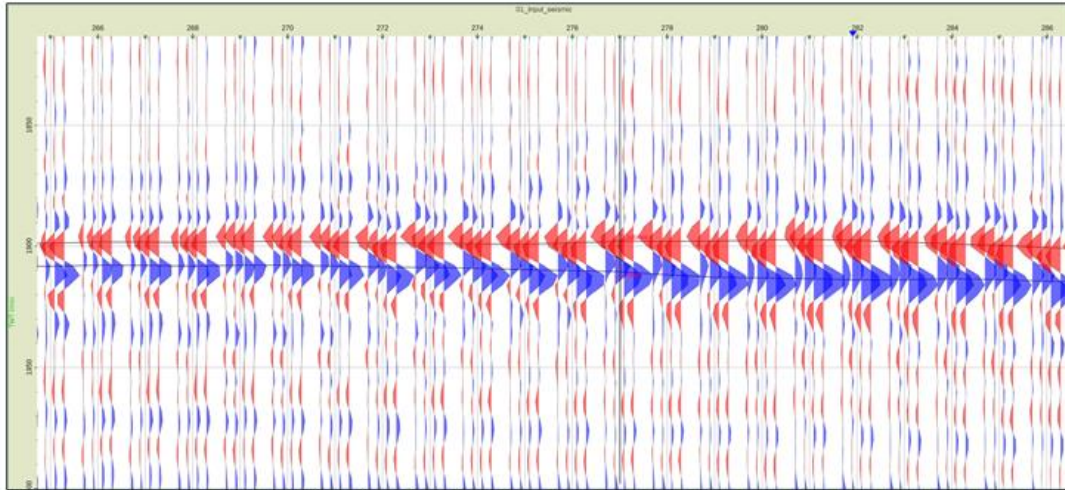
Yes:

- The well ties are quite similar before and after, but the higher frequency content is visible in the wavelet.
- For the simultaneous inversion, we shall use a different wavelet for each offset, so the spectral balancing is not necessary.
- However, we shall also be looking at workflows involving calculation of the AVO gradient, where the frequency difference from near to far may be important.

TIME ALIGNMENT

Time alignment matters because it can have quite a large effect on AVO gradient and the resulting simultaneous inversion. The data in the Norton dataset are in places not very well aligned:

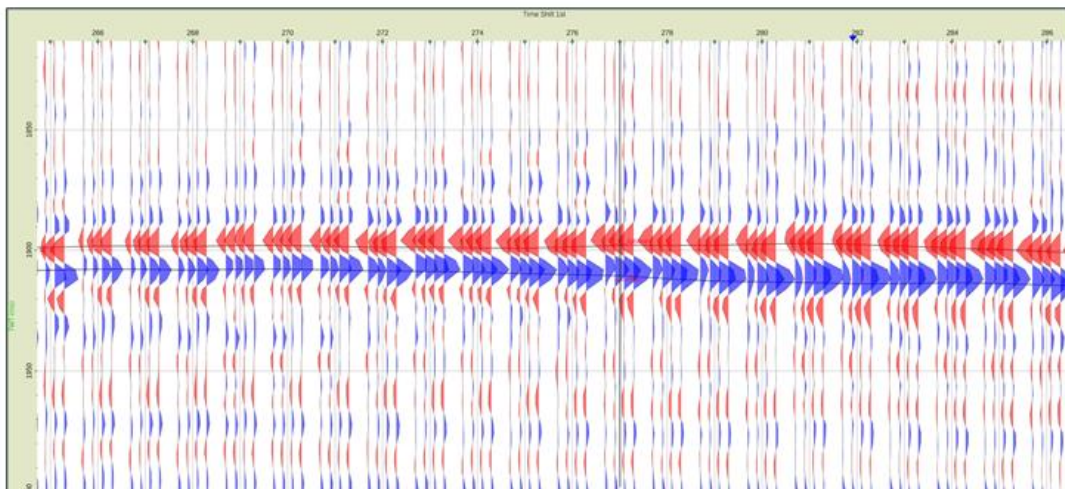
Before time alignment



The traces show considerable residual moveout at the objective level.

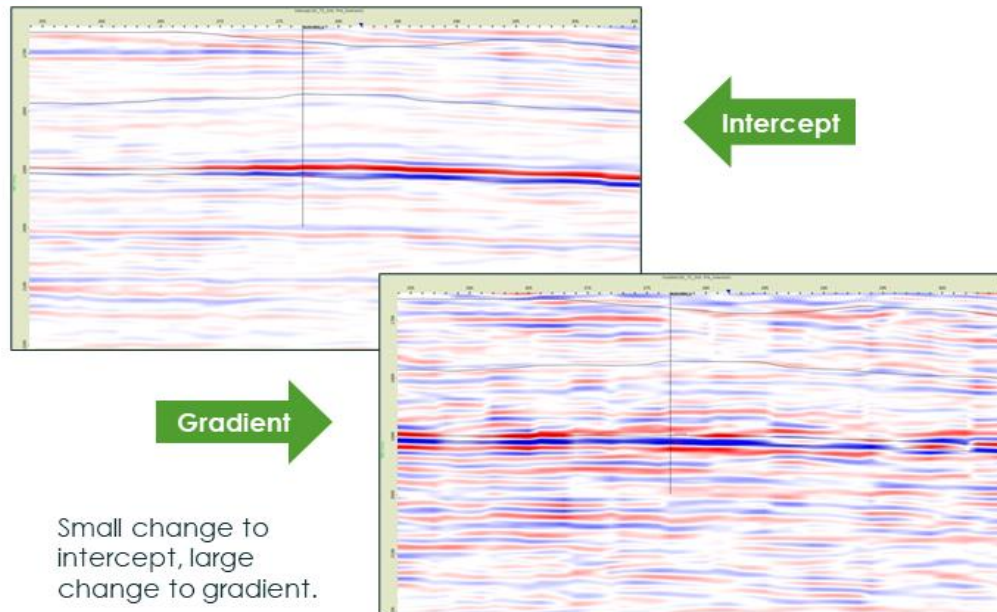
We are going to apply a workflow to align the traces better:

After time alignment

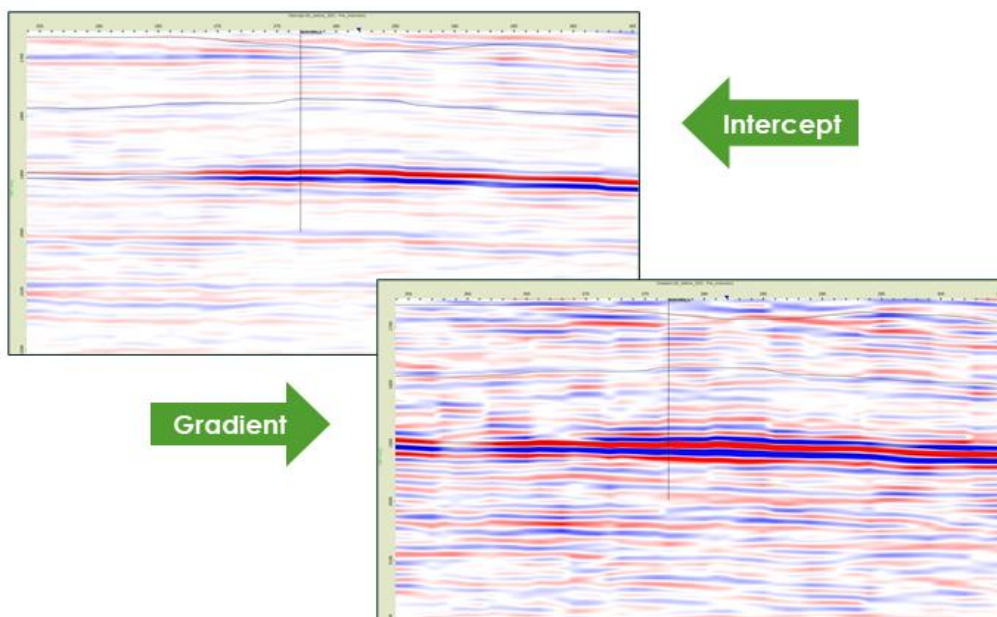


After alignment, residual moveout is much reduced (though still not completely removed in this case)..

Intercept and gradient: after alignment



Intercept and gradient: original data



Misalignment tends in general to increase the magnitude of the AVO gradient. This effect can be removed by aligning the traces correctly. Even so, the AVO gradient tends to be larger than it should be because of the systematic effect of noise (Hendrickson 1999). Noise removal (perhaps by dip-steered median filtering Höcker and Fehmers 2002) is beneficial, as are methods of gradient calculation that work over multiple time samples to stabilise the value (Whitcombe et al 2004).



Exercise 3: time alignment

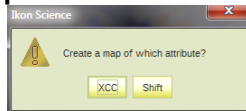
Before calculating a correction, it is recommended to measure the level of mis-alignment. To do this right-click in the attribute panel at the base of the 3D viewer and select 'Measure attribute'.

Select 'Correlation' in 'Measurement Category' menu and 'XCC r (with shift)' in 'Measurement Type'. Set the seismic input data to by selecting '01_Input_seismic_14 (01_Input_seismic)', as Quantity_1 and '01_Input_seismic_40 (01_input_seismic)' as Quantity_2. Parameterise correlation window to 80ms by selecting 'Jubilee' in Horizon 1 and Horizon 2 and set the offset to -40ms from Horizon 1 and 40ms from Horizon 2. Set the, the maximum shift to 16ms and the minimum cross correlation to 0.5. This attribute shows the time shift required on each trace to optimise the correlation between each trace and the ultra far stack.

Name the attribute 'XCC_TS_Jubilee' and click 'Ok' in the bottom of the dialogue.

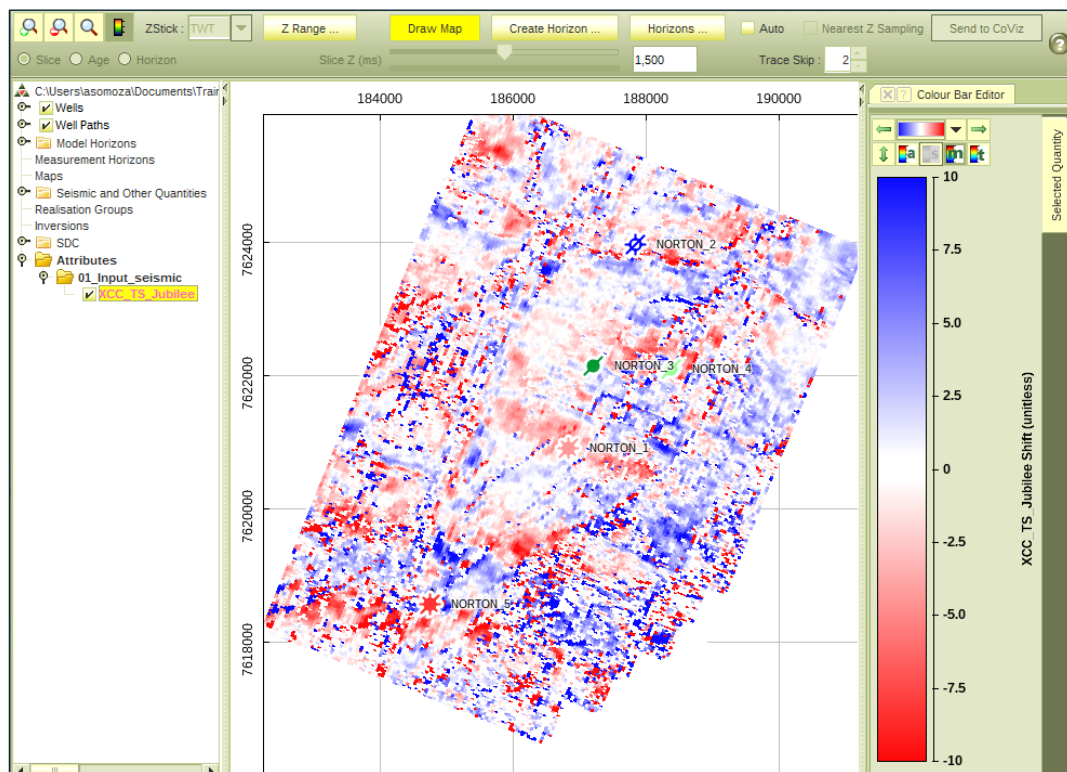
View the attribute lines in the attribute panel by selecting it in Attributes category of the data tree. Zoom in on an area of the seismic pseudo gathers in the RokDoc3D viewer, does the attribute line match what you see in the data?

To view a measure of alignment across the volume click on the map viewer icon  on the top panel. In the map viewer expand attributes in the data tree, expand 'Attributes' and select 'XCC_TS_Jubilee' (to display the shifts map). Set the 'trace skip value to '2' and click 'draw map'. In the popup dialogue click



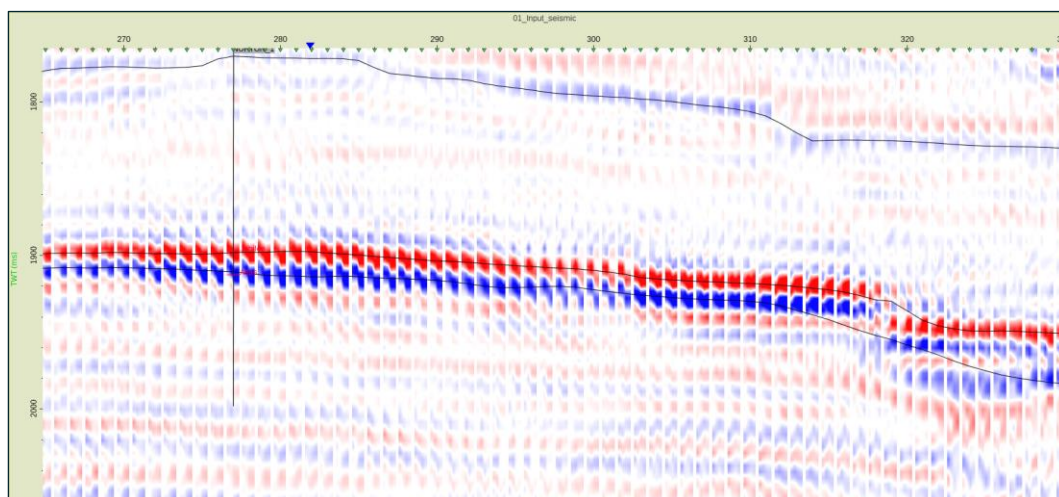
'Shift' button.

The map is calculated but initially displays on a colour scale that fails to show variation in the data. Click on the colour bar icon in the map viewer, right click on the colour bar and select 'set plot range'. Set the plot to a manual range of -10 to 10 and change colour scheme to symmetrical Blue White Red. Map could be saved by clicking the Create Horizon button.



Go back to the RokDoc 3D window. In the Seismic Data Conditioning panel, click the down arrow at 'Add new step', click on 'Time Shift Estimation' and click on the green '+' sign.

In the tree, toggle off '01_Input_seismic', right-click on 'Spec Bal' and toggle on 'Plot as colour'. Unzoom by clicking the zoom out icon  several times to get a display similar to this:

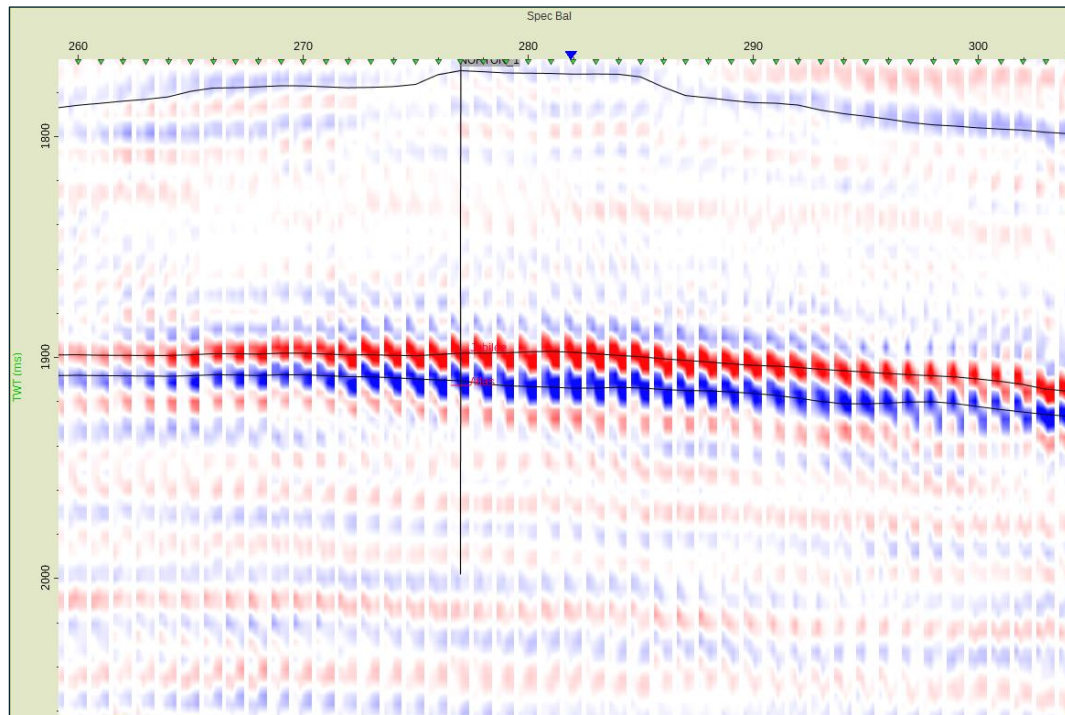


Set parameters as follows under 'Controls': 'Reference Stack' = 14.0 (data will be lined up to the nears); 'Time Window Length' = 100. Set the range to 'Jubilee' -400 to 'Atlas' +200. Click 'Calculate'. In the tree, toggle on 'Time Shift'.

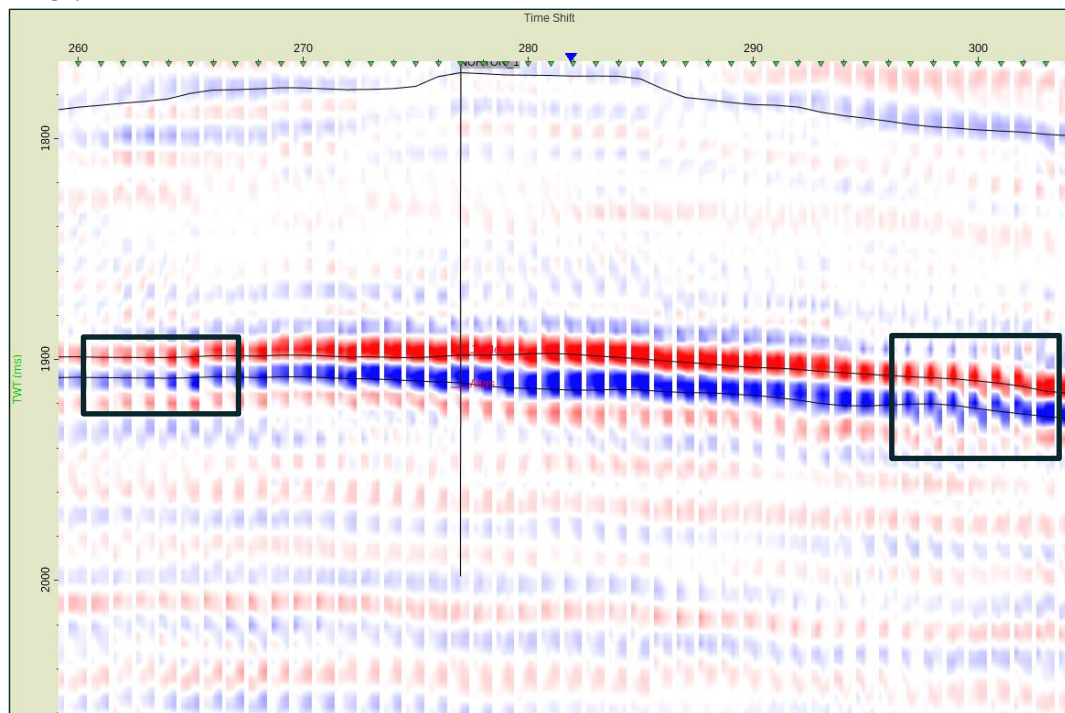


In the tree, toggle 'Time Shift' on and off to observe what the flattening has done. There is definite improvement in quite a few places, but alignment isn't perfect everywhere. How can we try to improve it?

Before:



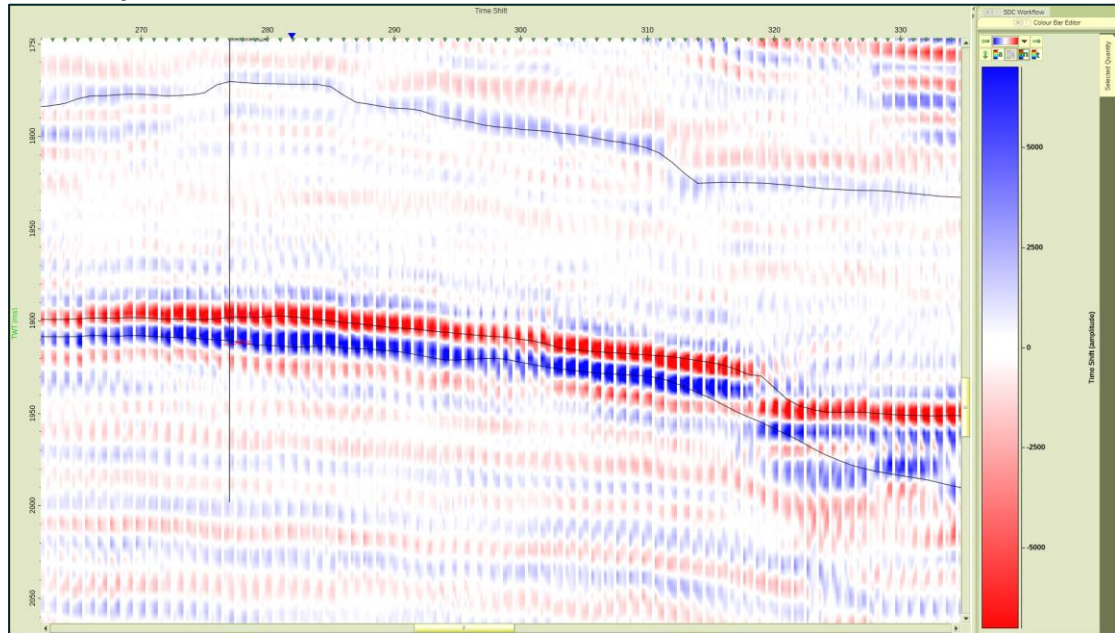
After:



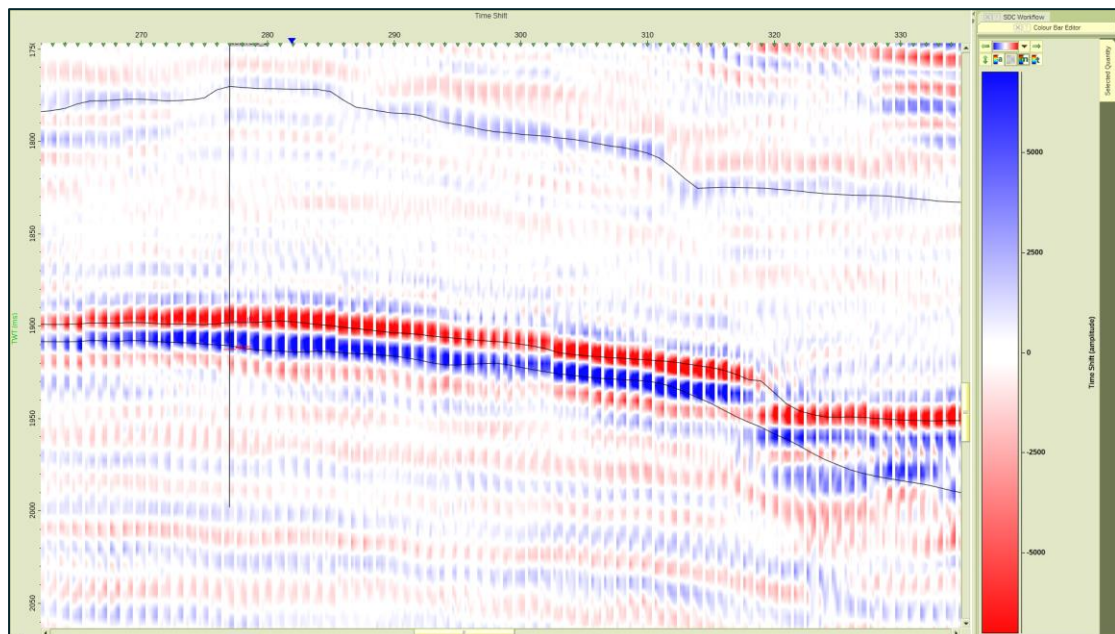


Naeini's method has difficulty estimating large time shifts, but as shown in his paper a second pass of shift application can be helpful. **Repeat last two steps to add a second pass of time shift with same parametrisation. Edit the name of the quantity in the table to be 'Time Shift 2nd'.**

After 1st pass:

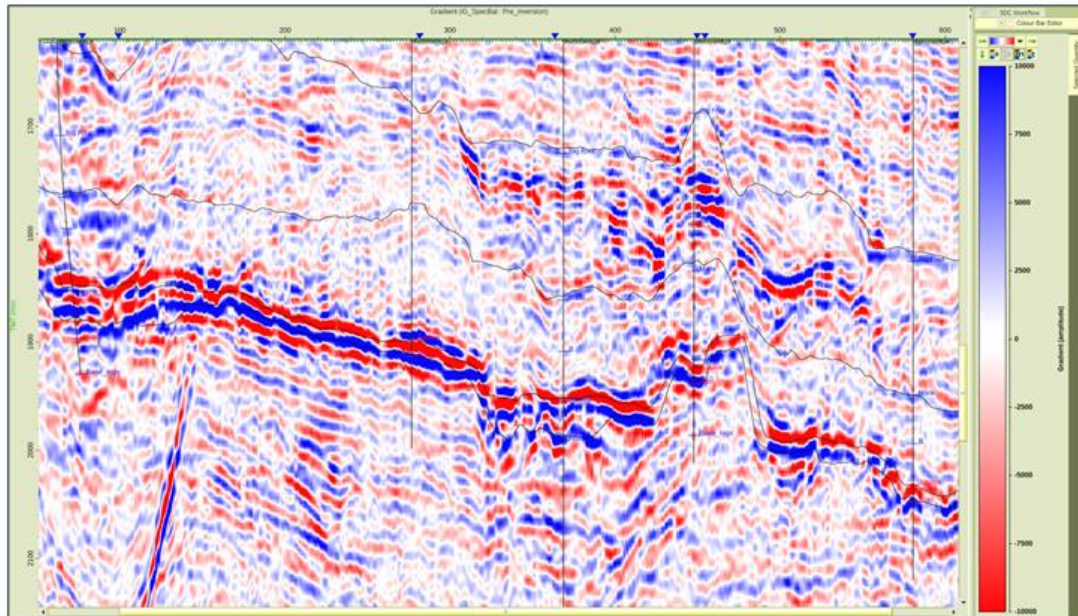


After 2nd pass:

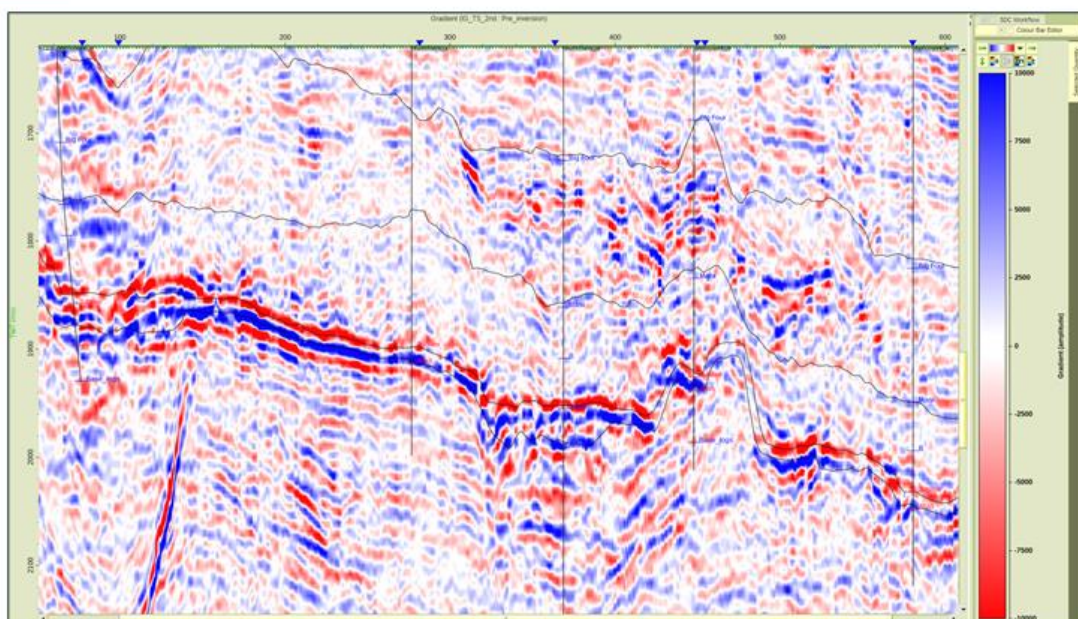


How much difference do the shifts make?

Gradient from gathers before time alignment



Gradient from gathers after 2nd pass time alignment





AMPLITUDE BALANCING



Amplitude balancing

- The seismic processor should have adjusted seismic amplitudes so that the same scaler from seismic trace to reflection coefficient applies across the entire offset range (ie the wavelet amplitude).
- However, the path length is longer for the far offsets so the amplitude will be lower due to spherical divergence and absorption. The processor corrects for these with a spherical spreading correction and a Q compensation process. The former is usually a simple function of offset, time and velocity; inaccuracies in the velocity field and neglect of VTI anisotropy may lead to erroneous amplitude correction. The latter depends on an estimate of the absorption coefficient Q which is difficult to measure accurately.
- The result is that amplitude variation with offset may have a systematic bias, which will obviously lead directly to incorrect AVO gradient values. Therefore, amplitude variation with offset needs to be checked against well data.



Exercise 4: amplitude balancing

We shall begin by following a workflow that compares RMS amplitudes in a window between synthetic and real seismic data. This is not strictly necessary to perform here, as there is a step in the simultaneous inversion setup that automates the procedure. We shall however calculate and apply scalars to the angle stacks using a standalone module so that they are corrected for use in calculation of AVO gradient, which is needed for alternative workflows involving intercept – gradient calculation and coloured inversion of chi angle projections that will be discussed later in the course.

In order to calculate the scalars for angle stacks software will compare synthetic traces to the seismic traces at each well and work out the RMS amplitude difference over a time window. This process requires a simple background model to be built in RokDoc 3D.

1. Click on 'Geological Modelling' > 'Make Zones', check that the bodies are listed in order of depth (see image below).

In the 'Make Zones' dialog, the user defines the surface above and below each body. Additionally, the user should specify the zone colour and the fill type. Other options available under 'Make Zones' dialog are Model Building parameters, generating TVDss horizons from TWT horizons and well markers, and performing depth conversion using horizons, interval velocity volumes, linear interpolation or a quadratic equation.

Make Zones

Stratigraphy | Model Building | Datums | Depth Conversion | Horizon Adjustment

Horizons

Horizon	Marker	Dominant	Interpolate Gaps	Snap Up
Seabed	Seabed	☐	☐	☐
Mercury	Mercury	☐	☐	☐
Ranger	Ranger	☐	☐	☐
Big Four	Big Four	☐	☐	☐
Manx	Manx	☐	☐	☐
Jubilee	Jubilee	☐	☐	☐
Atlas	Atlas	☐	☐	☐

Zones

Name	Horizon Above	Horizon Below	Colour	Stratigraphy
Top	Top	Seabed		Follow Bottom
Seabed	Seabed	Mercury		Proportional
Mercury	Mercury	Ranger		Proportional
Ranger	Ranger	Big Four		Proportional
Big Four	Big Four	Manx		Proportional
Manx	Manx	Jubilee		Proportional
Jubilee	Jubilee	Atlas		Proportional
Atlas	Atlas	Bottom		Follow Top

OK Cancel



2. Switch to the 'Datums' tab, set the 'Water depth below MSL' to be 'Seabed_D'. In the 'Depth Conversions' tab, set the underburden velocity to 3200m/s. Click on 'OK'. This tool allows you to specify the time-depth relationship for the 3D session in many different ways. However as all of our work will be done in the time domain we will leave this set to the default method. Note that the wells are displayed in time in the 3D session using their own independent time-depth relationships (derived at each well during the well tie process).
3. Back in 'Geological Modelling' > 'Make Base Property Model'. Check that each body is set to fill by interpolation using the $1/R^2$ method, to do this, select Vp and change the interpolation type under 'Fill Parameters' and select $1/R^2$, repeat with Vs and Rho. The interpolation type needs to be changed for Vp-Vs-Rho in each zone. To copy the properties across the other zones, highlight the zone 'Top' in the same table and press the copy button  at the top, then select the other zones and click on the paste button .
4. In the same 'Make Property Model' dialog, click on 'Select Well Data' at the top and click 'Select' in the next dialogue, logs for Vp, Vs and Rho should be the conditioned ones. Click on 'Ok' and 'Ok' again. You now have a model of Vp, Vs and Rho across the 3D session (and calculated derivative quantities such as AI). The scaling tool will use these 'body fill' quantities later in the workflow.

Select Markers, Logs, Minerals & Saturations											
Change Data Types...			Change Fluids...			Change Minerals...					
Well ▲	Colour	GR	Vp	Vs	Rho	Por	Water	Oil	Gas	Quartz	Shale
NORTON_1		GR	Vp_conditioned	Vs_conditioned	Rho_conditioned	PHIE_fill	COMP	So	Sg	COMP	Vshale
NORTON_2		GR	Vp_conditioned	Vs_conditioned	Rho_conditioned	PHIE_upd	Sw	COMP	Sg	COMP	VSH_upd
NORTON_3		GR	Vp_conditioned	Vs_conditioned	Rho_conditioned	PHIE_upd	SWE_upd	SOE_upd	COMP	COMP	VSH_upd
NORTON_4		GR	Vp_conditioned	Vs_conditioned	Rho_conditioned	PHIE_upd	SWE_upd	SOE_upd	COMP	COMP	VSH_upd
NORTON_5		GR	Vp_conditioned	Vs_conditioned	Rho_conditioned	PHIE_upd	Sw_upd	COMP	Sg_upd	COMP	VSH_upd

By default, the property model will be built using the well data selected at the moment of generating the 3D session. If the user would like to change the well data, then click on 'Select Well Data' in the 'Edit Base Property Model' dialog.

5. Click on 'Utilities' > 'Seismic Scaling' to launch the seismic scaling module.
6. Select the SDC 'Time Shift 2nd' flattened results as the input seismic. Select the appropriate averaged wavelet for each angle (near_final_average, mid_final_average, far_final_average and ultrafar_final_average) or a Wavelet Set. Check that the 'Base scenario model' quantities are selected (AI, SI and Rho), switch to the 'Seismic scaling' tab.

- Choose the option to calculate scaling at the wells and select a time window from Jubilee -400 to Atlas +200.

Note that now there are horizons with the suffix (**for_SDC & Inversion**) as the name of the 3D session. These horizons are conditioned horizons by the software during making zones. Software will automatically interpolate all the gaps and resolve horizon intersections.

- Set the Area Around wells to 9x9 and click 'Calculate Scaling'. PP Average Scalars' will be calculated automatically. Check that the values at each well look sensible. Note that the near scalar is absolute, but the remaining scalars are all relative to the near stack. Click 'Save' button to save the results.

Scaling across the Norton wells

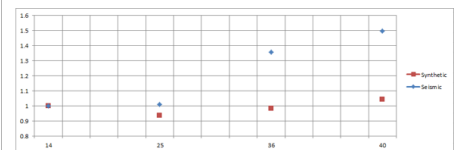


These scalars show that there is well-to-well variation. This could in principle be taken account of by using a laterally variable overall scalar. However, we don't really know how to do this, unless we understand what causes the variation.

These scalars imply that we need to scale the partial stacks.

Well Name	Angle 14.0	Angle 25.0	Angle 36.0	Angle 40.0
NORTON_1	1.467e-05	0.8574	0.6701	0.7427
NORTON_2	1.632e-05	0.8606	0.7073	0.6749
NORTON_3	1.588e-05	0.9024	0.7919	0.7575
NORTON_4	1.888e-05	1.2503	0.8038	0.8197
NORTON_5	2.729e-05	0.8544	0.7926	0.8437

Angle	Scalar
14.0	1.861e-05
25.0	0.9450
36.0	0.7532
40.0	0.7677



Amplitude change on Mid, Far and Ufar stacks relative to Near stack. Average from five wells.

- In the SDC workflow module pick the 'Scalars' option from the 'Pick new step' drop-down menu and click the green '+' button.
- Import the previously saved scalar set and **manually set the near scalar value to "1.0"**. click 'Calculate'

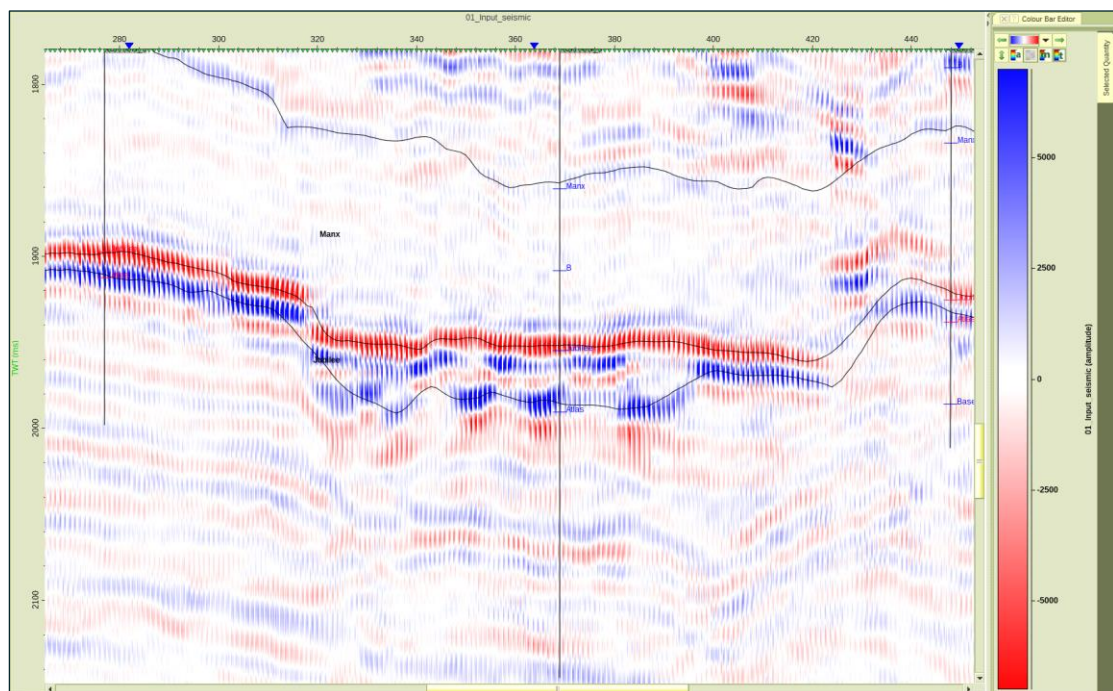


Import Scalar Set		
	Horizon	Scalars
Near	☐	☒ 1.0000
->	☐	☒ 0.9450
->	☐	☒ 0.7532
Far	☐	☒ 0.7677

Note that SDC tool is using absolute scalers for each stack. That is why we need to reset scaling coefficient for Near stack to 1 which means no scaling.

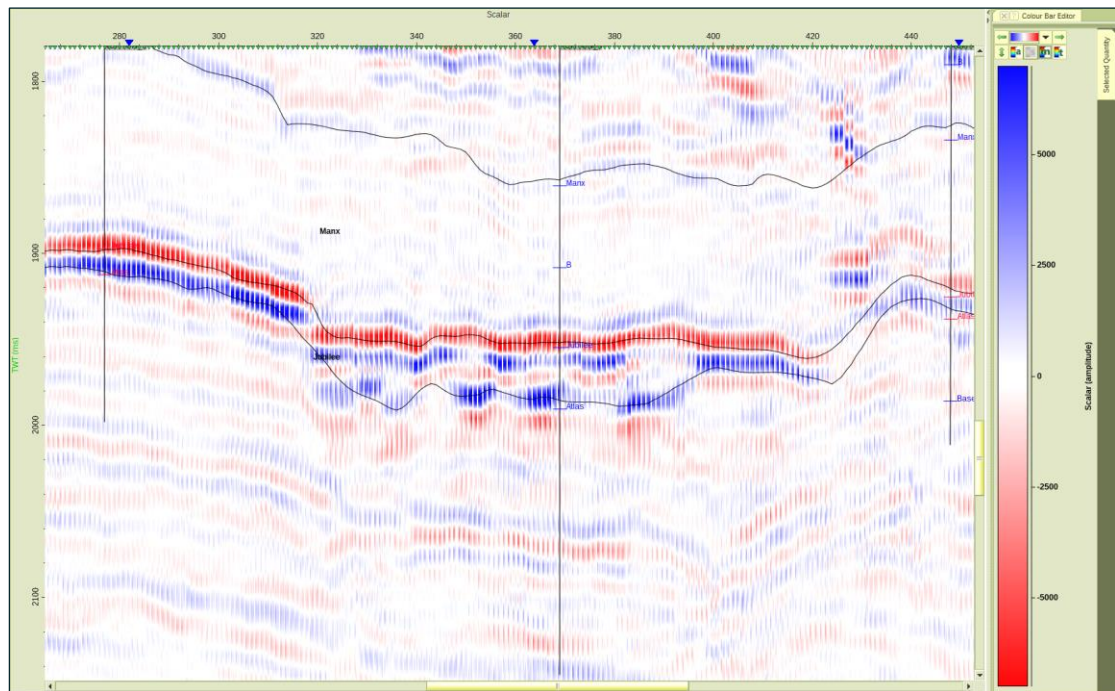
11. In the tree, expand the 'SDC' folder if necessary, then left-click on 'Scalar'. Now you can compare the data before and after SDC by moving around the section, toggling between '01_Input_seismic' and 'Scalar'.

Input data in SDC:





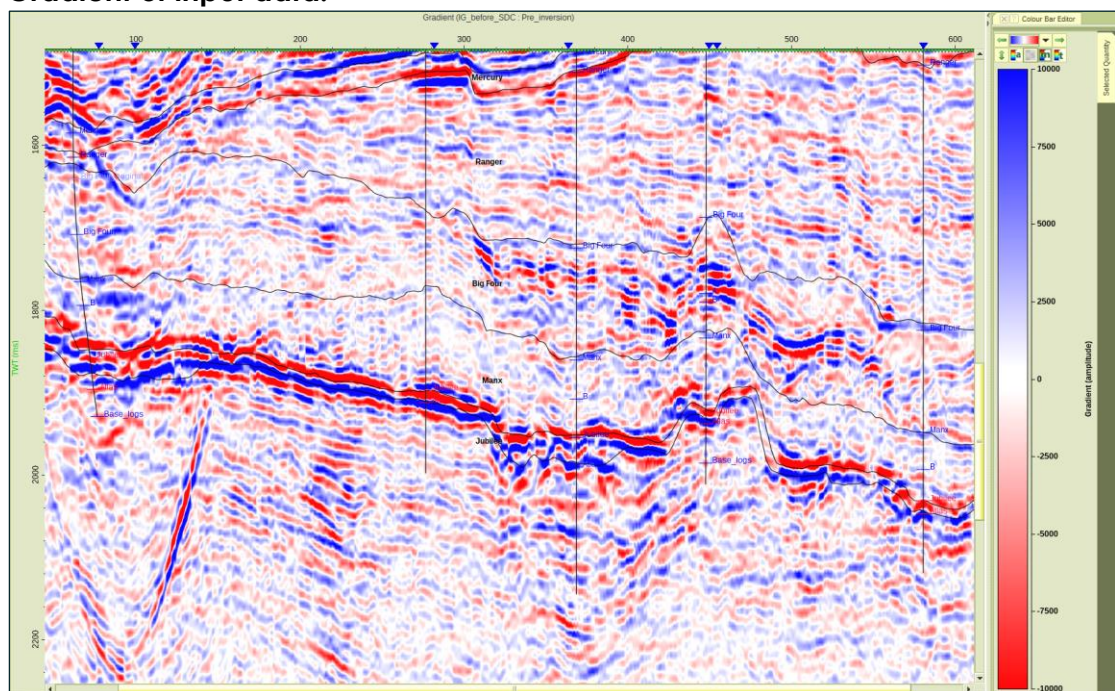
Data after SDC, including scaling:



To see what difference we have made to the AVO gradient, we can compare the gradient from the initial data with the gradient from the final conditioned data:

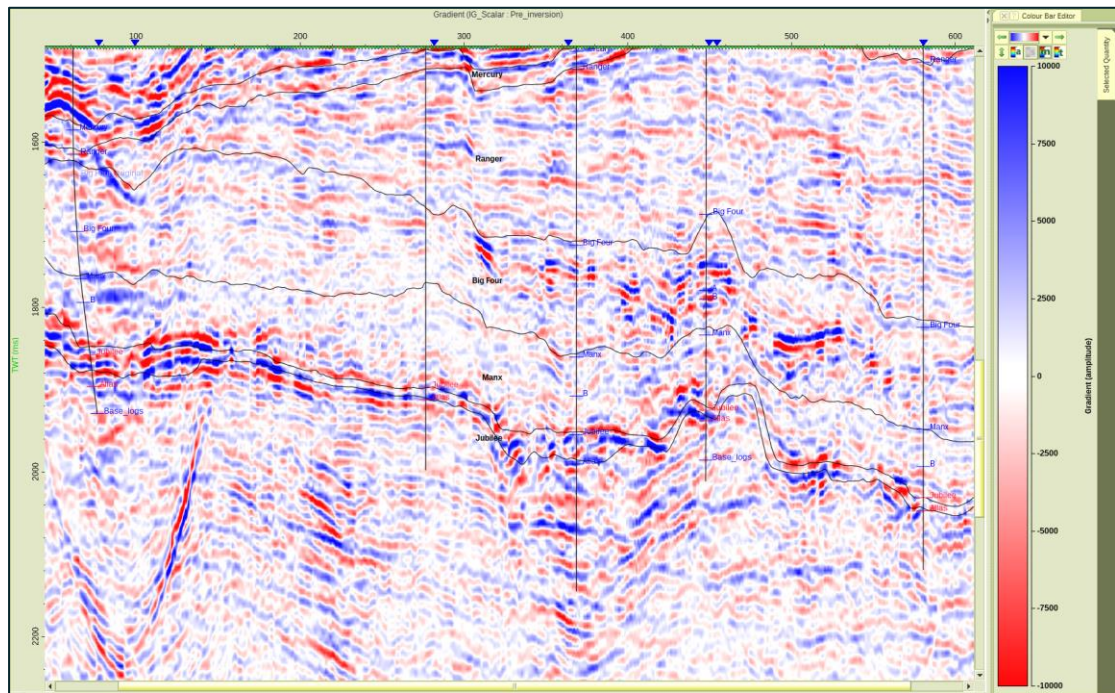
- 12. Add an 'Intercept & Gradient' step to the SDC workflow, use the linear fit method and the full seismic range and click 'Calculate'.**

Gradient of input data:





Gradient after SDC:

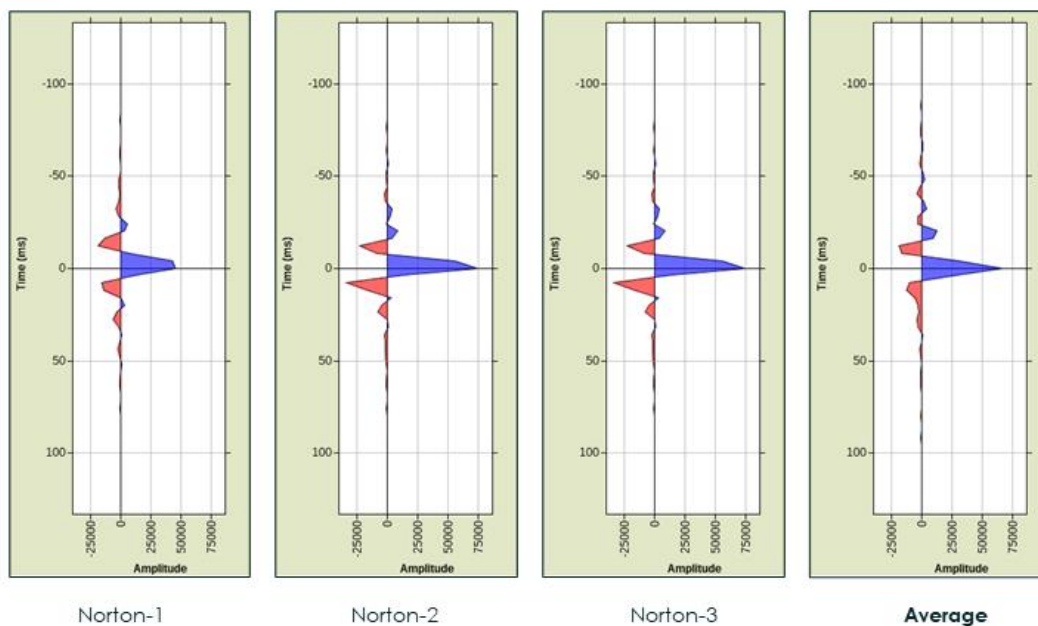


The gradient from the final data is much lower in amplitude, and is more consistent with the modest change of amplitude with offset that we expect from the synthetics.

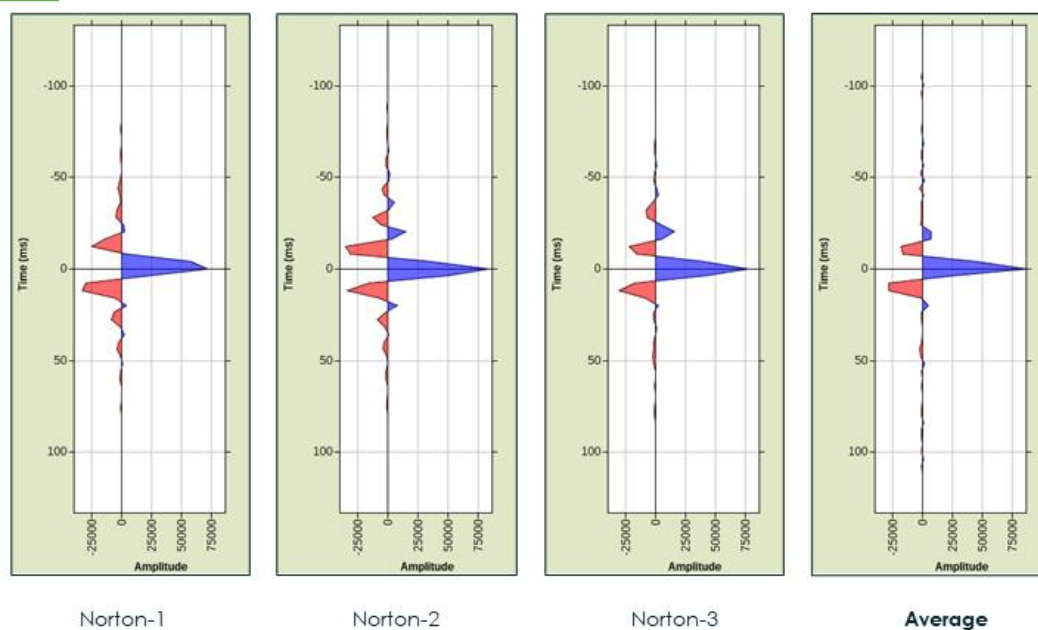
13. To realise the conditioned volumes as segy files right-click on the 'Scalar' group under SDC in the data tree and click 'Export 3D'. Ensure that the 'Add to session' and 'Create P stack quantities' checkboxes are ticked.
14. Add the Intercept and Gradient quantities to the export list. Click 'OK' and select an appropriate location for RokDoc to write the volumes out to. Leave the export to run.

Final well-ties

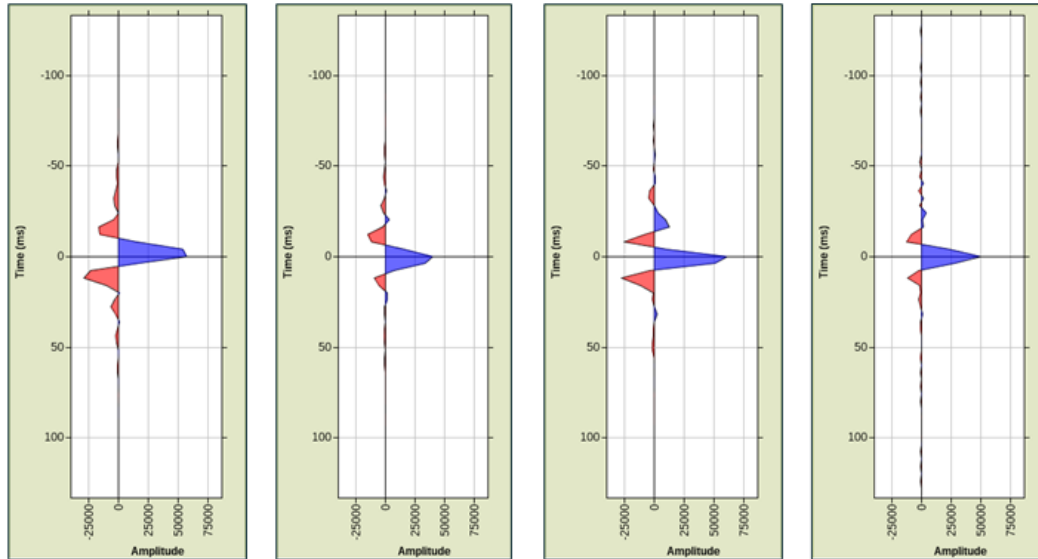
Final wavelet - near



Final wavelet - mid



Final wavelet - far



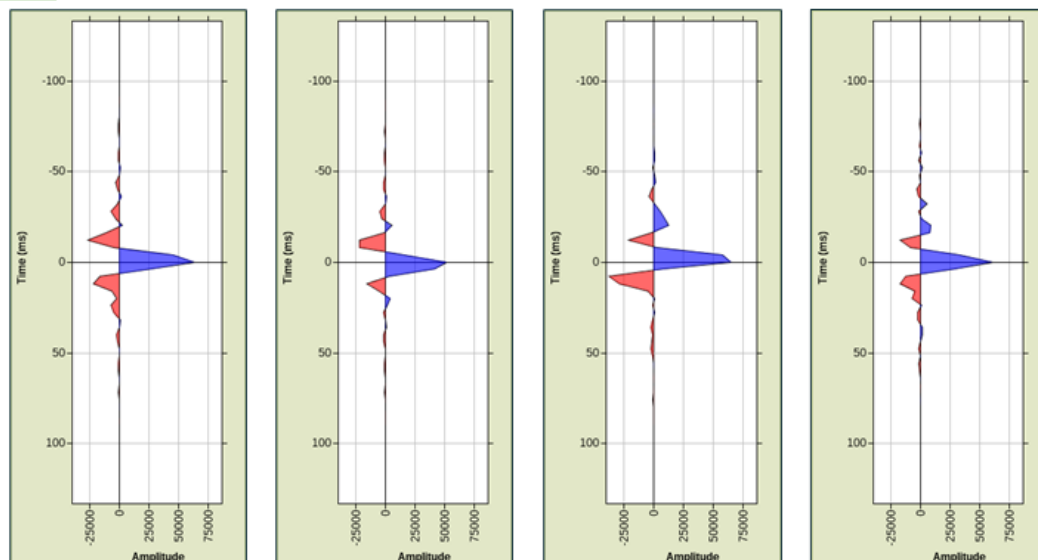
Norton-1

Norton-2

Norton-3

Average

Final wavelet – ultra far



Norton-1

Norton-2

Norton-3

Average



These wavelets look quite satisfactory. They are close to zero phase and have similar bandwidth across the offset range, as expected after spectral balancing. Note however that the amplitudes are not identical (higher amplitude on the mids, lower amplitude on the fars). In spite of the correction we have made to the average scaling, the well ties derive the wavelet that gives the best overall match between synthetic and seismic traces. At higher frequencies, this is disrupted by the effect of small inaccuracies in the time-depth relation. We are going to use these average wavelets as input to the simultaneous inversion, and they should be a good description of the wavelet shape; the scaling relative to the real traces still needs to be adjusted in the inversion setup.



Inversion: setting up the input data

We now come to setting up the inversion. The first thing is to set up the input data, then the background model.

1. **In the RokDoc-3D Viewer, close the 'Seismic Data Conditioning' tab if it is still open, then click on 'Inversion' > 'Simultaneous Inversion and Scaling' in the menu bar at the top of the window. In the 'Input Data' tab of the dialogue, click on the down arrow next to 'Angle Stack Data' and select the realised SDC output group.**

We are going to use the output from the SDC stage as our input. For the purposes of the course we are using a dataset which has been created by running the SDC process, as set up previously, over the entire volume; this is just to make sure everyone has the same starting dataset for the inversion. We could alternatively use any output from the SDC step and explore dynamically the effect of changing SDC parameters on the inversion result. We also have a set of conditioned angle stacks in case if you are not entirely happy with results you've got or in case if calculation takes too long to complete. You can add conditioned data to the 3D session from **Data > Add Pre/Partial Stack Data** menu. Chose **02_SDC_realized** form drop down menu and hit OK.

2. **Click on the down arrow next to 'Output Quantities' and select 'AI, SI, Rho'.** Note that several other combinations are available.
3. **Check that 'No. of impedances' is set to 3.** It is optional to include density in the calculation. As noted previously, it is likely to be poorly determined, but with good quality seismic and good well-ties, we might get some sensible results.
4. **To choose the wavelets, double-click in each 'Wavelet' box and select from the list: Angle 14.0, 'near_final_average'; Angle 25.0, 'mid_final_average'; Angle 36.0, 'far_final_average'; Angle 40.0, 'ultrafar_final_average'. Alternatively you can select wavelets by clicking 'Select Wavelet Set' button.** These are just the final average wavelets that we saw in the last section.



Exercise 5: making the background model

Prior setting up the data to make the background models, we have to re-calculate scalars. **Why do we have to calculate scalars again if we already scaled the seismic?** The first time we did it to correct the AVO gradient. This time we will calculate scalars to match inverted impedances to well impedances. This involves using different time window which more focused on inversion interval.

- From the data tree, right-click on the 'Wells' folder and select 'Select Well Data' then click 'Select' button again in 'Select wells' dialogue. In the 'Select Markers, Logs, Minerals & Saturations' dialog make sure that *_conditioned logs for each Vp, Vs and Rho were selected. You can change them by double-clicking on the logs name. The dialog should look like the image below.

Well	Colour	GR	Vp	Vs	Rho	Por	Water	Oil	Gas	Quartz	Shale
NORTON_1	Red	GR	Vp_conditioned	Vs_conditioned	Rho_conditioned	PHIE_fill	COMP	So	Sg	COMP	Vshale
NORTON_2	Blue	GR	Vp_conditioned	Vs_conditioned	Rho_conditioned	PHIE_upd	Sw	COMP	Sg	COMP	VSH_upd
NORTON_3	Green	GR	Vp_conditioned	Vs_conditioned	Rho_conditioned	PHIE_upd	SWE_upd	SOE_upd	COMP	COMP	VSH_upd
NORTON_4	Light Green	GR	Vp_conditioned	Vs_conditioned	Rho_conditioned	PHIE_upd	SWE_upd	SOE_upd	COMP	COMP	VSH_upd
NORTON_5	Red	GR	Vp_conditioned	Vs_conditioned	Rho_conditioned	PHIE_upd	Sw_upd	COMP	Sg_upd	COMP	VSH_upd

In this step you select the logs that are going to be used to create the model. The background model is now set to the in-situ fluid case. Because we want to check the relative scaling of the angle stacks, we are going to use a background model whose synthetic response can be directly compared with the real data. After we have saved the scalars, we will change the log selection to the brine case so that the final inversion is based on a brine-case model; any hydrocarbon effects in the inverted output are then definitely derived from the seismic data and not the starting model. **Click 'OK'.**

- On the top menu bar, click 'Geological Modelling' > 'Edit Base Property Model'. Check that interpolation method still set the to $1/R^2$ for Vp, Vs and Rho quantities for each zone but other options are available including collocated co-kriging which can be used to incorporate seismic interval velocity information. In order to speed up interpolation process such properties like GR, Por, Saturation and Mineral volumes can be set to constant instead of "Interpolate" like shown below.



Property	Fill Type	Fill Parameters
GR	Constant	Constant value 70
Vp	Interpolate	Interpolation type : 1/R²; (Background model : Constant value 2,560)
Vs	Interpolate	Interpolation type : 1/R²; (Background model : Constant value 1,130)
Rho	Interpolate	Interpolation type : 1/R²; (Background model : Constant value 2.35)
Por	Interpolate	Interpolation type : Ordinary Kriging, Default; (Background model : Constant value 0.25)
Water	Constant	Constant value 1.00
Oil	Constant	Constant value 0.00
Gas	Constant	Constant value 0.00
Quartz	Constant	Constant value 0.80
Shale	Constant	Constant value 0.20
Seabed		
Mercury		
Ranger		
Big Four		
Manx		
Jubilee		
Atlas		

- In the data tree, toggle off 'Zones' if still on, click the icon next to 'Base Property Model' to expand it, then expand 'Data' and 'Impedances' in the same way.
- Check that in the 'Simultaneous Inversion and Scaling' dialogue, in the 'Input Data' tab, the background impedance models are set to AI, SI and Rho from the Base Scenario Model.
- Open the 'Seismic Scaling' tab. Toggle on 'Compute Scalars' and Compute at: 'Wells'. Set the time limits to be -50 ms for the Jubilee horizon and 50 ms for the Atlas horizon. This means that the scalars will be optimised for the target level. Set the Area Around Wells to 3x3 CDPs. Click 'Calculate Scaling'. PP Average Scalars' will be calculated automatically.

OPTIONS

Scaling: ☐ No Scalars ☒ Compute Scalars ☐ Read Scalars

Compute at: ☒ Wells ☐ Line of Section

Area Around Wells: 3x3

SCALING WINDOW

Upper: ☒ Horizon ☒ Time Jubilee +/- (ms): -50

Lower: ☒ Horizon ☒ Time Atlas +/- (ms): 50



The result should look roughly like this:

▼ WELLS

PP Scale Factors

Use	Well Name	Angle 14.0	Angle 25.0	Angle 36.0	Angle 40.0
☒	NORTON_1	1.383e-05	0.9858	0.9700	0.9091
☒	NORTON_2	1.579e-05	1.0144	0.8112	0.6853
☒	NORTON_3	1.435e-05	0.8527	0.9472	0.8467
☒	NORTON_4	1.079e-05	1.8359	1.4620	1.4155
☒	NORTON_5	1.881e-05	0.8603	0.9936	0.8772

Calculate Scaling

▼ SCALARS

PP Average Scalars

Angle	Scalar
14.0	1.471e-05
25.0	1.1098
36.0	1.0368
40.0	0.9468

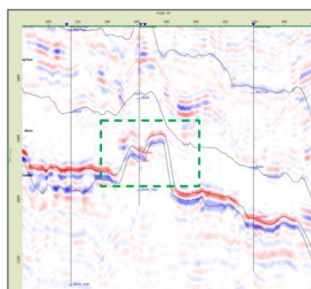
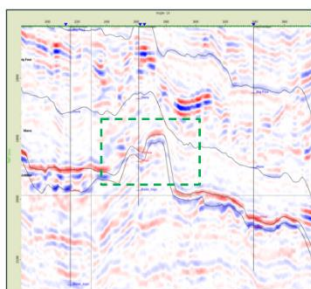
▼ OUTPUTS

Set Description... Suffix: SeismicScaling Save

Note that the near angle shows the overall scalar, the other angles show the scalar relative to the nears.

The anomalous results at well 4 are probably due to an imaging problem.

Norton-4 on near and far stacks



On the near trace display, Top and Base Jubilee are quite weak and instead there are prominent dipping events: imaging problem?

It is reasonable to omit this well from the scalar calculation.



10. Uncheck the 'Use' box for Norton 4 and calculate the scalars again.

▼ WELLS

PP Scale Factors

Use	Well Name	Angle 14.0	Angle 25.0	Angle 36.0	Angle 40.0
☒	NORTON_1	1.383e-05	0.9858	0.9700	0.9091
☒	NORTON_2	1.579e-05	1.0144	0.8112	0.6853
☒	NORTON_3	1.435e-05	0.8527	0.9472	0.8467
☐	NORTON_4	1.079e-05	1.8359	1.4620	1.4155
☒	NORTON_5	1.881e-05	0.8603	0.9936	0.8772

Calculate Scaling

▼ SCALARS

PP Average Scalars

Angle	Scalar
14.0	1.570e-05
25.0	0.9283
36.0	0.9305
40.0	0.8296

▼ OUTPUTS

Set Description... Suffix: SeismicScaling_Siminv_Insitu Save

11. Give the scalar set a name with a suffix 'Siminv_Insitu' and click on 'Save'.

Once the new scalars set are calculated, it's time to build the background models input in the inversion.

12. We need to set the model logs to the brine case. In the menu bar at the top of the screen, click 'Data > Select Well Data M > Select'. In the pop-up double-click on each entry for Vp, Vs and Rho in turn and select the *_WTR_conditioned logs. When complete, click 'OK'.

Select Markers, Logs, Minerals & Saturations

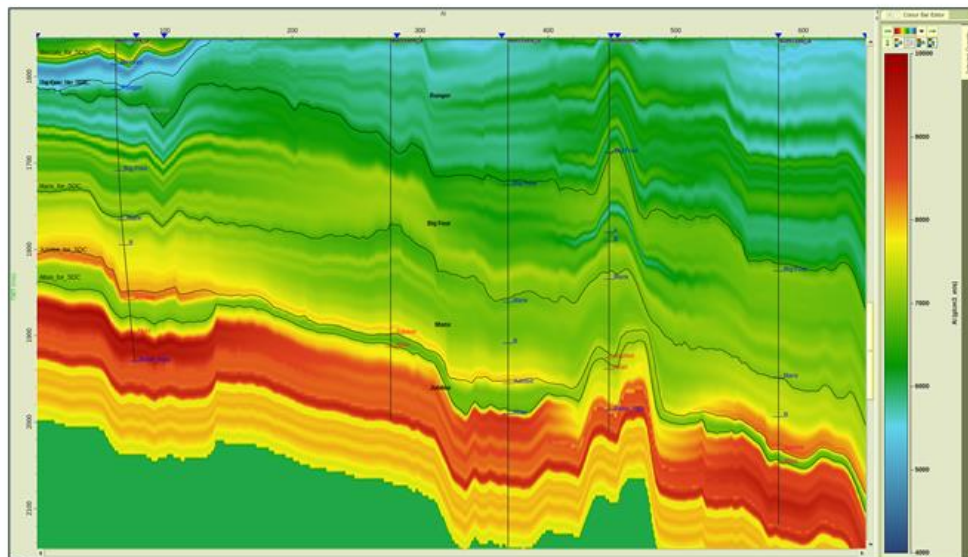
Change Data Types... Change Fluids... Change Minerals...

Well	Colour	GR	Vp	Vs	Rho	Por	Water	Oil	Gas	Quartz	Shale
NORTON_1	Red	GR	Vp_WTR_conditioned	Vs_WTR_conditioned	Rho_WTR_conditioned	PHIE_fill	COMP	So	Sg	COMP	Vshale
NORTON_2	Blue	GR	Vp_WTR_conditioned	Vs_WTR_conditioned	Rho_WTR_conditioned	PHIE_upd	Sw	COMP	Sg	COMP	VSH_upd
NORTON_3	Green	GR	Vp_WTR_conditioned	Vs_WTR_conditioned	Rho_WTR_conditioned	PHIE_upd	SWE_upd	SOE_upd	COMP	COMP	VSH_upd
NORTON_4	Green	GR	Vp_WTR_conditioned	Vs_WTR_conditioned	Rho_WTR_conditioned	PHIE_upd	SWE_upd	SOE_upd	COMP	COMP	VSH_upd
NORTON_5	Red	GR	Vp_WTR_conditioned	Vs_WTR_conditioned	Rho_WTR_conditioned	PHIE_upd	Sw_upd	COMP	Sg_upd	COMP	VSH_upd

Note that we have selected the brine case here as we do not want to bias our inversion results towards the hydrocarbon case, instead we want the seismic data to drive the impedances away from brine saturated values and towards hydrocarbon saturated values. Though this means we are starting with an intentionally incorrect model at some of the well locations, this will increase confidence in the results that indicate hydrocarbon presence as we know these are not being influenced by 'interpolated hydrocarbon' in the background model.

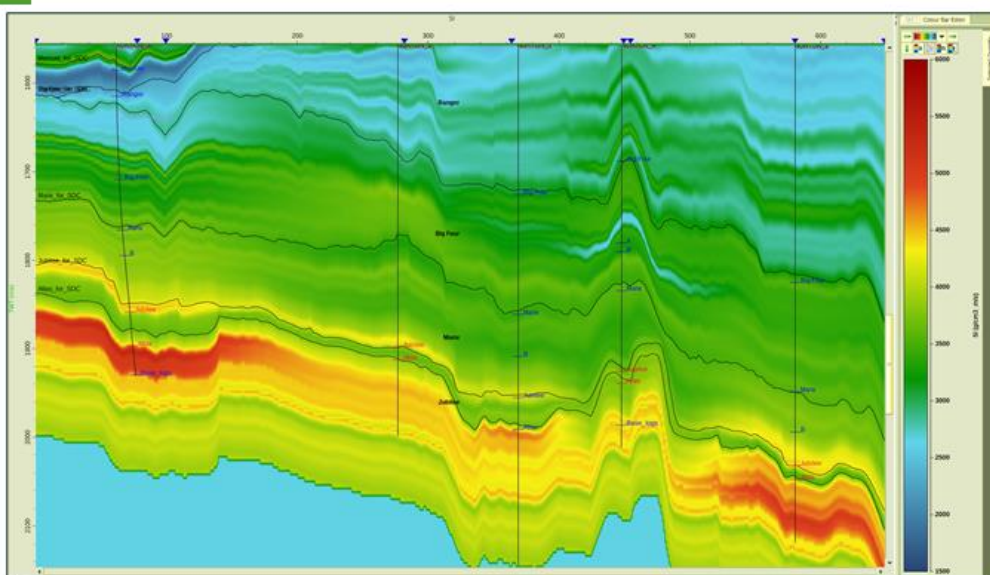
13. Elastic quantities in the 'Base Property Model' are now water-bearing quantities.

Inversion background model: AI



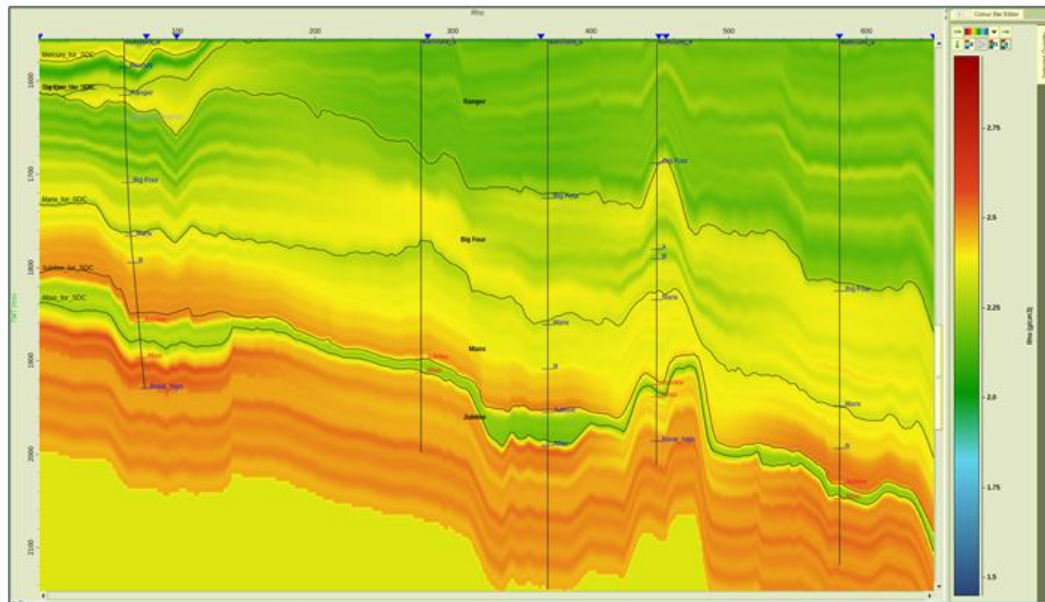
You need to check for sensible interpolation between wells, but also for any high-amplitude spikes on the logs such as are sometimes seen at top and base of the log. These must be edited out, because after high-cut filtering they may dominate the model response. (Log editing can be done in the RD Well Viewer).

Inversion background model: SI



This would not be a good starting model as it imposes too much structure on the solution, but for the inversion we shall use the brine logs and the model will be low-pass filtered to the range 8 – 12 Hz, which leaves only a very smooth representation of the impedance.

Inversion background model: Rho



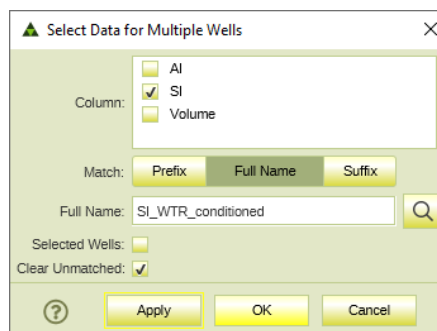
14. Go back to 'Simultaneous Inversion and Scaling' dialogue and check input **seismic wavelets and background model**. This is done to make sure that inversion module will refresh all the inputs after we've updated the background model.
15. Open the 'Seismic Scaling' tab. Toggle on 'Read Scalars', click the button, in the pop-up click on the scaler set you just created to select it, and click on 'OK'.
16. Open the 'Inversion Parameters' tab. Set the TWT limits to 'Jubilee' - 400 ms and 'Atlas' + 200 ms.



Exercise 6: log impedance background trends

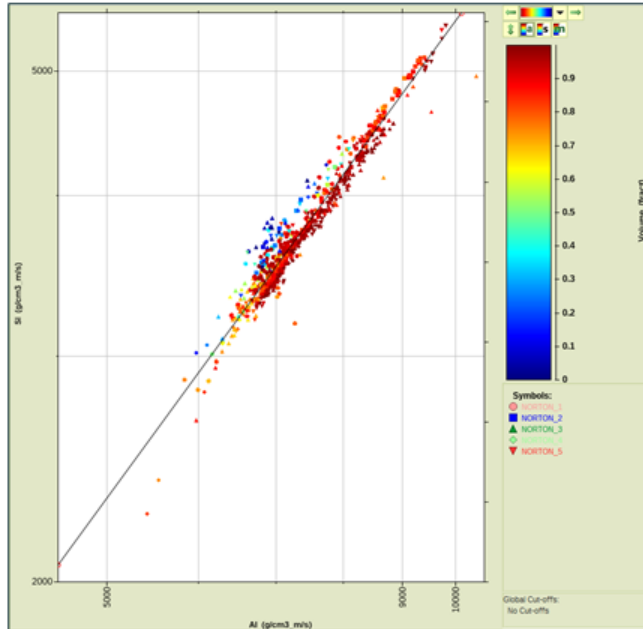
The simultaneous inversion does not invert for AI, SI and density directly but instead for deviations from background trends of $\ln SI$ against $\ln AI$ and of $\ln \rho$ against $\ln AI$.

1. In the same 'Inversion Parameters' tab > 'Background impedance Log/Log relationships' panel, click on 'Plot' in the 'AI:SI' line. A cross-plot appears. Click on the down arrow next to 'Colour' in the bottom left-hand corner and select 'Volume' instead of 'Rho'. Click on 'RPM Templates' in the bottom right and toggle off any that are switched on and click 'OK'.
2. We want to make the background trend from the brine logs. In the top-left 'Well Selection' box click on the search button , in the pop-up dialog, toggle on AI, then select the 'Full Name' button and select "AI_WTR_conditioned". Click 'Apply', the selections in the table will update. Repeat the same steps for SI.



3. In the top-right 'Well W.I. Selection' box check that 'Atlas', 'Big Four', 'Jubilee' and 'Manx' are toggled on, then in the top line toggle on 'Use'. Click 'Refresh' at bottom left. A trend-line has been supplied on the plot. This is a fairly satisfactory fit to the background shale trend, but it can be adjusted by clicking-and-dragging the red end points.
4. To reset the trend line to the best fit of the displayed data, right-click in the plot area and select 'Add' > 'Add projection line'. Here we're fitting and average trend between the brine sands (blue) and the shales (red) to not bias the inversion to one lithology.

Ln SI vs. Ln AI



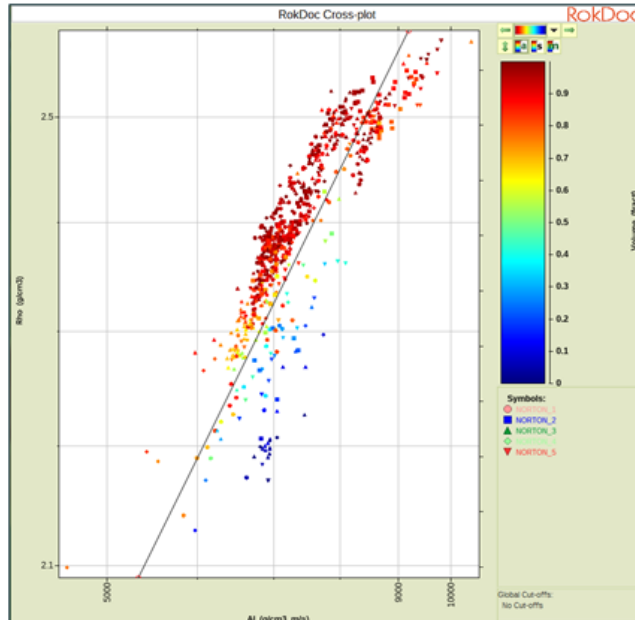
Sand trends (blue) are different from shale trends (red).

However, the best fit line looks to be a good background fit.

Name	Colour	Symbol	Use
Manx	Blue	Closed Vertical	☒
Jubilee	Red	Closed Circle	☒
Atlas	Green	Cross	☒
Big Four	Yellow	Closed Square	☒

5. Back in the 'Simultaneous Inversion and Scaling' dialogue, click 'Get Fit' for AI:SI. The slope and intercept values are transferred to the white boxes. Click on 'Close' in the bottom-right of the cross-plot.
6. Click on 'Plot' in the 'AI:Rho' line. Set up the cross-plot in the same way as for AI:SI, select brine-bearing logs, colour the data by Volume and select WI: Manx, Jubilee, Atlas and Big Four. In this case, fitting a trend is more challenging because the data do not collapse onto a one clear trend. Move the trend line to your liking, and click on 'Get Fit' to populate the white boxes. Click on 'Close' in the bottom-right of the cross-plot.
7. So that everyone is using the same values in the subsequent inversion, you can **manually enter slope/intercept values of 1.2511 and -1.2641 for AI:SI and 0.3829 and -1.1015 for AI: Rho.**

Ln ρ vs. Ln AI

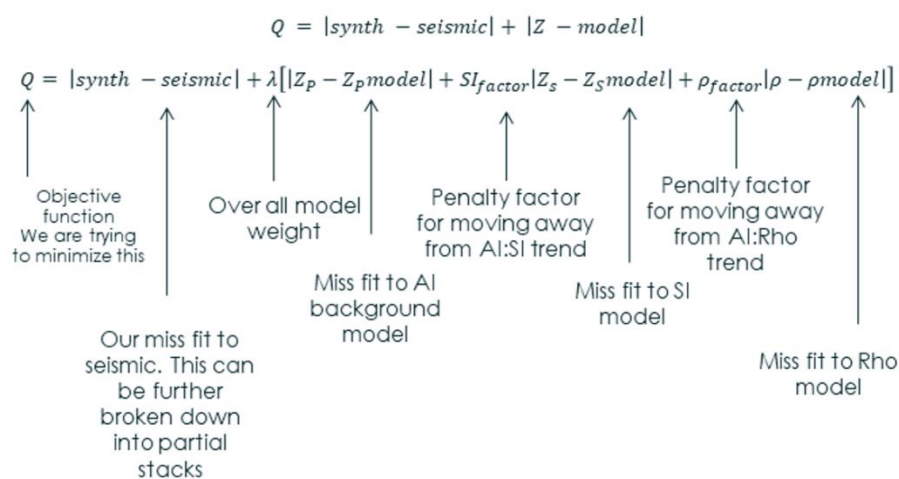


Sand trends (blue) are again different from shale trends (red).

However, the trend line shown looks to be a fair overall fit.

Name	Colour	Symbol	Use
Manx	Red	Closed Vertical	☒
Jubilee	Blue	Closed Circle	☒
Atlas	Green	Cross	☒
Big Four	Yellow	Closed Square	☒

Conceptual model of Simultaneous inversion

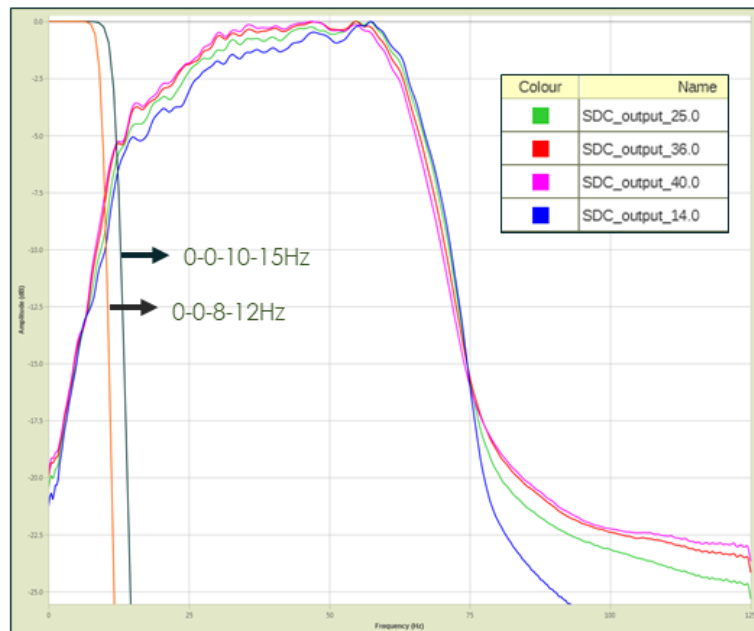


BACKGROUND MODEL FILTER SETTINGS

These settings will be investigated during testing of the inversion parameters. If the frequency limit is too high, the inversion result will probably be a good fit to the well data but the synthetic will not be a good fit to the input seismic; the converse is true if the frequency limit is too low.

The starting point is to work out what frequency range is needed to fill in the gap in the seismic low-frequency response. To work this out we can view the spectra plot for our seismic data. Looking at the spectra plot, 10-15Hz high cut filter should be fine. We will also experiment with other high cut filters later on and view the difference in the inversion results.

High-cut filter for background model



Spectral plot created using conditioned seismic between jubilee (-400ms) and Atlas (+200ms) – same as the inversion window.

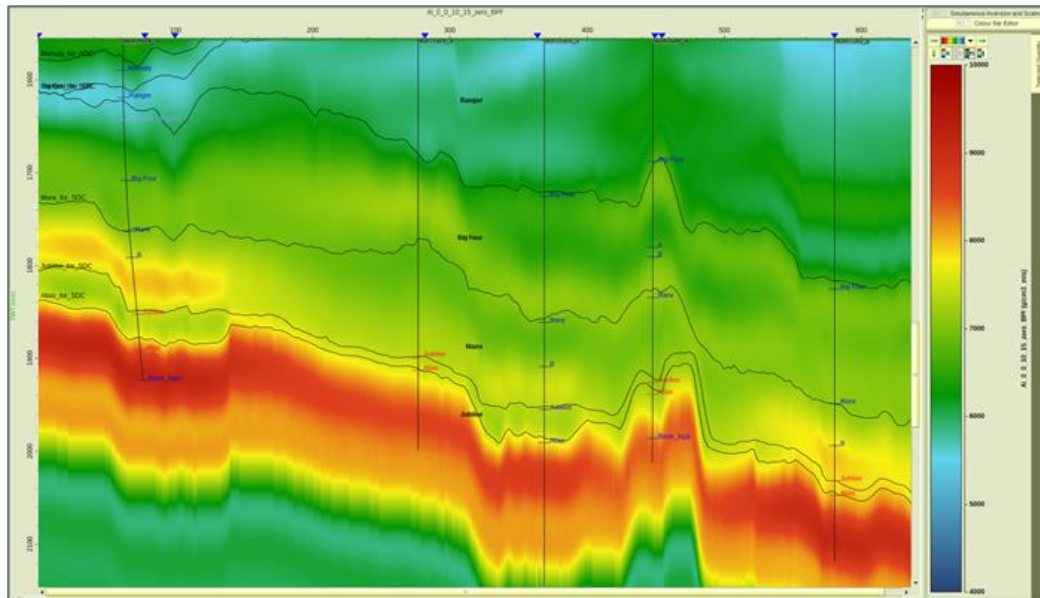
The plot is also showing a two wavelets:

Black line: wavelets created using 0-0-10-15Hz

Orange line: wavelet created using 0-0-8-12Hz

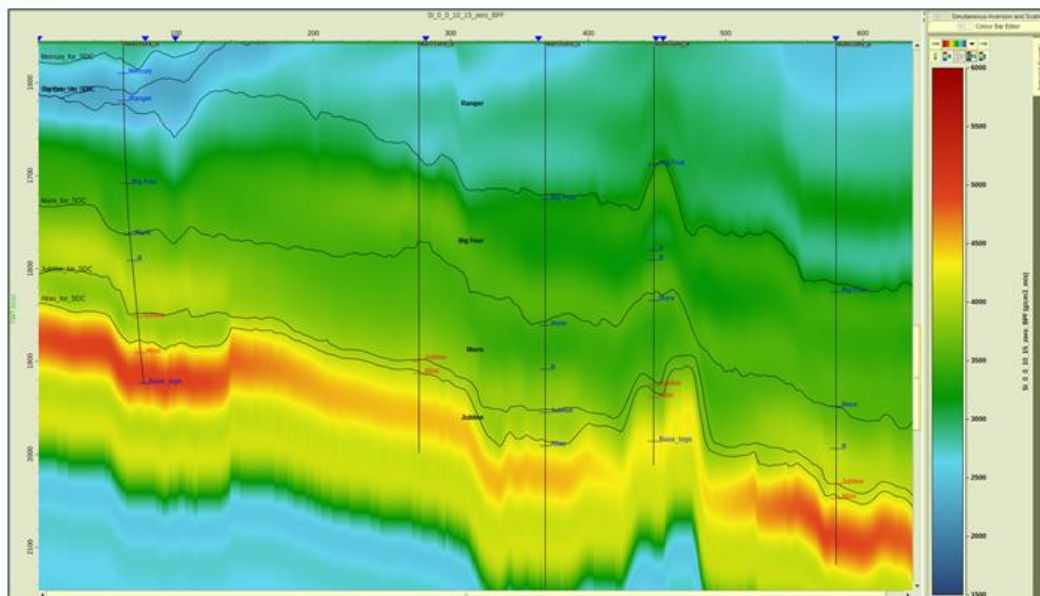
We can observe that 10-15Hz is a reasonable frequency cut-off; the cutoff should not be too steep to avoid ringing.

AI model filtered at 10-15Hz



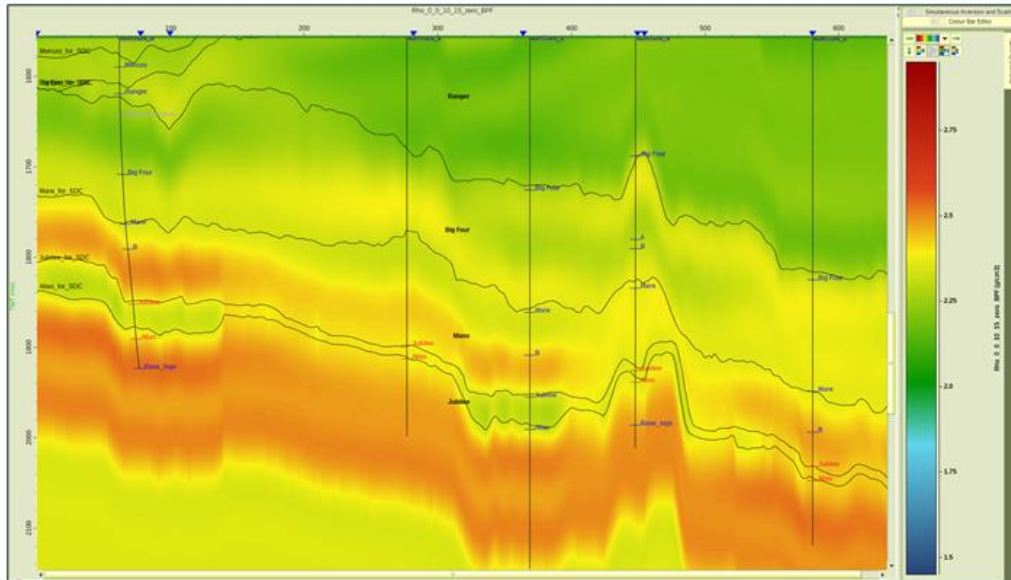
The model is quite bland at the objective level and is not going to impose structure on the inversion result.

SI model filtered at 10-15Hz



The model is again fairly bland at the Jubilee sand and is not going to impose much structure on the inversion result.

Density model filtered at 10-15Hz



There is a little lateral variability in the model, which will have to be kept in mind if any usable density is output from the inversion.



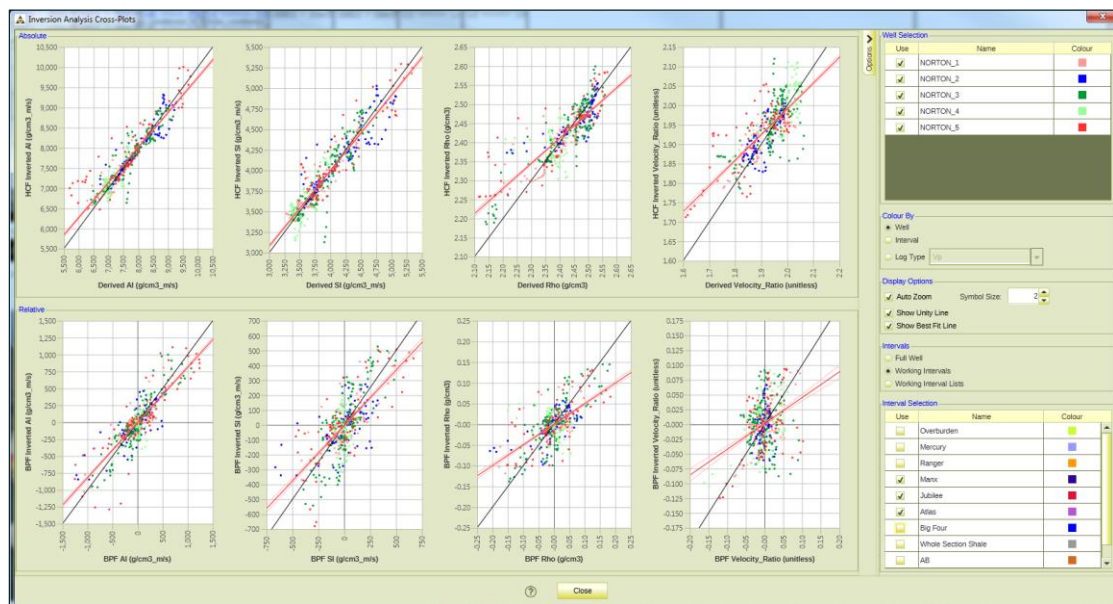
Exercise 7: Inversion parameter testing

1. In the 'Inversion Parameters' tab toggle on 'Filter Background Impedance Model' and set 'High Frequency Cut-off Start' to 10 Hz, 'End' to 15 Hz.

MODEL WEIGHT

This controls the balance between the deviation of impedance from the initial model and the synthetic error (difference between synthetic and actual trace). The algorithm tries to minimise both, but this weight controls the relative importance of the two. Finding a good value may involve trial and error over an extensive range.

2. **As a starting value, set 'Model Weight' to 0.05.** We shall adjust this to get a good fit to the well AI. **Set the 'Impedance Factor' to 1 for both SI and Rho.** These are additional model weight multipliers for these quantities and will be individually adjusted later to get the best fit to the well data. **Set the stack weights to 1.0 throughout.** These weights allow any particular stack to be given higher or lower weight, e.g. depending on the noise level.
3. In the 'QC and Calculate' tab, toggle on 'Synthetics', 'VpVs ratio' and 'Residual Traces'. Under "Filter Quantities" set the low cut to match the high cut filter for the LFBM (i.e. 10-15Hz). Expand the 'Invert Around wells'. In 'Select Wells' click on the '+' button then select 'Norton-1', 'Norton-2' and 'Norton-3' and click 'OK'. For parameter testing we will primarily use the Well QC view, but you may also wish to check the 'Cross Plots' and 'Statistics' options. The cross plots provide a useful comparison between the well log data and the inversion results in various different elastic spaces. The statistics can be useful for determining the optimum inversion when the differences are too small to be visually distinguished. If you choose to show statistics you should also screengrab the results or export them to a .csv file with an appropriate name for comparison with other inversion tests.
4. **Click on 'Invert Around Wells'.** Once calculation finished buttons 'Well Viewer' 'Cross-Plot' and 'Statistics' become active. You can click each one in turn to examine QC Cross-plots, statistics and inversion results around wells.



Inversion Analysis Statistics								
RMS Error								
Well Name	Use in Average	AI_Inv	AI_InvHCF	SI_Inv	SI_InvHCF	Rho_Inv	Rho_InvHCF	VpVs_Inv
NORTON_1	☒	328.29	274.36	170.80	137.58	0.05	0.04	
NORTON_2	☒	208.58	167.79	151.74	126.80	0.04	0.03	
NORTON_3	☒	262.09	204.03	172.10	141.97	0.04	0.04	
NORTON_4	☒	360.64	280.75	229.61	179.87	0.05	0.05	
NORTON_5	☒	648.35	588.08	369.24	334.04	0.10	0.09	
Average		361.59	303.00	218.70	184.05	0.06	0.05	
Cross Correlation								
Well Name	Use in Average	AI_Inv	AI_InvHCF	SI_Inv	SI_InvHCF	Rho_Inv	Rho_InvHCF	VpVs_Inv
NORTON_1	☒	0.94	0.96	0.96	0.97	0.91	0.94	
NORTON_2	☒	0.98	0.98	0.97	0.98	0.94	0.96	
NORTON_3	☒	0.96	0.98	0.95	0.97	0.95	0.96	
NORTON_4	☒	0.76	0.83	0.77	0.84	0.81	0.86	
NORTON_5	☒	0.82	0.85	0.88	0.90	0.69	0.74	
Average		0.89	0.92	0.91	0.93	0.86	0.89	
Proportion of Energy Predicted (PEP)								
Well Name	Use in Average	AI_Inv	AI_InvHCF	SI_Inv	SI_InvHCF	Rho_Inv	Rho_InvHCF	VpVs_Inv
NORTON_1	☒	1.00	1.00	1.00	1.00	1.00	1.00	
NORTON_2	☒	1.00	1.00	1.00	1.00	1.00	1.00	
NORTON_3	☒	1.00	1.00	1.00	1.00	1.00	1.00	
NORTON_4	☒	1.00	1.00	1.00	1.00	1.00	1.00	
NORTON_5	☒	0.99	0.99	0.99	0.99	1.00	1.00	
Average		1.00	1.00	1.00	1.00	1.00	1.00	
		Export Close						

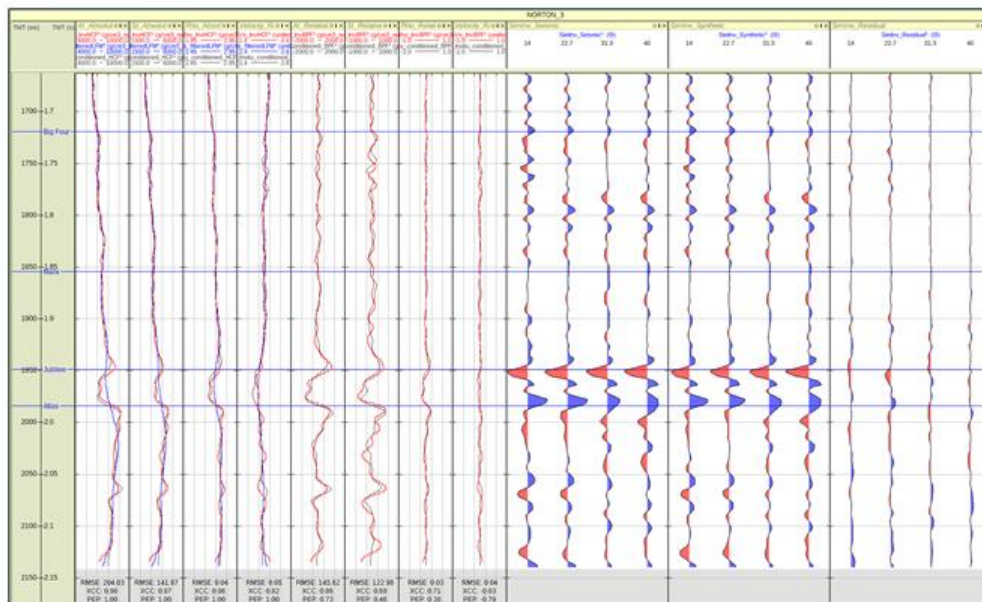
- In the well viewer, zoom in to cover the interval of interest: Ctrl-click in the TWT scale, hold down the mouse button and drag to the required TWT interval. The display shows the actual well logs in black, the inversion result in red, and the filtered background model in blue.



6. If required change the log scales: right-click on a log name in the track header, click on 'Set Log Plot Range', enter Left and Right values, make sure the box 'Apply to all logs in track' is checked, then click OK. Suitable ranges are 4000 – 10000 for AI, 2000 – 6000 for SI, 1.45 – 2.95 for density, and 1.5 – 2.5 for velocity ratio. Normally correct values should be pre-set in project settings.

The spectra plot is another useful QC tool. Ideally the input seismic and synthetic spectra should be a very close match. The residuals should have a fairly uniform low amplitude and a fairly flat distribution across the frequency range of the seismic as ideally the residuals will be random 'white noise'.

Norton-3 QC plot, low fz: 10-15Hz, model weight 0.05



Good inverted impedances match but poor Vp/Vs modelling. Also, good modelling of relative logs.

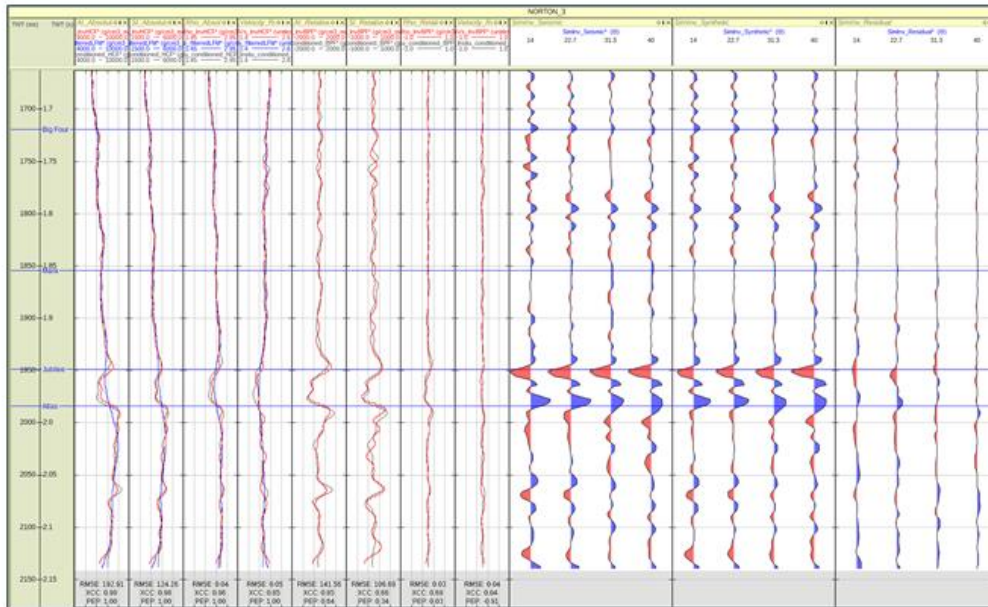
In the e-course, Norton-1 is used as an example for well QC.

7. We can try adjusting the model weight to see if we can improve the AI fit. In the 'Simultaneous Inversion' tab, alter the model weight to 0.02 and click 'Invert Around Wells'.



8. Repeat the process with a model weight of 0.1.

Norton-3 QC plot, low fz: 10-15Hz, model weight 0.1



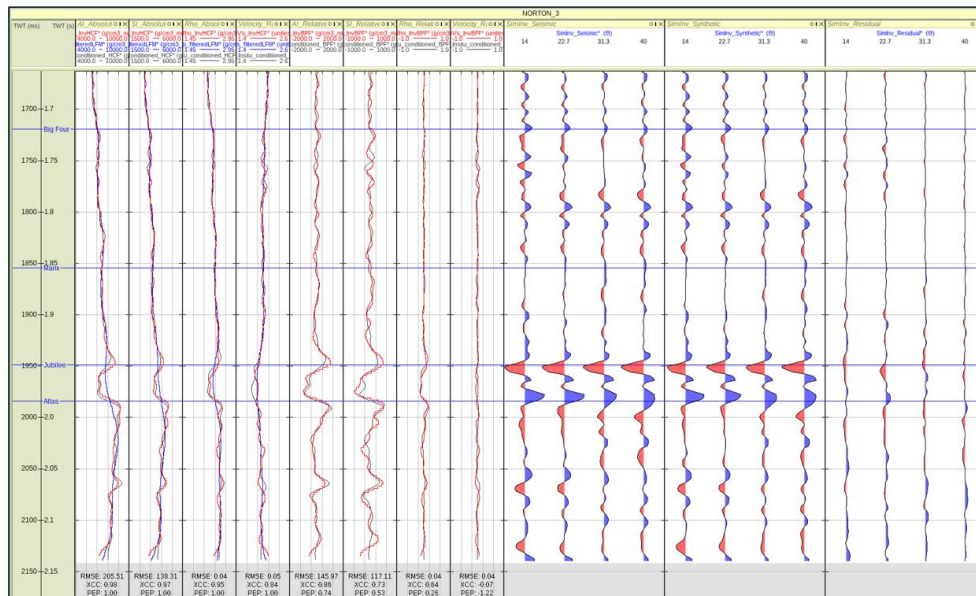
The relative inverted impedances are better good match than using model weight of 0.05. However, previous test showed better modelling of absolute inverted impedances.

Model weight of 0.05 gives the best result so far. In the real world we would refine this value further by trying even lower values. For the course, we will use this value and see if altering the impedance factors improves the result.

- In the 'Simultaneous Inversion' tab, change 'Model Weight' to 0.05 and set the Impedance Factor for SI to 2. 'Invert Around Wells' as before, you do not need to save every test, however it is often a good idea to save when there are clear improvements or when you settle on a final value. Does the SI look underconstrained or overconstrained by the low frequency model shown? Have any of the other results improved?



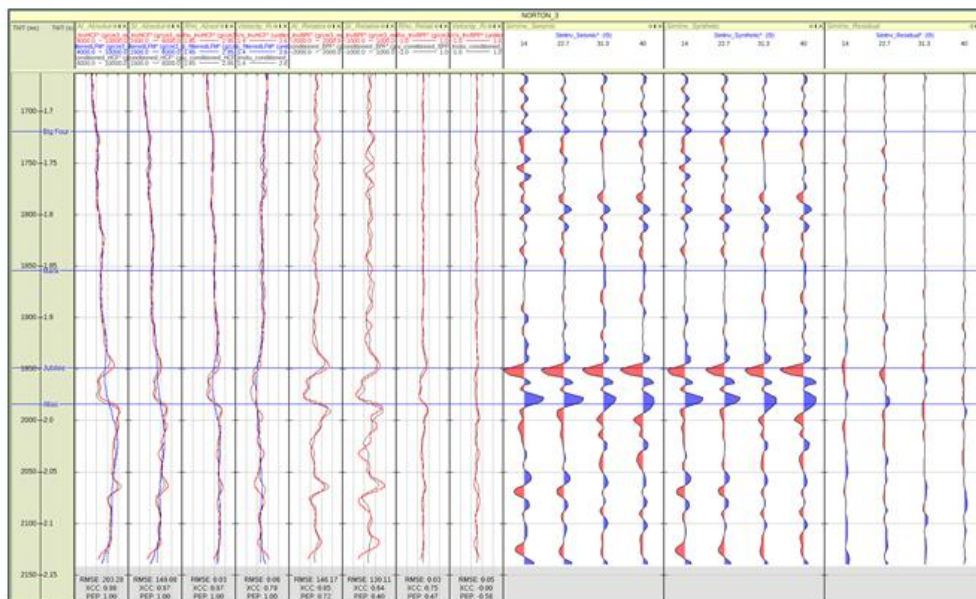
Norton-3 QC plot, low fz: 10-15Hz, model weight 0.05, SI factor: 2



Better modelling of SI, slight over-prediction of relative impedances, compared with SI factor of 1.

10. Set the SI impedance factor to 0.5 and try again.

Norton-3 QC plot, low fz: 10-15Hz, model weight 0.05, SI factor: 0.5

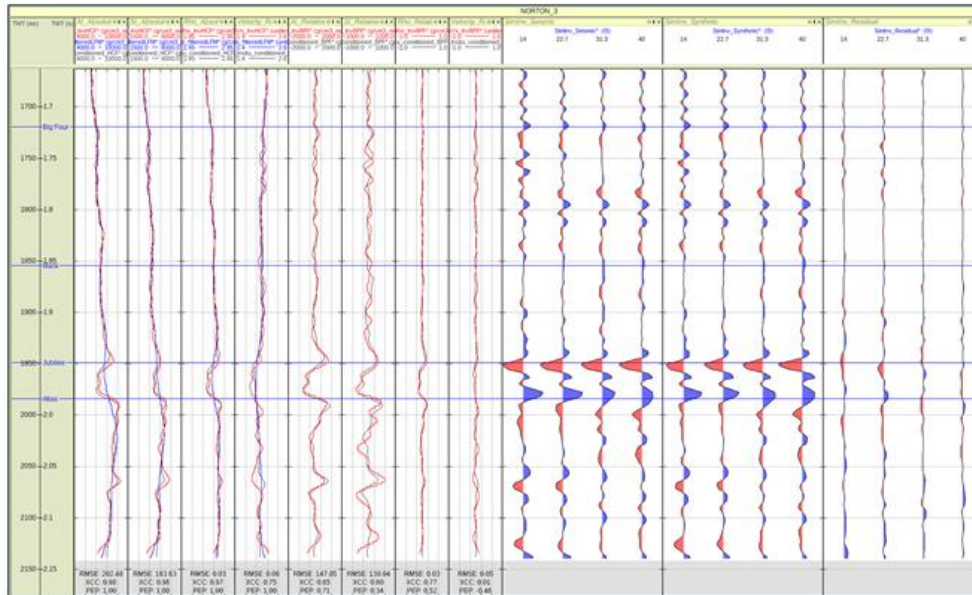


Overall, better match between inverted and insitu logs than using SI factor of 1.



11. Set the SI impedance factor to 0.2 and try again.

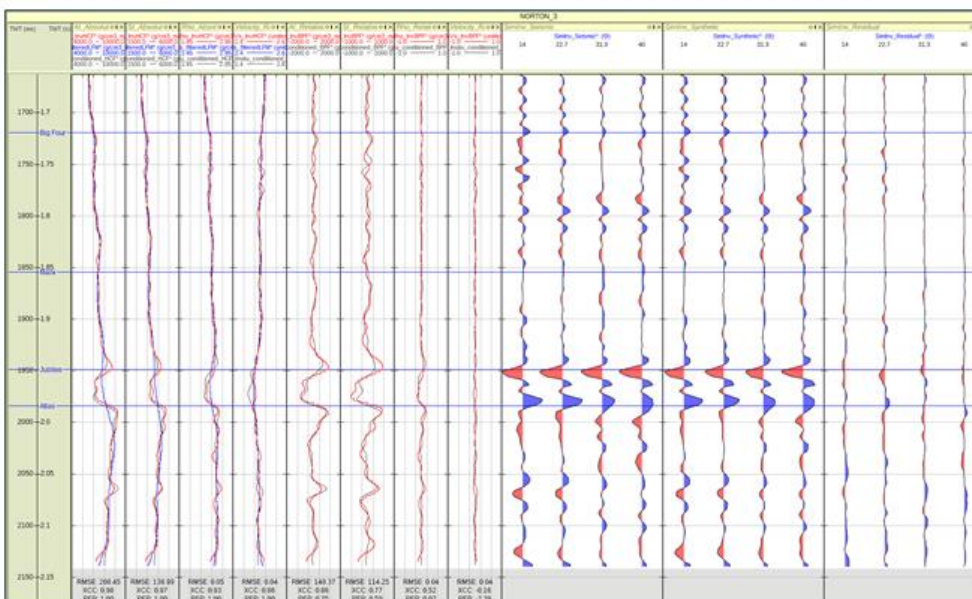
Norton-3 QC plot, low fz: 10-15Hz, model weight 0.05, SI factor: 0.2



SI factor of 0.5 gives better match than 0.2.

12. Set the SI factor to 5 and try again.

Norton-3 QC plot, low fz: 10-15Hz, model weight 0.05, SI factor: 5



SI factor of 1 gives better match between inverted absolute/relative and insitu logs.



We could investigate the density factor in the same way, but this is not important for our objective of fluid and lithology discrimination in the Jubilee sand. So instead, what about the seismic stack weighting or wavelet scaling?

13. Calculate the inversion using Model Weight of 0.05 and SI factor of 5. Save the recipe giving it a clear name e.g. 'MW0.05_SiF5_StkF1.'

In real life we would investigate the model weighting factors across all of the wells (except for any 'blind' test wells, which would not be used here) to find out what would give the best overall result, but for the course we shall use the factors we have settled on from the work so far.

SUBSTACK WEIGHTS

So far we have assumed it is best to use all four substacks at full weighting, but in fact this is doubtful. We have just seen that the mid trace at well 4 shows high residuals. So, what happens if we would give this trace a low weight, in effect ignoring it?

14. In the 'Inversion Parameters' tab try reducing the stack weighting on volume 2 (Mid stack) to 0.01. QC the results as before. Note that the residual will increase, this is the intended behaviour. Look for improvements in the residuals of the other stacks or in the impedance results

15. Try increasing the stack weight to 0.5, are the results better or worse?

16. Try reducing both the Far and Ultra Far stack weights to 0.6, while revert mids back to 1 does this give any improvement?

LOW FREQUENCY MODEL

It is easier to see what effect is caused by a change in the low-frequency model if we look at a test inversion along a line.

19. 'Calculate' and save the best inversion so far along the line of section.

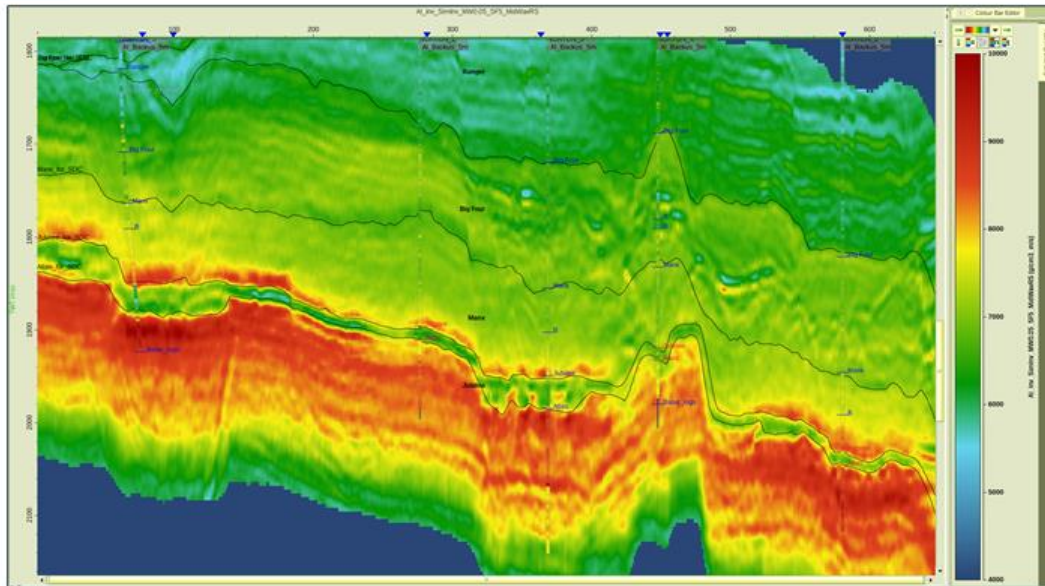
Expand the new entry in the tree 'Inversions' > 'Simultaneous Inversion' > 'Siminv' > 'Quantities' > 'Absolute'.

20. Review the results. Click the box next to 'AI_inv', click on the colour bar

icon at the top of the window . The colour bar appears at the right-hand side. Right-click in the bar, click 'Set Scale' and set the session AI range to 4000 – 10000 and click 'Turn on Session AI Range for all session AI data'. To maximise the seismic area click the right-facing arrow at the top of the vertical strip separating the seismic display from the colour bar .

21. You can navigate around the seismic view using scroll bars and zoom options on the top of the window.

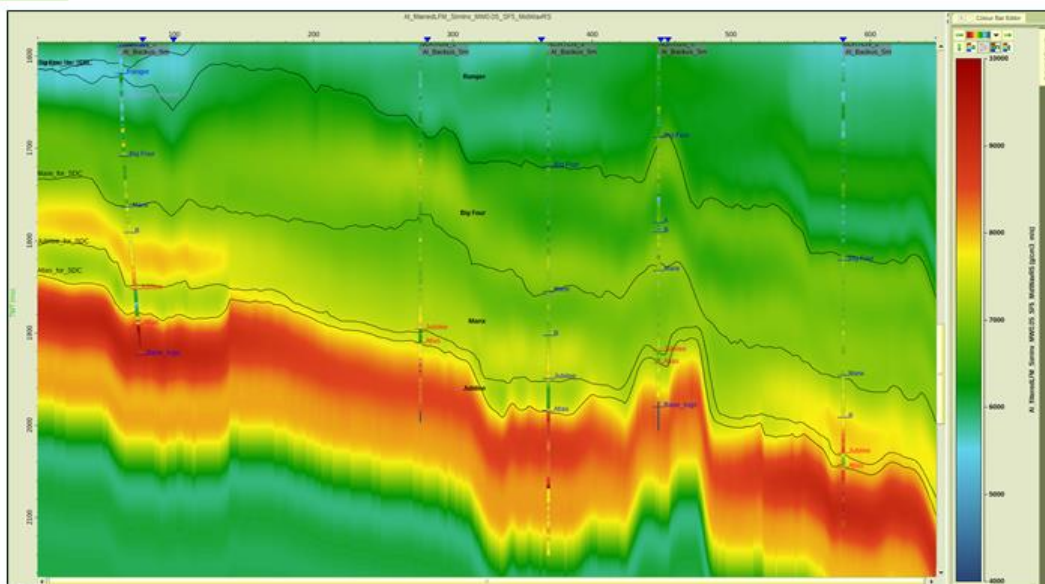
Inversion along test line: AI



The Jubilee sand is a clear low-AI layer.

22. To view the low frequency model, uncheck 'AI' in the data tree and check 'AI_filtered_LFM' instead. Set the colour bar in the same way as for the AI display. The low frequency model is not imposing much structure on the impedance at Jubilee sand, which shows low AI relative to overburden and underburden.

Inversion along test line: AI low fz model

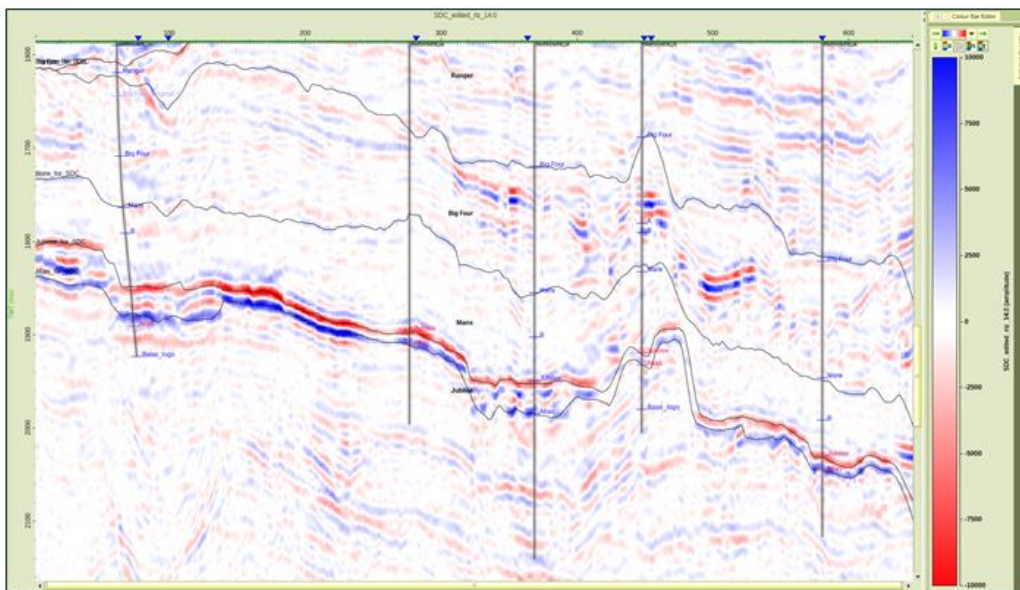


The LFM is not imposing much structure on the AI results.

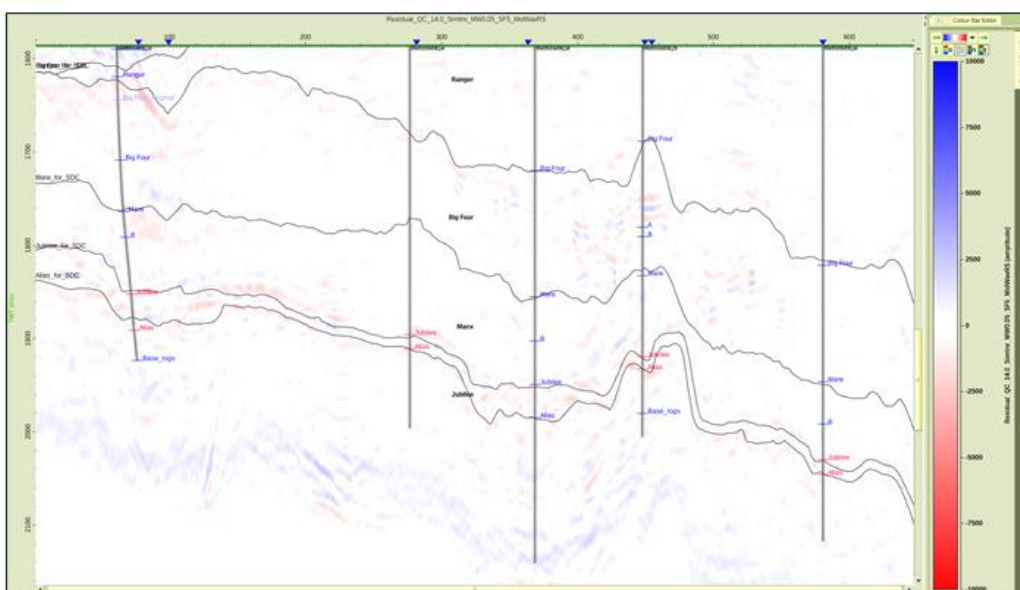
23. In the same way inspect the 'SI' and 'Rho' and their corresponding LFM's. The result for the Rho is noisier than for AI and SI.

24. Importantly, we can now check the seismic residuals (difference between synthetic and real seismic) to see how well the inversion is honouring the seismic data. Right click on 'Synthetic_QC_14.0' in the tree, click 'Set Scale', verify that the scale is set to -10000 to 10000 and click 'OK' then click the check box in the tree to display the synthetic.
25. Click the check box to display 'Residual_QC_14.0', again verify that the scale is set to -10000 to 10000. The residuals are small.

Test inversion: near trace synthetic



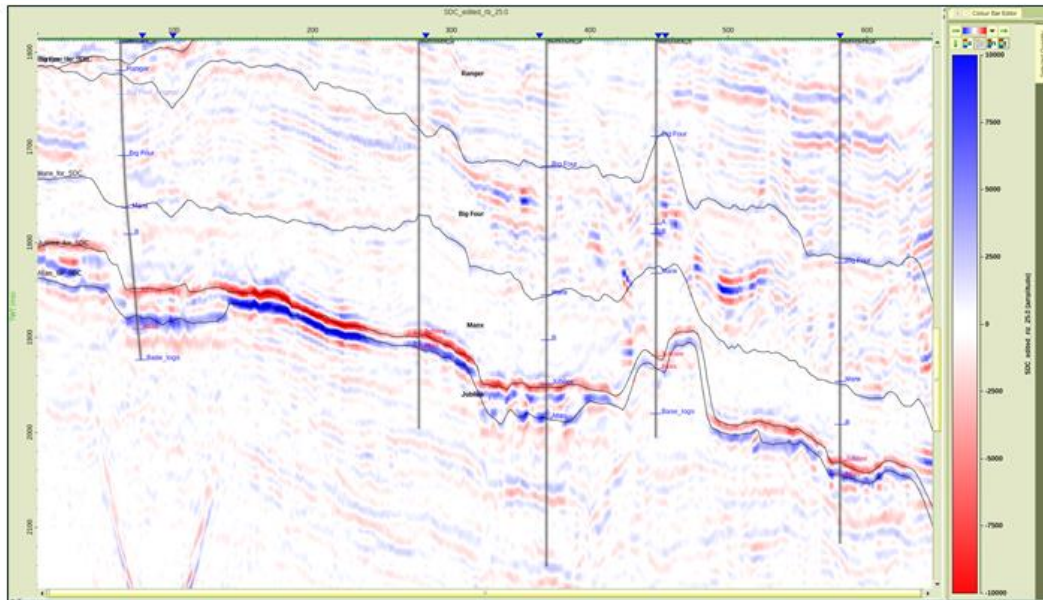
Test inversion: near trace residual



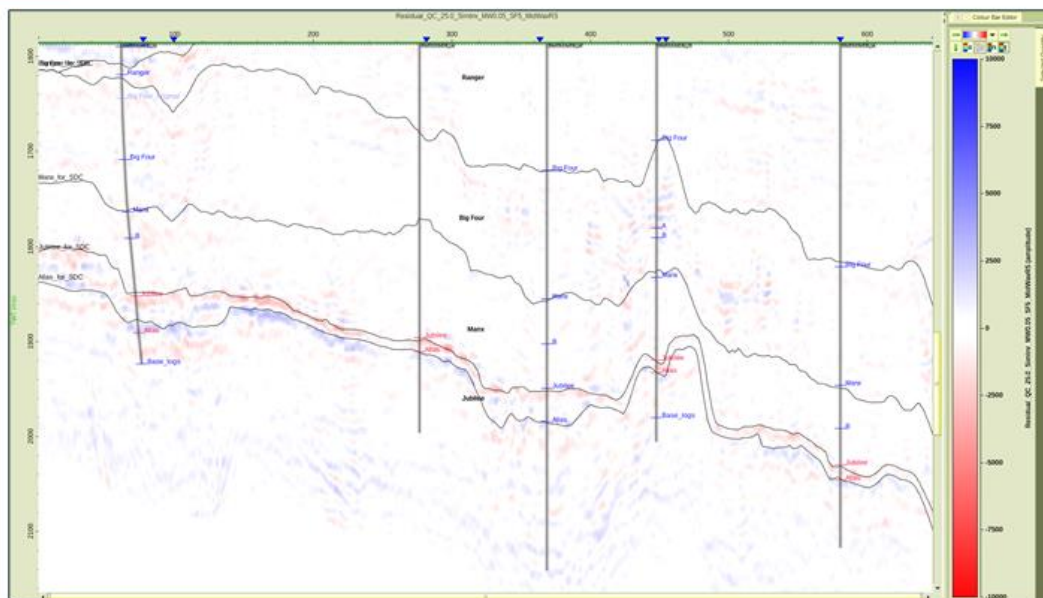
Residuals are low around Jubilee, at the end of the inversion window we can observe low frequency noise.

26. Repeat the process to display mid, far and ultrafar synthetics and residuals.

Test inversion: mid trace synthetic

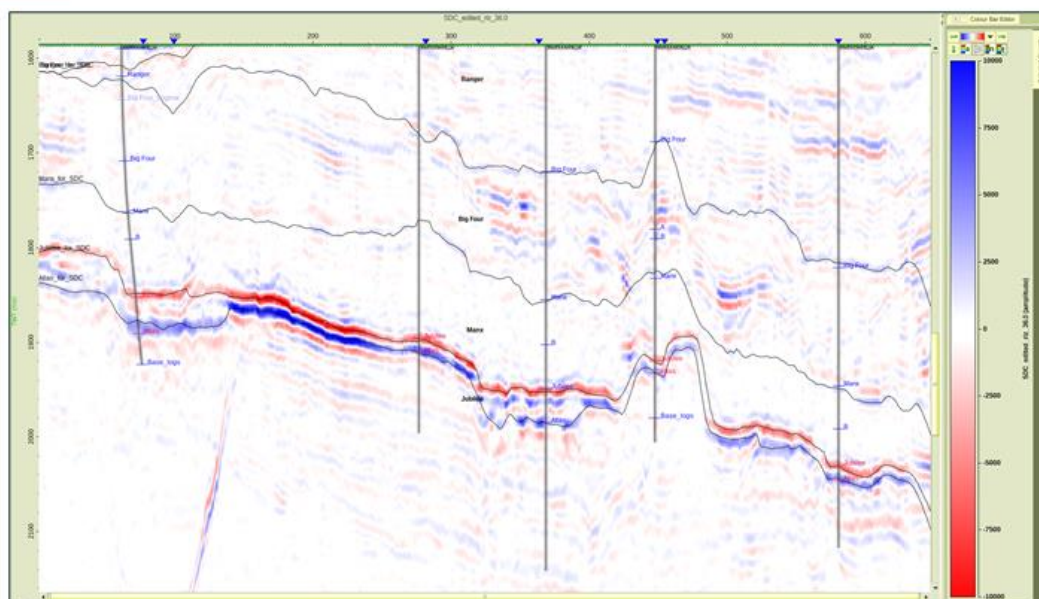


Test inversion: mid trace residual

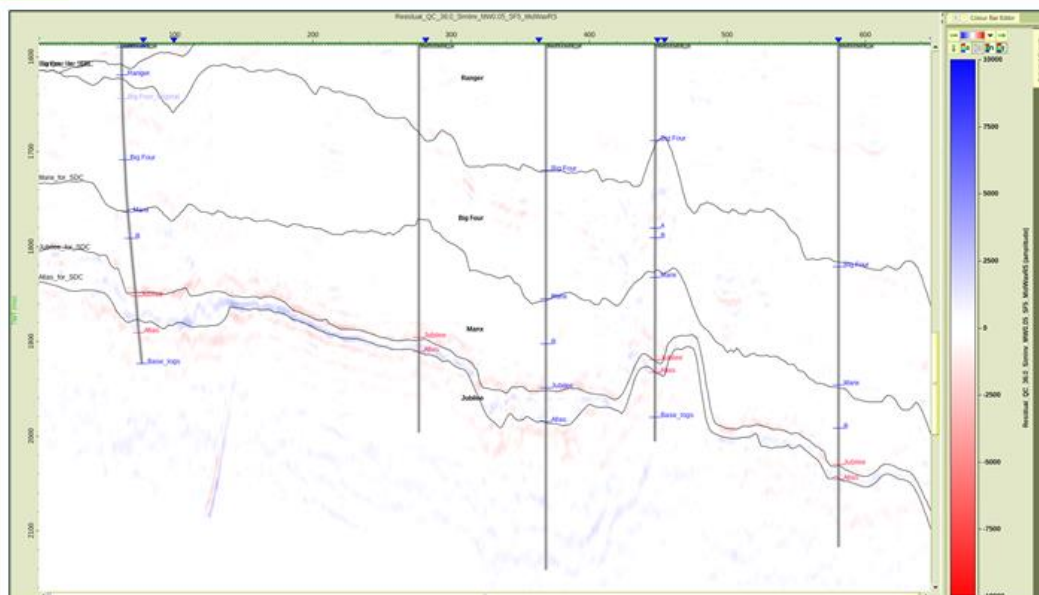


Residuals are larger around Jubilee. It can be observe low frequency noise.

Test inversion: far trace synthetic

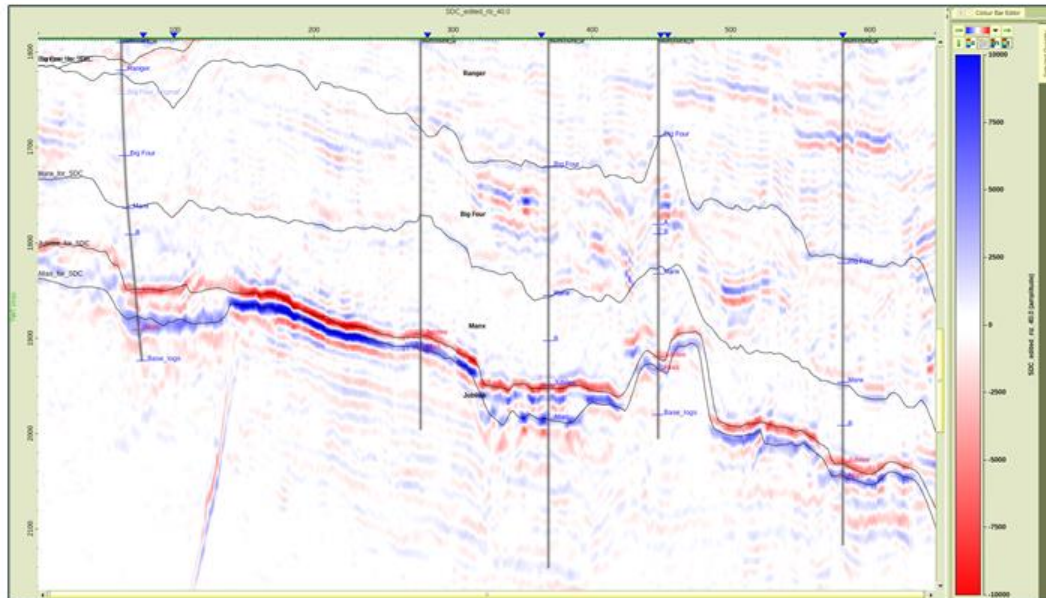


Test inversion: far trace residual

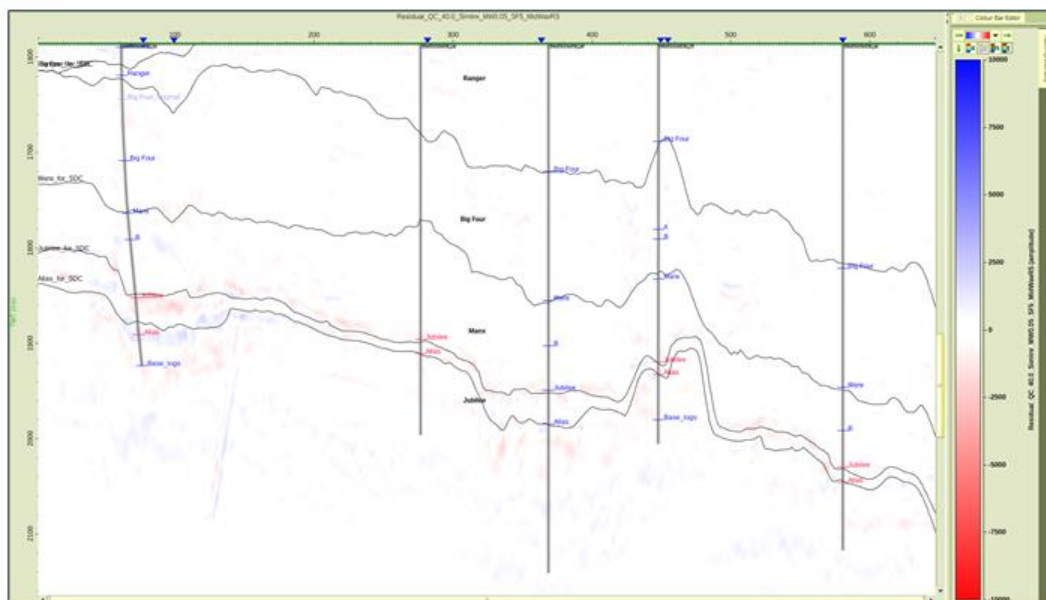


The residuals are lower outside the Jubilee. It is likely that the gas effect is not captured fully by the inversion causing higher residuals in gas areas.

Test inversion: ultrafar trace synthetic



Test inversion: ultrafar trace residual



Similar to the far stack residuals, these residuals are very low outside the Jubilee but slightly higher in gas affected areas.

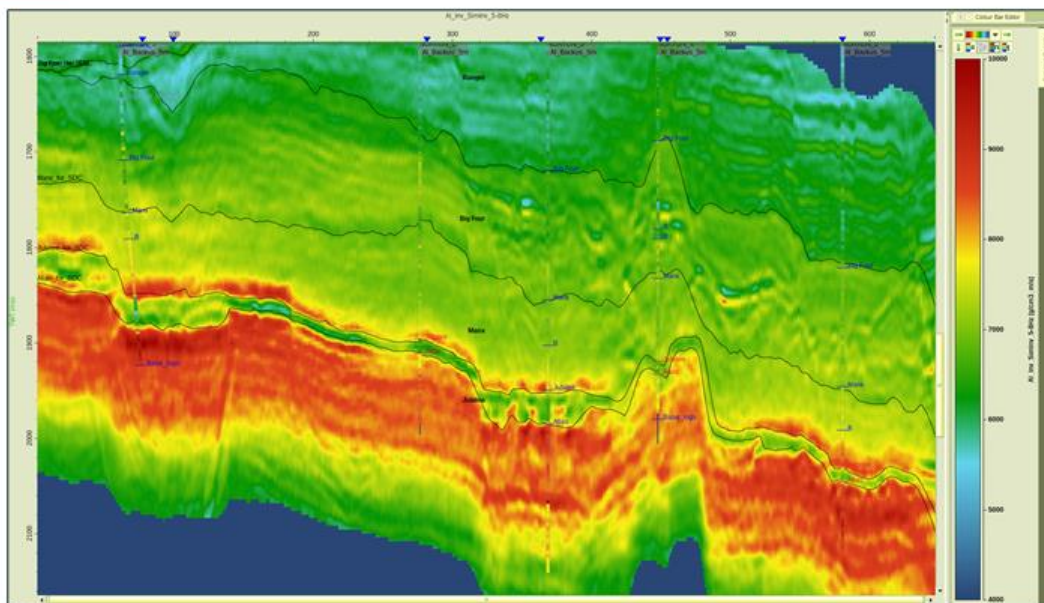


In general the residuals are quite low, except for the near stack. However close to Norton-5 we see that the residuals become brighter. This could be because the inversion is not able to deviate far enough away from the background model to capture the large fluid effect caused by the gas. If we were to continue testing we might want to experiment with dropping the Model Weight further. This likely cause the inversion results to degrade elsewhere by may improve results in the Jubilee Sand.

How sensitive is the result to the low frequency model?

27. **Bring back the simultaneous inversion dialogue by clicking on the left arrow at the top of the vertical strip at the right-hand edge of the seismic window. Under 'QC and Calculate' toggle on Poissin's ratio, LambdaRho and MuRho, click 'Calculate Inversion' enter the suffix '10-15' and click 'Save'.** The inversion quantities in the tree turn black to indicate they have been saved (red type denotes unsaved quantities).
28. **Change the high frequency cutoff from 10 – 15 Hz to 5 – 8 Hz and click 'Calculate Inversion'. Display some of the inverted quantities as for the previous inversion.** The impedances are only slightly changed, and the residuals are almost unchanged.

Test inversion: AI 5-8Hz background model

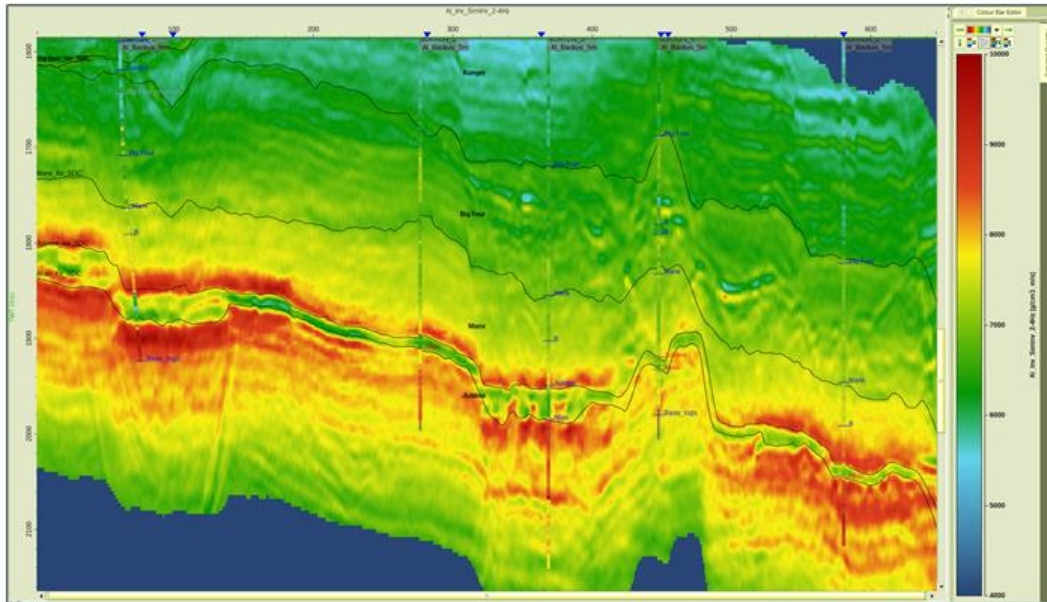


Similar result to AI using 10-15Hz background model.

There is little change from the previous result with a 10 – 15 Hz model.

29. Now change the cut-off frequencies to 2 - 4Hz and repeat.

Test inversion: AI 2-4Hz background model



Noticeably different from previous tests, especially in Jubilee sand.

There is some change in the impedances, though not much in the Jubilee sand, but almost no change in the residual. Within broad limits, the choice of high-cut filter is not important. For the course we shall use the 10-15 Hz version already saved.

In this manual some aspects of the inversion testing such as model weights, stack weights and high-cut filter are being discussed in more detail. While in the e-course, many other practical aspects of RokDoc inversion module are being demonstrated.

To realise the volumes (which makes them available to the project outside this RokDoc-3D session) or export the traces as SEG Y:

30. **Right-click on 'AI_Inv_10-15' and click on 'Export 2D' or 3D. Select 'AI_Inv_10-15' and 'SI_Inv_10-15' for export and remove the rest of the inversion outputs from the right panel. Enter the suffix as 'realise' and click 'OK'. The AI and SI appear in the tree under 'Seismic and Other Quantities' > 'Seismic'. Note that here you could realise traces over a 3D volume using 'Export 3D' and save them as a segy file to make them available to other applications.**



Comments on parameter testing

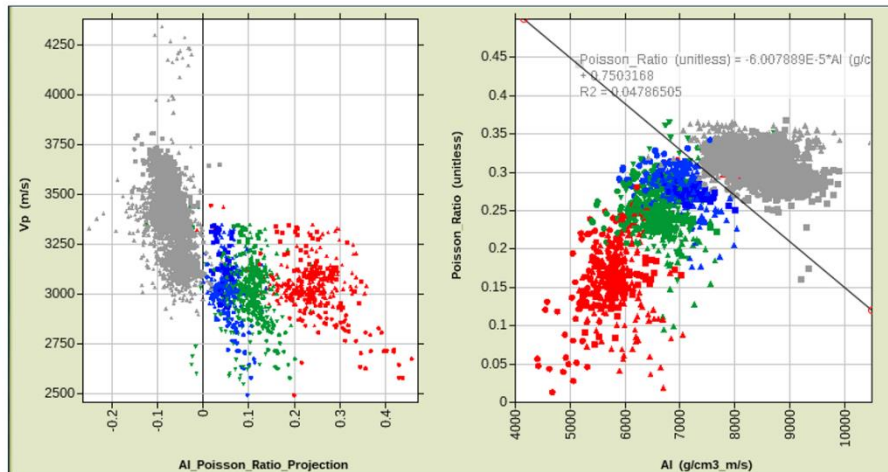
- Can be very time-consuming if there are many wells
- Concentrate on the interval of interest and what you want to get out of the inversion (here, Jubilee sand and V_p/V_s)
- Look carefully at the match to well impedance and the synthetic residuals and think about possible causes of problems:
 - Well log quality
 - Seismic noise
 - Wavelet phase/amplitude
 - Lateral seismic amplitude variations



Exercise 8: Making derivative outputs



AI-PR



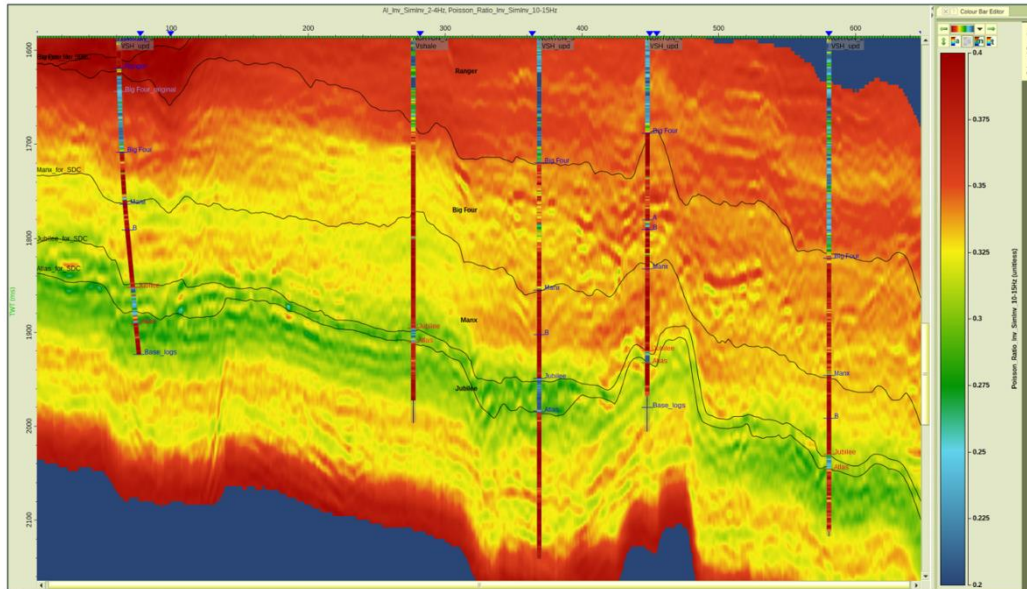
AI – PR plot from Norton-1, -2 and -3 data, as shown earlier in the course. Left-hand plot shows data projected parallel to the line on the AI – PR plot. We want to calculate the quantity $PR + 6.0078 \times 10^{-5} AI - 0.7503$.

1. We already have calculated PR and other derivatives. If not then **go back 'QC and Calculate' tab of simultaneous inversion, toggle on 'Poisson's ratio' and 'LambdaRho' and click 'Calculate Inversion'** Poisson Ratio and LambdaRho appear in the Siminv Quantities in the tree. **and save with a suitable suffix.**

Display the Poisson's ratio with a colour scale of 0.2 to 0.4. Right-click on any well trajectory and click on 'Show Logs for all Wells', toggle on 'Show logs of type'; and select Volume, then toggle on 'Active log' and click on 'OK'. Right-click on any well displayed again and select 'Well Display Settings'. Set the well path width to 0.1 inches. Toggle on 'User Defined Log Min and Max Range' and set the Min to zero and the Max to 1. Toggle on colour display, click 'Copy to All' and click 'OK'.

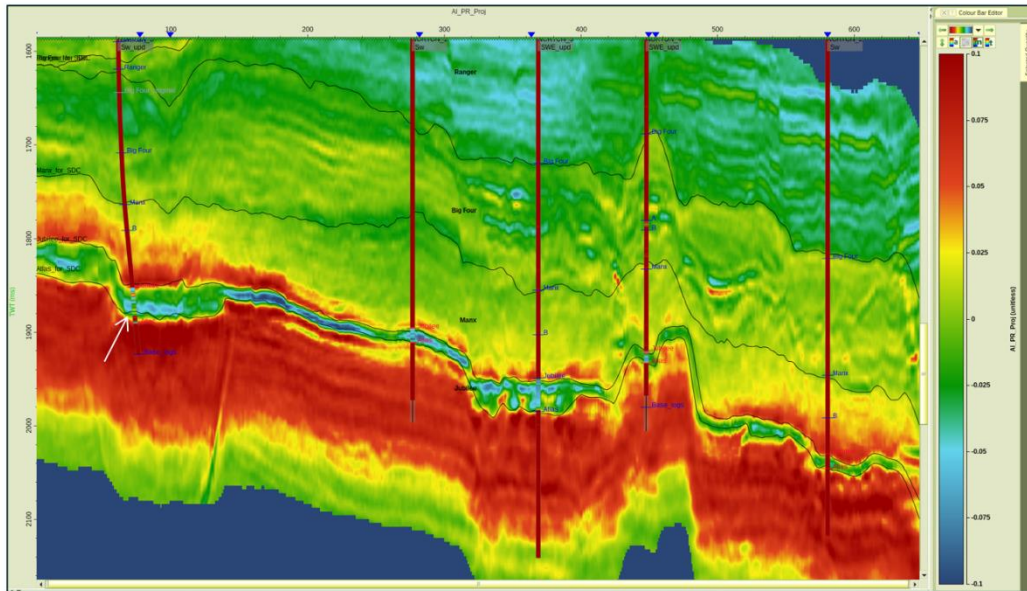


Poisson's ratio section



3. In the RokDoc-3D window, click 'Utilities' > 'Calculator' in the menu bar at the top of the window. Enter the formula $(\text{Poisson_Ratio} + 6.0078\text{E-}5 * \text{AI} - 0.7503)$ into the calculator from the slide above; use the 'Quantity' key to specify PR (SImlnv_10-15Hz) and AI (SimInv_10-15Hz), and the 'Constant' key to enter a number. The quantities you need are 'Poisson_Ratio_Inv_...' and 'AI_Inv...'. Note that this discriminator is negative for hydrocarbon sands and positive for shale.
4. On the 'Output Quantities' line enter the name as 'AI-PR_Proj', double click on 'Type' and select Ratio, click on 'Calculate' and then on 'Save Recipe'. Close the calculator. The result appears under 'Seismic and Other Quantities' > 'Calculator' in the tree. Click on the box next to 'AI-PR_Proj' in the tree to display it. Scale the colour bar to -0.1 and 0.1 and zoom out to show the entire length of the section.
5. Right click in the graph area below the section and click 'Measure Attribute'. In the popup set 'Measurement Category' to 'Value'. Set 'Measurement Type' to 'Max(-ve)' and set 'Quantity 1' to 'AI-PR-Proj'. Set Horizon 1 to 'Jubilee' and Horizon 2 to 'Atlas'. Give attribute a name and click 'OK'. Right click on the graph line, click on 'Set Range' and set the minimum to 0.2 and the maximum to -0.4. This choice flips the direction of the y axis so that up corresponds to hydrocarbon and down corresponds to brine sand and shale.

AI – PR projection



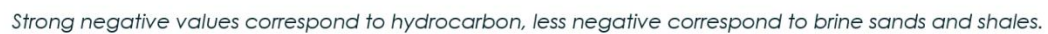
There is a GOC in well 5 at about 1880 msec. The other wells have oil fill except well 2 which has brine sand. There is a fair correlation of gas fill to the AI – PR projection, but the distinction between oil and brine sand is not clear.

The data show good discrimination for sand versus shale, and for gas versus oil, but not for oil versus brine.

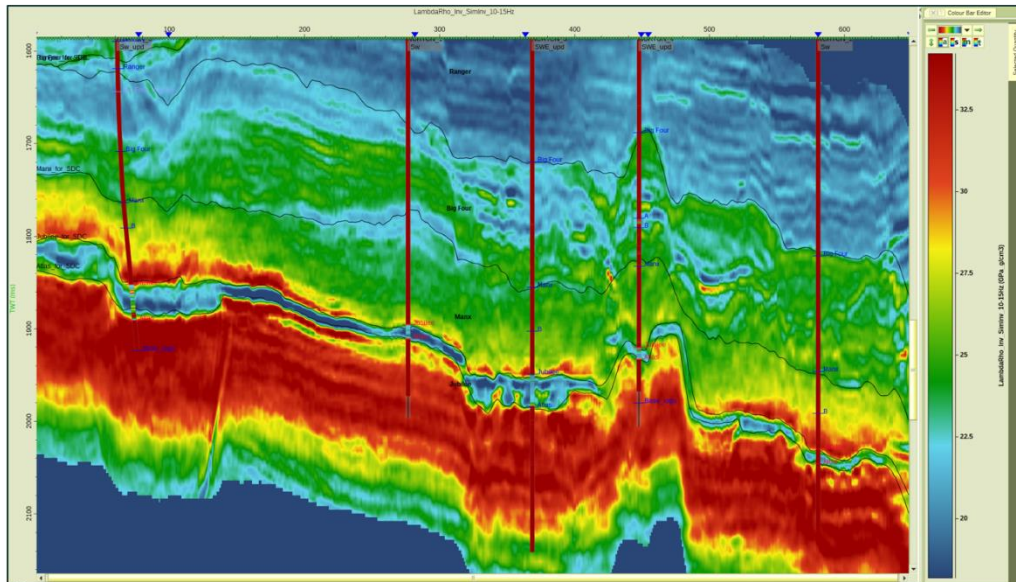
- The measured attribute in the previous step can be converted to an



attribute map by clicking on  at the top of the RokDoc-3D session. In the Map Viewer > data tree > Expand the 'Attribute' folder, then 'Calculation' folder and toggle on the attribute 'Max(-ve) AI_PR_Proj'. Set the 'Trace Skip' to 4 and click on 'Draw Map'.


$$\lambda\rho - \mu\rho$$


7. Toggle on the calculated $\lambda\rho$ in the tree, and autoscale the colour bar.

 $\lambda\rho$


It is a good sand discriminator and fairly good fluid indicator.

8. Create an attribute of LambdaRho Average over the reservoir zone: Right-click in the attribute panel and select 'Measure attribute'. Set 'Measurement category' to 'Average' and 'Measurement type' to 'Ave Amp'. Check that the quantity is set to LambdaRho_inv... . Set the horizons to the 'Jubilee' and 'Atlas'. Name the attribute LR_ave and click 'Ok'.
9. Check that the attribute looks sensible in the attribute panel. Open a map viewer and draw a map of the attribute at a trace skip value of 2. What geological features can you see, where would you predict hydrocarbons were present?
10. Right-click on $\lambda\rho$ in the data tree and select 'Export 3D' to export your final inversion results in 3D. Toggle on AI, SI, PR and VpVs quantities to the export list. Check the 'Add to session' box and click 'Submit' to start the export (this will take some time as it is calculating the inversion for the full 3D volume).



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